

Gastroenterology

SYMPTOMATOLOGY AND EVALUATION OF GASTROINTESTINAL DISEASE

Anorexia (Table 5.1)

Q. List common causes of loss of appetite (anorexia).

VOMITING

Q. List common causes of persistent vomiting.

- **Vomiting** is the forceful oral expulsion of gastric contents. It is a complex reflex and involves both autonomic and somatic neural pathways.
- **Nausea** is a feeling of wanting to vomit and often precedes actual vomiting.
- **Retching** is a strong involuntary unproductive effort to vomit. It is associated with contraction of abdominal muscles but without expulsion of stomach contents through the mouth.

Mechanism of vomiting: Synchronous contraction of the diaphragm, intercostal muscles, and abdominal muscles raises intra-abdominal pressure. This is combined with relaxation of the lower esophageal sphincter and causes forcible ejection of contents of the stomach.

Common Causes of Nausea and Vomiting (Table 5.2)

TABLE 5.1: Common causes of anorexia.

Infections: Viral fever and tuberculosis	Renal disease: Chronic renal failure
Endocrine diseases: Hypothyroidism, hyperparathyroidism, Addison's disease, and panhypopituitarism	Malignancies: Any malignant tumors, e.g., carcinoma stomach, pancreas. Leukemias and lymphomas
Liver disease: Hepatitis and cirrhosis	Psychiatric causes: Depression, and anorexia nervosa

TABLE 5.2: Common causes of nausea and vomiting.

Abdominal causes	
Mechanical obstruction Gastric outlet obstruction and small bowel obstruction*	Other intra-abdominal causes Acute appendicitis, acute cholecystitis, acute hepatitis, acute mesenteric ischemia, eosinophilic gastroenteritis, and gastric and duodenal ulcer disease*
Motility disorders ➤ Functional dyspepsia gastroesophageal reflux disease (GERD)* ➤ Gastroparesis*	
Drugs	
Aspirin and other nonsteroidal anti-inflammatory drugs	Cardiovascular drugs: Antiarrhythmics, antihypertensives, beta-blockers, calcium channel blockers, digoxin, and diuretics
Antidiabetic agents	
Antigout drugs	Central nervous system drugs: Antiparkinsonian drugs (levodopa and other dopamine agonists, anticonvulsants)
Antimicrobials: Acyclovir, antituberculosis drugs, erythromycin, sulfonamides, and tetracycline	
Cancer chemotherapy: Cisplatin, cytarabine, dacarbazine, etoposide, 5-fluorouracil, and methotrexate	Others: Theophylline
Infectious causes	
Acute gastroenteritis: Viral and bacterial	Nongastrointestinal (systemic) infections
Metabolic and endocrine causes	
Acute intermittent porphyria, diabetes mellitus, diabetic ketoacidosis, hyperparathyroidism, and other causes of hypercalcemia*	Addison's disease, hyperthyroidism, hyponatremia, hypoparathyroidism, and pregnancy
Central nervous system causes	
Demyelinating disorders, disorders of the autonomic system, hydrocephalus, meningitis, and migraine headaches*	Intracerebral lesions with edema (projectile vomiting): Abscess, hemorrhage, infarction, and neoplasm*

Contd...

Contd...

Labyrinthine disorders

Labyrinthitis*

Meniere's disease*

Motion sickness

Other causes

Anxiety and depression*

Cardiac disease:

Cyclic vomiting syndrome*

Congestive heart failure

Ethanol abuse*

Myocardial infarction

Functional disorders and bulimia nervosa*

Myocardial ischemia

*Indicates causes of persistent vomiting

Rumination Syndrome/MerycismSee **Box 5.1****Complications of Chronic Vomiting***Emetic injuries to the esophagus and stomach:*

- Lesions may range from mild erythema to erosions and ulcerations
- Abrupt retching or vomiting episodes may produce longitudinal mucosal and even transmural lacerations at the level of the gastroesophageal junction. When the lacerations are associated with acute bleeding and hematemesis, the clinical condition is described as the **Mallory-Weiss syndrome**.
- **Boerhaave's syndrome** refers to spontaneous rupture of the esophageal wall, with free perforation and secondary mediastinitis.
- **Dental caries** and erosions may result from chronic vomiting.

Spasm of the glottis and aspiration pneumonia:

In patients with diminished consciousness, or in an older person or patient with a depressed cough reflex, may be associated with aspiration of gastric contents into the bronchi, resulting in acute asphyxia and a subsequent risk of **aspiration pneumonia**.

Fluid, electrolyte, and metabolic alterations:

Hypochloremic alkalosis is usually the first metabolic abnormality to develop and is attributable to loss of fluid and hydrogen and chloride ions. Hypokalemia, hypernatremia, and dehydration can occur.

Nutritional deficiencies

Nutritional deficiencies may result from reduced caloric intake or loss of nutrients in the vomitus.

Q. Discuss treatment of persistent vomiting.**Treatment****1. Supportive measure: Correction of fluid and electrolyte balance****2. Medication:**

- **Phenothiazines and related drugs:** Prochlorperazine 5–10 mg thrice daily
- **Dopamine antagonists:** Metoclopramide 10 mg 30 minutes before meals and at bedtime. Side effects include drowsiness and extrapyramidal effects. Domperidone (10 mg thrice daily) has no central nervous system (CNS) side effects
- **Antihistaminic agents**, e.g., diphenhydramine
- **Serotonin 5-HT₃ receptor antagonists:** Useful in chemotherapy associated emesis. Examples ondansetron and granisetron
- **Neurokinin-1(NK-1) receptor antagonist**, e.g., aprepitant (oral) and fosaprepitant (parenteral) indicated only in chemotherapy-induced nausea and vomiting
- **Motilin receptor agonists:** Erythromycin intravenously in boluses of 200–400 mg every 4–5 hours
- **Muscarinic receptor agonist:** Bethanechol
- **Synthetic cannabinoids:** Nabilone and dronabinol
- **Glucocorticoids:** Especially in raised intracranial tension (ICT), and chemotherapy-induced emesis
- **Gastric electrical stimulation** in refractory cases

Hiccough**Q. List the common causes of hiccough.**

The symptoms of hiccups (hiccoughs, singultus) is an involuntary, intermittent, and spasmodic contraction of the diaphragm and intercostal muscles and glottic closure. Its causes are listed in **Table 5.3**.

Box 5.1: Features of rumination syndrome.

- Rumination is a functional disorder resembles vomiting but does not involve an integrated somatovisceral response coordinated by the emetic center
- It consists of the repetitive effortless regurgitation of small amounts of recently ingested food into the mouth followed by re-chewing and re-swallowing or expulsion

TABLE 5.3: Causes of hiccough.

➤ Hasty ingestion of food and fluids	➤ Diabetic ketoacidosis, renal failure, respiratory failure, and electrolyte imbalance
➤ Hyponatremia	➤ Local irritation of the diaphragm: Due to gaseous distension of stomach and intestines, subphrenic abscess, peritonitis, and acute myocardial infarction
➤ Uremia	➤ Psychogenic
➤ Irritation of the phrenic nerve: Due to compression by tumors, esophagitis, pericarditis, mediastinitis, and surgery of thorax and abdomen	➤ Unknown
➤ Cerebrovascular accidents (especially lateral medullary syndrome), encephalitis, and brain tumors	

Symptomatic treatment

- Advised to drink cold water, swallowing a teaspoon of dry sugar
- Apply pressure over the eyeballs
- Perform Valsalva maneuver
- Rebreathing into a paper bag
- Drugs (**Box 5.2**)
- Local infiltration of phrenic nerve with procaine
- Acupuncture

Constipation

Q. Define constipation (Rome IV criteria). List the common causes.

Definition: Constipation is defined as persistent, difficult, infrequent passage of hard stools, or seemingly incomplete defecation/evacuation. Many patients may also complain of excessive straining or lower abdominal fullness/discomfort.

Criteria for Functional Constipation (Box 5.3)

Risk factors: Advanced age, female gender, low level of physical activity, low socioeconomic status, nonwhite ethnicity, and use of certain medications.

Causes of Constipation

See Table 5.4.

Investigations

It is neither possible nor appropriate to investigate all patients with constipation.

Initial visit:

- Routine biochemistry: Serum calcium, serum glucose, and thyroid function tests
- Complete blood count
- Examination of stool including occult blood
- Digital rectal examination
- Proctoscopy and sigmoidoscopy or colonoscopy: To detect anorectal disease or exclude carcinoma colon (in patients older than 50 years or those with alarm situations such as blood in stool or anemia or weight loss or new onset of symptoms).

If these investigations are normal, a 1-month trial of dietary fiber and/or laxatives is advised.

Next visit: If symptoms persist, barium studies or CT colonography is indicated to look for structural disease. Colonic transit studies, anorectal manometry, and defecography are performed in only in resistant cases without a structural disease.

Q. Discuss management of constipation.

Treatment

- Regular exercise and adequate fluid intake
- Treat the underlying cause or eliminate offending medication
- **Fiber supplementation:** If there is no secondary cause, increasing the fiber content of the diet and fluid intake. A fiber supplement such as wheat bran or psyllium and mucilaginous seeds and seed coats (e.g., ispaghula husk) with water two to four times per day
- **Laxatives (Table 5.5):** They should be restricted to severe cases. Osmotic laxatives act by increasing colonic inflow of fluid and electrolytes by osmotic activity. This softens the stool and stimulates colonic contractility. The stimulatory laxatives act by stimulating colonic contractility and by causing intestinal secretion. Prucalopride is effective in refractory constipation

Box 5.2: Drugs used for the treatment of hiccough.

- Chlorpromazine 25–50 mg orally or intramuscularly
- Domperidone 10 mg thrice daily
- Metoclopramide 10 mg thrice daily
- Xylocaine viscus 15 mL thrice a day
- Baclofen 5–10 mg thrice a day
- Nifedipine, haloperidol, phenytoin, olanzapine, nefopam, and gabapentin can be used

Box 5.3: Rome IV criteria for functional constipation.

1. Two or more of the following six must be present*
 - a. Straining during at least 25% of defecations
 - b. Lumpy or hard stools in at least 25% of defecations
 - c. Sensation of incomplete evacuation for at least 25% of defecations
 - d. Sensation of anorectal obstruction/blockage for at least 25% of defecations
 - e. Manual maneuvers to facilitate at least 25% of defecations (e.g., digital evacuation, support of the pelvic floor)
 - f. Fewer than three defecations/week
2. Loose stools are rarely present without the use of laxatives
3. There are insufficient criteria for irritable bowel syndrome (IBS)

*Criteria fulfilled for the previous 3 months with symptom onset at least 6 months prior to diagnosis.

TABLE 5.4: Causes of constipation.

Primary causes of constipation	Secondary causes of constipation	
Gastrointestinal disorders: ➤ Dietary: Lack of fiber and/or fluid intake ➤ Motility: Irritable bowel syndrome, slow-transit constipation ➤ Structural: Carcinoma colon, diverticular disease, and Hirschsprung's disease ➤ Defecation: Anorectal disease such as Crohn's, fissures, and hemorrhoids	Medications: ➤ Antacids ➤ Anticholinergic agents (e.g., antiparkinsonian drugs, antipsychotics, antispasmodics, and tricyclic antidepressants) ➤ Anticonvulsants (e.g., carbamazepine, phenobarbital, and phenytoin) ➤ Calcium-channel blockers (e.g., verapamil) ➤ Diuretics (e.g., furosemide) ➤ Iron supplements ➤ Nonsteroidal anti-inflammatory drugs (e.g., ibuprofen) ➤ Mu-opioid agonists (e.g., fentanyl, loperamide, and morphine)	Metabolic and endocrinologic disorders: ➤ Diabetes mellitus ➤ Heavy metal poisoning (e.g., arsenic, lead, and mercury) ➤ Hypercalcemia ➤ Hyperthyroidism ➤ Hypokalemia ➤ Hypothyroidism Neurologic and myopathic disorders: ➤ Amyloidosis ➤ Autonomic neuropathy ➤ Dermatomyositis ➤ Multiple sclerosis ➤ Parkinsonism ➤ Progressive systemic sclerosis ➤ Shy-Drager syndrome ➤ Spinal cord injury

TABLE 5.5: Various laxatives.

Bulk laxatives	Emollient laxatives (stool softeners)	Stimulant laxatives (stimulate motility and intestinal secretion)	Osmotic laxatives	Others
➤ Soluble (ispaghula, psyllium) ➤ Insoluble (methyl cellulose)	➤ Docusate salts ➤ Mineral oil ➤ Other agents (e.g., polyethylene glycol)	➤ Phenolphthalein ➤ Bisacodyl ➤ Anthraquinones—senna and dantron (only for the terminally ill) ➤ Sodium picosulfate ➤ Castor oil	➤ Magnesium sulfate ➤ Lactulose/lactitol ➤ Macrogols	➤ Guanylate cyclase-C receptor agonists (linaclotide and plecanatide) ➤ Lubiprostone (chloride channel activator) ➤ Misoprostol ➤ Colchicine ➤ Prucalopride (5HT₄ receptor agonist)

Chronic Blood and Mucus in the Stools

Q. MI3.2 Identify the common etiologic agents of diarrhea and dysentery. How will you investigate such a case?

Causes

See Table 5.6.

Investigations

- **Stool examination**
 - **Macroscopy:** Fresh blood/altered blood/foul smelling/bulky/floats in water/mucus
 - **Microscopy (Table 5.7)**
- **Stool culture and sensitivity:** Grows the organism
- **Proctoscopy:** To detect any ulcers or tumors of rectum and hemorrhoids
- **Sigmoidoscopy:** To detect any ulcers/tumors in the sigmoid colon
- **Colonoscopy:** To visualize colon
- **Barium enema:** To identify any growth or filling defects, strictures, ulcers, and diverticula
- **Biopsy:** Of an ulcer or growth

TABLE 5.6: Causes of chronic blood and mucus in the stools.

➤ Dysentery – Amebic dysentery – Bacillary dysentery	➤ Inflammatory bowel disease – Ulcerative colitis – Crohn's disease ➤ Intestinal tuberculosis	➤ Carcinoma: Large bowel (rectum, sigmoid and colon) ➤ Diverticulitis ➤ Mesenteric vascular disease ➤ Necrotizing enterocolitis
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TABLE 5.7: Microscopic components that can appear in stool.

➤ Ova ➤ Trophozoites ➤ Red blood cells ➤ Bacteria	➤ Cysts ➤ Pus cells ➤ Macrophages
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Occult Blood in the Stool

- Stool may appear normal (i.e., no melena) in patients with gastroduodenal bleeding of up to 50–100 mL/day.
- Blood loss in the stool of a normal individual varies from 0.5 mL/day to 1.5 mL/day. Occult blood tests begin to become positive usually when blood loss is around 2 mL/day.
- **Test for detection of occult blood in stool:**
 - **Guaic test:** Gastrointestinal tract bleeding is often intermittent. Hence, this test should be performed for several successive days. This test is negative in a person on iron or bismuth.
 - **Other tests:** Fecal immunochemical tests and heme-porphyrin test.

Conditions with Occult Blood in Stool (Table 5.8)**TABLE 5.8:** Conditions with occult blood in stool.

Upper gastrointestinal bleeding	Esophageal varices, esophagitis, peptic ulcer disease, gastritis, and malignancy
Lower gastrointestinal bleeding	Hemorrhoids, malignancy, diverticulitis, inflammatory bowel disease, and celiac sprue
Drugs	Aspirin or steroids (rare)
Others	Hookworm infestation and mesenteric vascular disease

Q. IM16.6 Distinguish between diarrhea and dysentery based on clinical features.

Sl. N.	Character	Diarrhea	Dysentery
1.	Definition	Diarrhea refers to a condition that involves the frequent passing of loose or watery stools (atleast three times a day.)	Dysentery is an intestinal inflammation, especially in the colon, that can lead to severe diarrhea with mucus or blood in the feces
2.	Stool type	Watery stool with no blood and mucus	Mucoid stool that may be accompanied by blood
3.	Affects	The small bowel	Affects the colon
4.	Target cells	Diarrheal infection targets only intestinal lumen and upper epithelial cells	Dysentery not only targets upper epithelial cells but colon ulceration also results
5.	Cell death	There is no cell death in diarrhea and the infection is only caused due to the release of toxins by an infecting agent	When a person gets dysentery, the upper epithelial cells are attacked and destroyed by the pathogen or disease causing agent
6.	Type of infectious agent involved	Diarrhea is mostly viral	Dysentery is mostly bacterial
7.	Causative agents	Norovirus, Rotavirus, Sapovirus, <i>Escherichia coli</i> (<i>E. coli</i>), <i>Giardia lamblia</i> etc.	<i>Shigella</i> , especially <i>S. flexneri</i> and <i>S. dysenteriae</i> type I, <i>Campylobacter jejuni</i> , especially in infants, and less frequently, <i>Salmonella</i> , Enteroinvasive <i>Escherichia coli</i> are closely related to <i>Shigella</i> and may cause severe dysentery. <i>Entamoeba histolytica</i> cause dysentery in order children and adults, but rarely in children <5 years of age
8.	Suggestive of	May be a condition in itself or a symptom of another condition	Dysentery is a condition in itself
9.	Major causes	➤ A viral infection ➤ Alcohol abuse ➤ Allergies to certain foods ➤ Diabetes ➤ Diseases of the intestines ➤ Eating foods that upset the digestive system ➤ Infection by bacteria or other organisms ➤ Laxative abuse ➤ Medications ➤ Overactive thyroid (hyperthyroidism) ➤ Radiation therapy ➤ Running ➤ Some cancers ➤ Surgery on the digestive system ➤ Trouble absorbing certain nutrients, also called malabsorption	➤ Bacterial infection ➤ Viral infection ➤ Parasitic worms ➤ Protozoa
10.	Symptoms	Watery motions, weakness, dehydration, abdominal pain and cramping (may or may not)	Motion with blood and mucous, abdominal pain, weakness, cramping, vomiting, dehydration
11.	Cramps and pain	The symptoms may or may not be accompanied by cramps or pain	The patient usually complains of cramps and pain in the lower abdominal area
12.	Fever	Fever is less common in diarrhea	Fever is more common in dysentery
13.	Effects	The effects of diarrhea are not that serious, apart from the risk of dehydration	Dysentery can cause a lot of complications, if left untreated

Weight Loss (Table 5.9)

TABLE 5.9: Common causes of weight loss.

Types of weight loss:

- **Physiological:** Due to dieting, exercise, starvation, or decreased nutritional intake (e.g., old age)
- **Pathological/involuntary**
 - Endocrine and metabolic disorders: Diabetes mellitus, hyperthyroidism, pheochromocytoma, Addison's disease, and panhypopituitarism
 - Gastrointestinal disorders: Malabsorption, tropical sprue, chronic pancreatitis, inflammatory bowel disease (ulcerative colitis and Crohn's disease), and parasitic infestations
 - Chronic infections: Tuberculosis, human immunodeficiency virus (HIV), fungal infections, amebic abscess
 - Malignancy: Stomach, colon, pancreas, liver, lung, lymphoma, and leukemia
 - Psychiatric illness: Anorexia nervosa, depression, and schizophrenia
 - Renal disease: Chronic renal failure and infective endocarditis
 - Cardiac disorders: Chronic congestive heart failure
 - Respiratory disorders: Emphysema, empyema, chronic obstructive pulmonary disease (COPD)
 - Rheumatological: Rheumatoid arthritis
 - Idiopathic

Glossitis

Q. List the causes of glossitis.

Definition: Glossitis is defined as inflammation of the tongue.

Clinical Features

- Patients may complain of lingual pain (glossodynia) or burning sensation (glossopyrosis).
- Atrophic glossitis is a sign of protein-calorie malnutrition and muscle atrophy and is commonly found in older adults.
- Median rhomboid glossitis manifests as an asymptomatic, well-defined erythematous patch in the mid-posterior dorsum of the tongue.

Causes

B-complex deficiency, megaloblastic anemia (pernicious anemia), oral candidiasis, herpes, cirrhosis, iron deficiency anemia, pellagra, scarlet fever, syphilis, chemical irritants, drug reactions, amyloidosis, sarcoidosis, infections, and vesiculo-erosive diseases.

Dyspepsia

Q. List the causes and differential diagnosis of dyspepsia.

Dyspepsia is derived from the Greek words *dys* and *pepe* and literally means “difficult digestion”.

- Dyspepsia is a collective description of a variety of gastrointestinal symptoms (**Table 5.10**)
- **Ulcer dyspepsia:** Dyspeptic symptoms associated with peptic ulcer
- **Nonulcer dyspepsia (functional dyspepsia):** No cause can be found
- **Flatulent dyspepsia:** Usually due to a functional disorder

Symptoms include early satiety, flatulence, bloating, and belching predominate.

TABLE 5.10: Symptoms of dyspepsia.

Postprandial fullness	Bloating in the upper abdomen
Early satiation	Nausea
Epigastric pain	Vomiting
Epigastric burning	Belching

Causes (Table 5.11)

TABLE 5.11: Causes of dyspepsia.

Luminal gastrointestinal tract	Medications
➤ Chronic gastric or intestinal ischemia ➤ Food intolerance ➤ Functional dyspepsia ➤ Gastroesophageal reflux disease ➤ Gastric or esophageal neoplasms ➤ Gastric infections (e.g., cytomegalovirus, fungus, tuberculosis, and syphilis) ➤ Gastroparesis (e.g., diabetes mellitus, postvagotomy, scleroderma, chronic intestinal pseudo-obstruction, postviral, and idiopathic) ➤ Irritable bowel syndrome ➤ Peptic ulcer disease ➤ Parasites (e.g., <i>Giardia lamblia</i> and <i>Strongyloides stercoralis</i>)	Acarbose, aspirin, other nonsteroidal anti-inflammatory drugs (including cyclooxygenase-2 selective agents), colchicine, digitalis preparations, estrogens, ethanol, glucocorticoids, iron, levodopa, niacin, narcotics, nitrates, orlistat, potassium chloride, quinidine, sildenafil, and theophylline Pancreaticobiliary disorders ➤ Biliary pain: Cholelithiasis, choledocholithiasis, and sphincter of Oddi dysfunction ➤ Chronic pancreatitis ➤ Pancreatic neoplasms Systemic conditions Adrenal insufficiency, congestive heart failure, diabetes mellitus, hyperparathyroidism, myocardial ischemia, pregnancy, renal insufficiency, and thyroid disease

Nonulcer Dyspepsia (Functional Dyspepsia, Nervous Dyspepsia; Nonorganic Dyspepsia)

Q. Discuss the etiology clinical features, investigations, and management of nonulcer dyspepsia.

Definition: Chronic dyspepsia in the absence of organic disease.

Second most common functional gastrointestinal disorder (after irritable bowel syndrome).

Etiology

- It is dyspepsia in the absence of organic disease and even on detailed investigation, no cause can be found.
- Etiology is poorly understood but probably due to a spectrum of mucosal, motility and psychiatric disorders.
- Symptoms are probably due to disturbances in the motor function of the gastrointestinal tract similar to that occurring in the irritable bowel syndrome. Both irritable bowel syndrome and nonulcer dyspepsia often exist together in the same patient.
- *Helicobacter pylori* infection should be excluded, because it may be responsible for symptoms in few patients.

Clinical Features

- Usually occurs in young (<40 years of age) females.
- Symptoms and subtypes are described in **Table 5.12**.
- Morning symptoms of pain and nausea on waking are characteristic.
- Features suggestive of irritable bowel syndrome such as pellet-like stools and feeling of incomplete evacuation after defecation may be observed.
- History may reveal stress factors such as worries, financial problems, employment, and family affairs.
- On examination, no diagnostic signs, except for inappropriate abdominal tenderness on abdominal palpation.
- All the organic causes of dyspepsia such as peptic ulcer disease (PUD), drug ingestion, depression, pregnancy, alcohol abuse, etc., should be excluded.
- In older patients intra-abdominal malignancy should be excluded which may present with alarming features such as weight loss, anorexia, dysphagia, and hematemesis or melena.

TABLE 5.12: Rome IV criteria for functional dyspepsia.

Includes <i>one or more</i> of the following:
1. Bothersome postprandial fullness
2. Bothersome early satiation
3. Bothersome epigastric pain
4. Bothersome epigastric burning
And
No evidence of structural disease (including at upper endoscopy) that is likely to explain the symptoms
Subgroups according to Rome IV criteria
➤ Epigastric pain syndrome (EPS): Pain centered in the upper abdomen as the predominant symptom
➤ Postprandial distress syndrome (PDS): Unpleasant or troublesome nonpainful sensation (discomfort) centered in the upper abdomen is the predominant symptom. This may be accompanied by upper abdominal fullness, early satiety, bloating, and nausea
Any one of the first two criteria (1 and/or 2, at least 3 days/week) is for PDS while any one of the last two criteria (3 and/or 4, at least 1 day/week) is for EPS

Investigations

The history often suggests the diagnosis.

- *Helicobacter pylori* infection should be serologically excluded in all patients.
- Exclusion of organic causes by following investigations:
 - Blood count, erythrocyte sedimentation rate (ESR) and occult blood in stools
 - Liver function tests (to exclude alcoholism)
 - Pregnancy test
 - Barium meal
 - Ultrasound scan may detect gallstones.
- Indications for endoscopy in patients with chronic dyspepsia/red flags are listed in **Box 5.4**.

Management (Fig. 5.1)

- Explanation, reassurance, and lifestyle changes
- Advice to avoid cigarette smoking and alcohol use
- Symptoms if associated with an identifiable cause of stress resolve with appropriate counseling
- Psychological factors influencing on gut function should be explained and may require psychological treatment
- Idiosyncratic and restrictive diets are not beneficial, but reducing intake of fat and coffee may help
- If endoscopy is noncontributory, empirical treatment is advised
 - Prokinetic drugs, such as metoclopramide (10 mg three times daily) or domperidone (10–20 mg three times daily), may be given before meals for nausea, vomiting, or bloating
 - Mosapride or itopride may also be tried
 - H₂-receptor antagonists or proton-pump inhibitors if night pain or heartburn is troublesome
 - *Helicobacter pylori* eradication therapy, if test is positive and may be effective in some patients with functional dyspepsia
 - Selective serotonin reuptake inhibitors may be effective in some patients
 - Low dose tricyclic agents (e.g., amitriptyline) may be of help in up to two-thirds
 - SSRI (selective serotonin re-uptake inhibitor) medication is tried in refractory cases

Box 5.4: Indications for endoscopy in patients with chronic dyspepsia/red flags.

- Age >60 years
- Dysphagia/previous peptic ulcer disease and odynophagia
- Clinically significant weight loss (>5% usual body weight over 6–12 months)
- Protracted vomiting
- Melena
- Anemia
- Palpable mass
- Jaundice
- Family history of stomach cancer

Distinguishing Features of Nonulcer Dyspepsia and Ulcer Dyspepsia (Table 5.13)

TABLE 5.13: Distinguishing features of nonulcer dyspepsia and ulcer dyspepsia.

Characteristics	Nonulcer dyspepsia	Ulcer dyspepsia
Nature of pain	Not episodic	Episodic (periodicity)
Duration of pain	Throughout the day	Occurs only on empty stomach
Relief of pain	Not affected by antacids	Relieved by antacids
Relation of pain with vomiting	Pain is not relieved by vomiting and patient cannot eat afterwards	Pain relieved by vomiting and patient can eat immediately
Relation pain to food	Pain provoked by food	Pain relieved by food
Location of abdominal pain	Diffuse referred to more than one site	Localized referred to epigastrium
Relation of pain to sleep	Pain at night, waking the patient from sleep is rare	Pain at night, waking the patient from sleep is common

TABLE 5.14: Various causes of upper gastrointestinal (UGI) bleeding.

Esophageal causes

- Esophageal varices (e.g., due to portal hypertension) usually occur from the lower 5 cm of esophagus
- Erosive esophagitis
- Esophageal carcinoma
- Mallory-Weiss syndrome

Gastroduodenal causes

- Peptic ulcer (gastric and duodenal) disease
- Erosive gastritis (e.g., after ingestion of NSAIDs and alcohol) or duodenitis
- Stress ulcers occurring with shock, sepsis, or severe trauma
- **Curling ulcers** in the proximal duodenum with **severe burns or trauma**
- Carcinoma stomach
- Gastric antral vascular ectasia (GAVE), also described as *watermelon stomach*, *Dieulafoy's lesion*
- Angiodysplasia or vascular malformations (*Heydes syndrome* in aortic stenosis)
- Rupture of aortic aneurysm or aortic duodenal fistula following aortic graft. Cushing ulcer associated with elevated intracranial pressure

Other cause

- Coagulation defects
- *Cameron's lesions*: Linear erosions or ulcerations in the proximal stomach at the end of a large hiatal hernia, near the diaphragmatic pinch
- Hemobilia
- Hemosuccus pancreaticus

GASTROINTESTINAL BLEEDING

Upper Gastrointestinal Bleeding

Q. IM15.1 Enumerate, describe and discuss the etiology of upper and lower GI bleeding.

Hematemesis is the bleeding proximal to the duodenojejunal junction (ligament of Treitz).

Etiology (Table 5.14 and Fig. 5.1)

Peptic ulcers are the most common cause of upper gastrointestinal (UGI) bleeding (up to about 50%).

Clinical Features (Table 5.15)

Q. 11.2 Discuss the appropriate history and clinical features that identify the source of bleeding, quantity, volume loss, etiology, and risk factors of UGI bleed.

Presentation: Patient usually presents with either hematemesis (vomiting of blood or coffee-ground material) and/or melena (black, tarry stool).

- **Hematemesis: Color of the vomitus** depends on how long the blood has been in the stomach and the severity of bleeding.
 - **Red** with clots when bleeding is severe and bright red indicates rapid and sizeable bleeding.
 - **Black** ("coffee grounds") when less severe and bleeding is small.

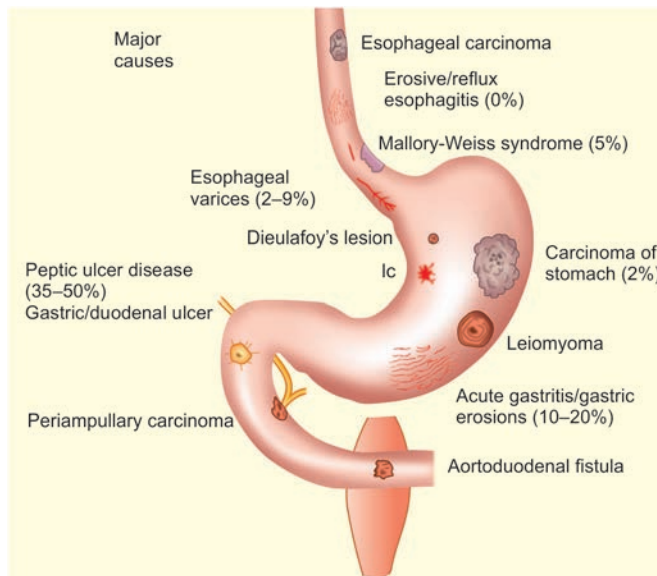


FIG. 5.1: Causes of acute upper gastrointestinal hemorrhage.

TABLE 5.15: Features suggesting probable source of upper GI bleeding.

History	Probable source of bleeding
Features suggestive of peptic ulcer	Peptic ulcer disease
Jaundice, pedal edema, abdominal distension, splenomegaly, ascites, and dilated abdominal veins	Esophageal varices
Features of liver cell failure, e.g., spider nevi, palmar erythema, jaundice, and gynecomastia	
Alcohol or drug (NSAIDs) ingestion, trauma, burns, or sepsis	Gastric erosions
Heavy alcohol intake or vomiting	Mallory-Weiss tear
Dysphagia and weight loss prior to bleeding	Malignant (stomach or esophageal cancer)/
Tenderness or a mass	intra-abdominal tumor

- **Melena:** It is the passage of **black, tarry, foul-smelling stools** containing altered blood. The characteristic color and smell are due to the action of digestive enzymes and bacteria on hemoglobin. It usually occurs when more than 60 mL blood is lost and blood has been present in the UGI tract for at least 14 hours (and as long as 3–5 days). Occasionally, it develops due to hemorrhage from the right side of the colon.

- Severe/massive acute UGI bleeding can sometimes cause maroon or bright red stool due to passage of frank blood per-rectum (**hematochezia**).

- **Occasionally**, presentation with **symptoms of blood loss only**.

These include:

- **Acute loss with intravascular volume depletion:** Dizziness, extreme pallor, and shock.
- **Chronic loss:** With symptoms of anemia

Differences between hematemesis and hemoptysis are presented in **Table 5.16**.

Q. Discuss evaluation and treatment/management of upper gastrointestinal bleeding.

History

Physical Examination: Look for

- Vital signs, orthostatic hypotension
- Abdominal tenderness
- Skin, oral examination
- Stigmata of liver disease
- **Rectal examination:**
 - Objective description of stool/blood
 - Assess for mass, hemorrhoid

Q. IM15.6 Distinguish between upper and lower gastrointestinal bleeding based on the clinical features.

Differences between upper and lower GI bleed are presented in **Table 5.17**.

Basic Investigations

- **Complete blood count**
 - Chronic bleeding leads to anemia, but the hemoglobin concentration and the hematocrit level may be normal after sudden, major bleeding until hemodilution occurs.
 - Low platelet count: Suggests chronic liver disease, dilution, or hematologic disorder.
- **Blood urea nitrogen (BUN) and creatinine:** The blood urea nitrogen level increases to greater extent than the

TABLE 5.16: Differences between hematemesis and hemoptysis.

Features	Hemoptysis	Hematemesis
Definition	Coughing out of blood	Vomiting out of blood
Content and color	Mixed with sputum and bright red in color	Mixed with food particles and coffee-ground in color
Contents	Frothy, associated sputum	No froth, associated food particles
Associated symptoms	Cough, dyspnea	Nausea, vomiting, retching, and abdominal discomfort
Melena	Rare	Common
Reaction	Alkaline (blue litmus remains unchanged)	Acidic (blue litmus remains unchanged)

TABLE 5.17: Differences between upper and lower GI bleed.

Features	Upper GI tract bleed	Lower GI tract bleed
Site of bleeding	Above ligament of Treitz	Below ligament of Treitz
Clinical presentation	Hematemesis or melena	Hematochezia/rarely melena
Nasogastric aspiration shows	Blood	Clear fluid
Bowel sounds	Hyperactive	Normal
BUN (blood urea nitrogen)/creatinine ratio	Increased	Normal

creatinine level because of increased intestinal absorption of urea after breakdown of blood proteins. An elevated blood urea with normal creatinine concentration implies severe bleeding.

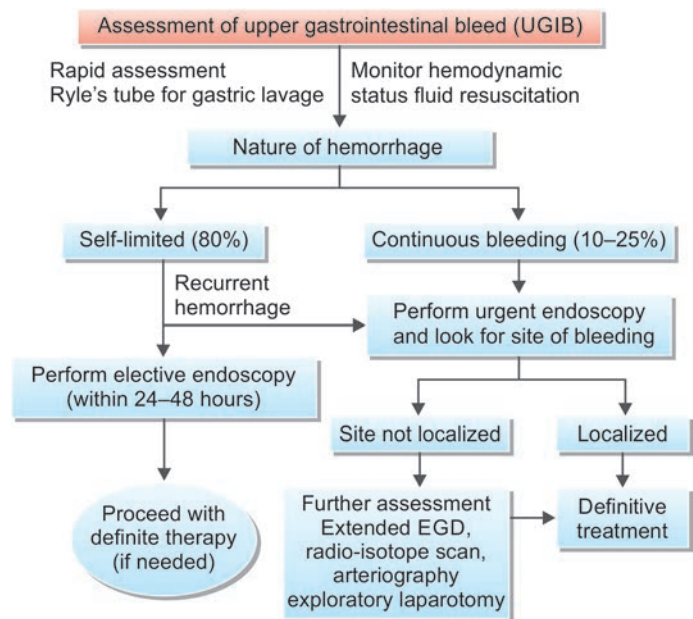
- **Liver function tests:** They may show evidence of chronic liver disease.
- **Prothrombin time:** It may be prolonged in patients with chronic liver disease or in coagulation disorders or patients on anticoagulated therapy.
- **Cross-matching:** Potential infusion of packed red blood cells.

Q. Write short essay on management and drugs used in upper gastrointestinal bleeding.

Assessment of upper GIT bleed is presented in **Flowchart 5.1**.

Q. IM15.11 Develop, document and present a treatment plan that includes fluid resuscitation, blood and blood component transfusion, and specific therapy for arresting blood loss.

FLOWCHART 5.1: Assessment of upper gastrointestinal bleed (UGIB).



Management

Medical resuscitation and replenishment of intravascular volume: In case of massive bleeding, resuscitation measures should be initiated simultaneously with the initial assessment

- **Intravenous access:** The first step is to gain intravenous access using at least one large-bore (14- or 16-gauge) intravenous catheters for essentially all patients so that normal saline can be infused as fast as necessary to maintain hemodynamic stability
- **Fluid resuscitation:** Adequate resuscitation and hemodynamic stabilization are essential prior to endoscopy (saline). Vasopressors if persistent hypotension
- **Blood product transfusions:** Initiate blood transfusions if the hemoglobin is <7 g/dL to maintain the hemoglobin at a level of ≥ 9 g/dL. Patients with active bleeding and a low platelet count ($<50,000/\mu\text{L}$) should be transfused with platelets. Patients with a coagulopathy with a prolonged prothrombin time with INR >1.5 should be transfused with fresh frozen plasma (FFP)

Initial clinical assessment:

- **Circulatory status:** Severe bleeding causes tachycardia, hypotension, and oliguria. Closely observe with hourly pulse, blood pressure, postural hypotension urine output, and level of consciousness
- **Evidence of liver disease:** May be present in patients with decompensated cirrhosis
- **Comorbidity:** Such as cardiorespiratory, cerebrovascular, or renal may be worsened by acute bleeding and also increase the hazards of endoscopy and surgical operations

Gastric lavage: This procedure has been abandoned in present day

- **Procedure:** Gastric lavage is performed by instilling 500 mL of ice-cold or tap water every 30–60 minutes.
- **Advantages** of nasogastric or orogastric tube help to localize the site of upper GI bleeding. Determine the type of material whether red blood or coffee-ground. Assess the rate, severity, persistency, and recurrence of the bleeding. Clear the blood from stomach for better endoscopic visualization prior to endoscopy. Remove blood → reduces the risk of encephalopathy in patients with liver disease. Dilute acid-pepsin in stomach → reduces bleeding from erosions. Minimize the risk of aspiration
- **Prognostic score:** Rockall score is based upon age, the presence of shock, comorbidity, diagnosis, and endoscopic stigmata of recent hemorrhage

Endoscopy

Pre-endoscopic pharmacotherapy

For nonvariceal UGIB (upper gastrointestinal bleeding)

- IV proton pump inhibitor: 80 mg bolus, 8 mg/h drip (esomeprazole)
- **Rationale:** Suppress acid, facilitate clot formation, and stabilization

Goal of endoscopic therapy: To stop acute bleeding and reduce the risk of recurrent bleeding

- Early endoscopy (within 24 hours) is recommended for most patients with acute UGIB
- Achieves prompt diagnosis, provides risk stratification and hemostasis therapy in high-risk patients

Endoscopic hemostasis therapy

It includes (1) epinephrine injection, (2) thermal electrocoagulation, and (3) mechanical (hemoclips). Combination therapy is superior to monotherapy Nonvariceal UGIB: Postendoscopy management (**Box 5.5**)

Box 5.5: Postendoscopy management of nonvariceal UGIB.

- Patients with ulcers requiring endoscopic therapy should receive PPI $\times$ 72 hours
- Determine *H. pylori* status in all ulcer patients
- Discharge patients on PPI (once to twice daily), duration dictated by underlying etiology and need for nonsteroidal anti-inflammatory drugs (NSAIDs)/aspirin
- In patients with cardiovascular disease on low dose aspirin: Restart as soon as bleeding has resolved

Management of upper GI tract bleeding is summarized in **Flowchart 5.2**.

Q. PE 26.11, IM15.10 Enumerate the indications for endoscopy, colonoscopy, and other imaging modalities in the investigation of Upper GI bleed.

Endoscopy: It is the diagnostic modality of choice for acute upper GI bleeding. Endoscopy has a high sensitivity and specificity for locating and identifying bleeding lesions in the upper GI tract. Therapeutic endoscopy can in most cases achieve acute hemostasis and prevent recurrent bleeding once a bleeding lesion has been identified. Early endoscopy, within 24 hours is recommended for most patients with acute upper GI bleeding. For patients with suspected variceal bleeding, endoscopy should be performed within 12 hours of presentation.

Colonoscopy: It is indicated in patients with hematochezia and a negative upper endoscopy. It is also frequently performed in patients with melena and a negative upper endoscopy, to rule out right-sided colonic lesions which may also present with melena.

Imaging: Plain radiographs may be useful to demonstrate bowel obstruction/perforation in patients who present with pain abdomen and distention. Barium studies are contraindicated in the acute setting of UGI bleed as it may interfere with the subsequent endoscopy/surgery. USG abdomen is particularly useful in patients with suspected liver disease with clinical signs of portal hypertension.

Variceal Bleeding

Occurs in one-third of patients with cirrhosis. In one-third initial bleeding episodes are fatal. Among survivors, one-third will rebleed within 6 weeks. Only one-third will survive 1 year or more. Management of variceal bleed is discussed in Chapter 6.

Lower Gastrointestinal Bleed

Q. List causes of lower gastrointestinal bleeding.

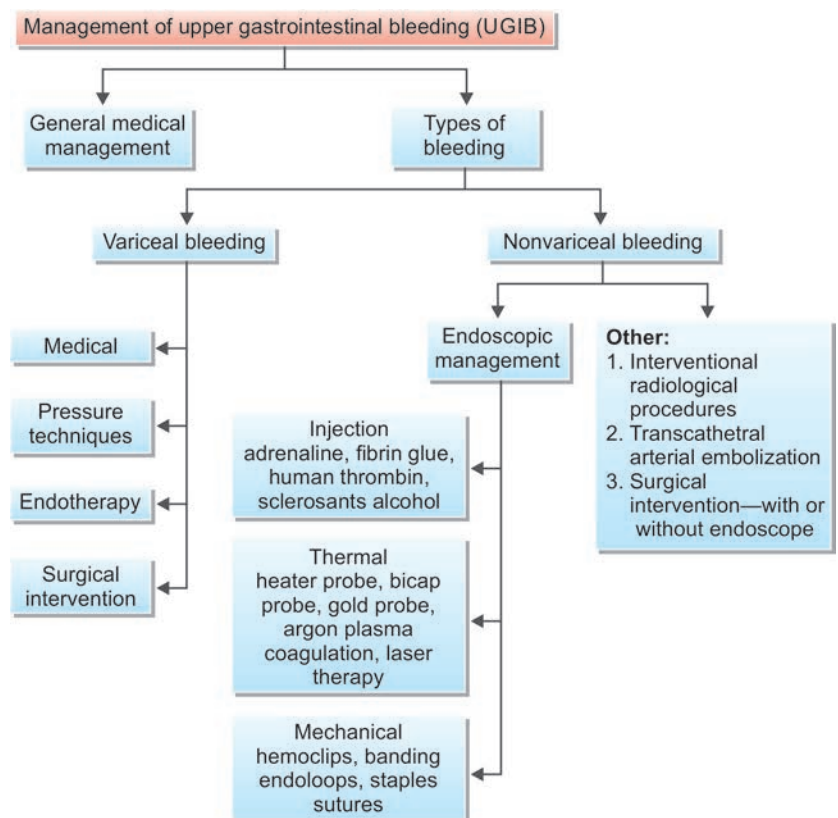
Lower GI bleeding generally signifies bleeding from the colon or anorectum. In patients with severe **hematochezia**, first consider possibility of UGIB. About 10–15% of patients with presumed LGIB are found to have upper GIB.

Etiology (Table 5.18 and Fig. 5.2)

TABLE 5.18: Causes of lower gastrointestinal bleeding.

Diverticulosis (main cause)	Angioectasias
Hemorrhoids anal fissures, ulcer	Colitis (IBD, infectious, and ischemic)
Neoplasms	Postpolypectomy
Dieulafoy's lesion	Radiation colitis, small bowel bleed
	Unknown causes—10%

FLOWCHART 5.2: Management of upper GI tract bleeding.



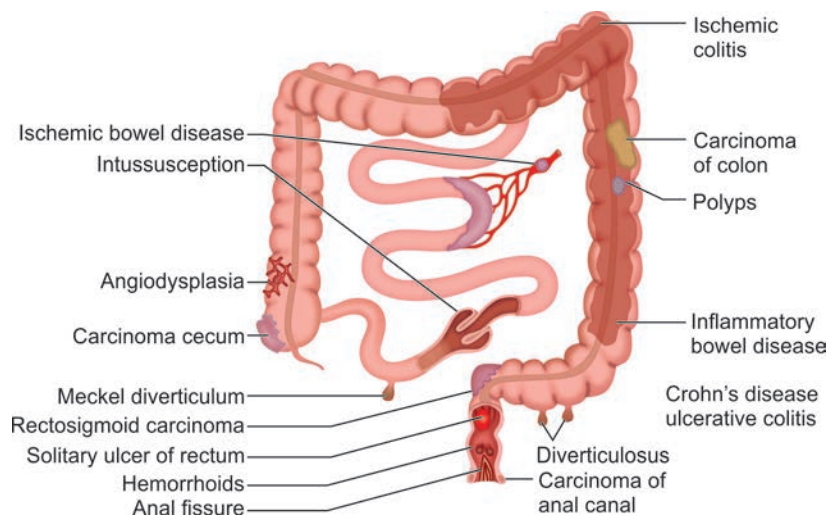
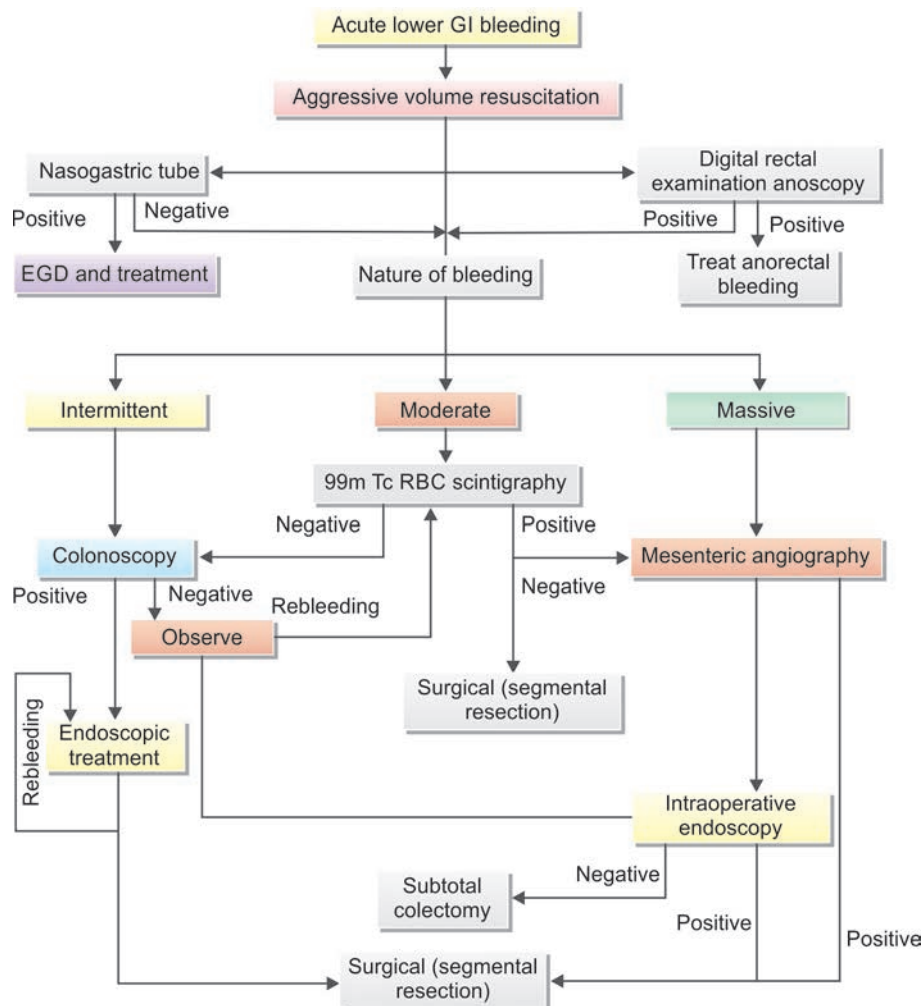


FIG. 5.2: Causes of lower gastrointestinal bleed (LGIb).

Management of LGI Bleed

See Flowchart 5.3.

FLOWCHART 5.3: Management of lower gastrointestinal bleeding.



APPROACH TO DIARRHEA

Q. IM16.1 Describe and discuss the etiology of acute and chronic diarrhea including infectious and noninfectious causes.

Q. IM16.2 Describe and discuss the acute systemic consequences of diarrhea including its impact on fluid balance.

Definition: Daily bowel movements of three or more times are considered to be abnormal. The upper limit of stool weight is 200 g daily. Although stool weight usually considered as a “scientific” definition of diarrhea, diarrhea should not be defined solely in terms of fecal weight.

- **Acute diarrhea** is defined as abrupt onset of increased frequency and/or fluidity of bowel movements.
- **Chronic diarrhea** is defined as passage of loose stools with or without increased stool frequency for more than 4 weeks.

Large-volume versus Small-volume Diarrhea

- Normal rectosigmoid colon functions as a storage reservoir.
- **Left colonic disorders:** Inflammatory or motility disorders involving the left colon compromises this rectosigmoid reservoir capacity and results in **frequent small-volume bowel movements**.
- **Right colonic or small bowel disorders:** In diarrhea due to disorders of the right colon or small bowel and if the rectosigmoid reservoir is intact, individual **bowel movements are less frequent and larger**.

Frequent, small, painful stools may point to a source in the distal colon, whereas painless large-volume stools suggest a right colonic or small bowel source.

Secretory versus Osmotic Diarrhea (Table 5.19)

- **Secretory diarrhea:** It results from malabsorption or secretion of electrolytes (secretory diarrhea).
- **Osmotic diarrhea:** It results from intestinal malabsorption of ingested nonelectrolytes.
- Osmotic diarrhea constitutes small number of cases whereas secretory diarrhea forms the much larger number of cases.
- In secretory diarrhea, sodium, potassium, and accompanying anions account almost entirely for stool osmolality. In contrast, in osmotic diarrhea poorly absorbable solutes within the lumen of the intestine account for much of the osmotic activity of stool water.

Watery versus Fatty versus Inflammatory Diarrhea

- **Watery diarrhea:** It implies a defect primarily in water absorption as a result of increased electrolyte secretion or reduced electrolyte absorption (secretory diarrhea) or ingestion of a poorly absorbed substance (osmotic diarrhea).
- **Fatty diarrhea:** It implies defective absorption of fat and perhaps other nutrients in the small intestine.
- **Inflammatory diarrhea:** It implies the presence of one of a limited number of inflammatory or neoplastic diseases involving the gastrointestinal tract.

TABLE 5.19: Secretory versus osmotic diarrhea.

Type of diarrhea	Causes	Examples
Secretory diarrhea	Exogenous secretagogues	Enterotoxins (e.g., cholera)
	Endogenous secretagogues	Neuroendocrine tumors (e.g., carcinoid syndrome)
	Absence of ion transporter	Congenital chloridorrhea
	Loss of intestinal surface area	Intestinal resection, diffuse intestinal mucosal disease
	Intestinal ischemia	Diffuse mesenteric atherosclerosis
	Rapid intestinal transit	Intestinal hurry following vagotomy
Osmotic diarrhea	Ingestion of poorly absorbed agent	Magnesium ingestion
	Loss of nutrient transporter	Lactase deficiency

Causes of Diarrhea in the Tropics

See Table 5.20.

TABLE 5.20: Causes of diarrhea in tropics.

Gastroenteritis induced by toxin	Gastroenteritis induced by changes in mucosa
➤ Performed toxins: <i>Staphylococcus aureus</i> and <i>Bacillus cereus</i> ➤ Enterotoxins produced in the intestine, <i>Clostridium perfringens</i>, <i>Vibrio cholerae</i>, Enterotoxigenic <i>Escherichia coli</i>, Enterohemorrhagic <i>E. coli</i>, and <i>Clostridium difficile</i>	➤ Mucosal changes without invasion: Rotavirus, Norwalk agent ➤ Invasion of mucosa/inflammatory (invasion or cytotoxin) dysentery – <i>Salmonella</i> species, <i>Shigella</i> species, Enterohemorrhagic <i>E. coli</i>, Enteroinvasive <i>E. coli</i>, <i>Campylobacter jejuni</i>, <i>Yersinia enterocolitica</i>, and <i>Entamoeba histolytica</i> ➤ Other causes: Mushrooms, heavy metals (arsenic), monosodium glutamate

Causes of Chronic Diarrhea (Table 5.21)

Q. Write short note on causes of chronic diarrhea.

TABLE 5.21: Causes of chronic diarrhea.

Fatty diarrhea	Watery diarrhea
Malabsorption syndromes - Mucosal diseases (e.g., celiac disease and Whipple's disease) - Mesenteric ischemia - Short bowel syndrome - Small intestinal bacterial overgrowth Maldigestion - Inadequate luminal bile acid concentration - Pancreatic exocrine insufficiency	- Osmotic diarrhea - Carbohydrate malabsorption - Osmotic laxatives - Secretory diarrhea - Bacterial toxins - Congenital syndromes (e.g., congenital chloridorrhea) - Disordered motility, regulation - Diabetic autonomic neuropathy - Irritable bowel syndrome - Postsympathectomy diarrhea - Postvagotomy diarrhea - Diverticulitis - Endocrinopathies: Addison's disease, carcinoid syndrome, gastrinoma, hyperthyroidism, mastocytosis, medullary carcinoma of thyroid, pheochromocytoma, somatostatinoma, and VIPoma - Laxative abuse (stimulant laxatives) - Medications and toxins
Inflammatory diarrhea	
- Diverticulitis - Infectious diseases - Invasive bacterial infections (e.g., tuberculosis and yersiniosis) - Invasive parasitic infections (e.g., amebiasis and strongyloidiasis) - Pseudomembranous colitis (<i>Clostridium difficile</i> infection) - Ulcerating viral infections (e.g., cytomegalovirus, herpes simplex virus) - Inflammatory bowel diseases: Crohn's disease, ulcerative colitis - Ischemic colitis - Neoplasia: Carcinoma of colon, lymphoma - Radiation colitis	

Approach to the Patient with Acute Diarrhea

Q. Discuss the evaluation and management of acute diarrhea.

Assessment of the Patient

- Degree of dehydration
- Evidence of specific cause
- Necessity of any diagnostic tests
- Requirement for any treatment

History (Table 5.22) Physical Examination

- Examine for signs of dehydration to assess the severity of the diarrhea. These include examination of pulse, blood pressure (including postural change), skin turgor, dryness of mucous membranes (e.g., mouth), mental status, and breathing.
- Severity of dehydration (**Table 5.23**)
- Electrolyte imbalances: Assess muscle strength and muscle reflexes which may be reduced in hypokalemia.
- Examination of abdomen to exclude any surgical cause (e.g., intestinal obstruction).

TABLE 5.22: Features to be noted during taking history in a patient with acute diarrhea.

- Duration of diarrhea
- Travel or use of antibiotics
- Having food outside in the recent past and time since consumption
- Any other family member or close associated affected
- Associated features:* Pain abdomen, fever, vomiting, and tenesmus
- Stool:* Frequency of loose stools, volume, appearance of stools (e.g., rice water), presence of blood or mucus in the stools
- Urine output
- Any alteration in the level of consciousness
- Occupation of the patient

TABLE 5.23: Severity of dehydration.

Features	Mild	Moderate	Severe
Urine output	Decreased	Decreased	Markedly decreased
Physical examination			
Level of consciousness	Normal	Normal	Depressed
Mental status	Normal/irritable	Lethargic	Comatose
Skin	Normal	Cool	Cool and mottled
Skin turgor	Normal	Reduced with skin tenting	Markedly reduced
Mouth/oral mucosa	Dry	Markedly dry	Parched
Eyes	Normal	Sunken (sunken fontanel in infants)	Markedly sunken
Pulse rate	Normal or mild increase	Tachycardia, feeble	Marked tachycardia
Blood pressure	Normal	Postural/orthostatic fall	Hypotension and frank shock
Respiration	Normal	Normal	Acidotic
Investigations			
Urine specific gravity	<1.020	>1.020	>1.035
Blood urea	Normal	Normal to raised	High

Laboratory Investigations

Laboratory investigations usually do not help in the management of acute diarrhea.

- **Total white blood cell (WBC) count:** Presence of high leukocyte count with shift to left suggests invasive bacterial infection.
- **Electrolytes and acid-base status:** It should be done in patients with severe dehydration. Severe diarrhea produces metabolic acidosis.
- **Blood cultures:** To be performed when bacteremia or a systemic infection is suspected.
- **Stool examination:** Grossly bloody or mucus in the stool suggests an inflammatory process. In severe cases stool should be examined for the presence of leukocytes, red cells, and cysts or trophozoites/parasites. In cholera *V. cholerae* show the characteristic darting motility. Stool with leukocytes considers inflammatory causes (*Shigella*, *Salmonella*, *Campylobacter*, *E. coli*, *Entamoeba*, and *Clostridium difficile*).
- **Stool culture:** Specific indications for stool cultures include bloody stools, stools that test positive for occult blood or leukocytes, prolonged course of diarrhea that has not been treated with antibiotics, immunocompromised host, or for epidemiologic purposes, such as cases involving food handlers.

Q. Describe and discuss the acute systemic consequences of acute diarrhea including its impact on fluid balance.

Systemic complications of acute diarrheal disease are presented in Table 5.24.

Effect on fluid balance: Usually, more than 90% of the fluid entering the small intestine is absorbed. Only about 1 liter reaches the large intestine. Further absorption occurs in the large intestine and only 100–200 milliliters of water is excreted each day in formed stools. When there is increased secretion and/or decreased absorption in the small intestine, large volumes enter the large intestine. Diarrhea occurs when this volume exceeds the absorptive capacity of the large intestine. Diarrhea stools contain large amounts of water, sodium, chloride, bicarbonate, and potassium. These losses cause dehydration (due to the loss of water and sodium chloride), metabolic acidosis (due to the loss of bicarbonate), and hypokalemia (due to potassium depletion). Among these, dehydration (negative fluid balance) is the most dangerous because it can cause decreased blood volume (hypovolemia), cardiovascular collapse, acute renal failure, mental confusion, and death if not treated promptly.

Small Bowel versus Large Bowel Diarrhea

See Table 5.25.

TABLE 5.24: Systemic complications of acute diarrheal disease.

Complication	Associated organism
Sepsis	<i>Shigella</i> species, nontyphoidal <i>Salmonella enterica</i> , and <i>Campylobacter fetus</i>
Hemolytic-uremic syndrome	<i>Shigella</i> species, Shiga toxin-producing <i>Escherichia coli</i>
Guillain-Barré syndrome	<i>Campylobacter jejuni</i>
Reactive arthritis	<i>Campylobacter</i> species, <i>Salmonella</i> species, and <i>Shigella flexneri</i>

TABLE 5.25: Differences between small bowel and large bowel diarrhea.

Features	Small bowel diarrhea	Large bowel diarrhea
Characteristic of diarrhea	Large-volume, watery diarrhea	Frequent, small volume, often painful bowel movements
Associated clinical features	Abdominal cramping, bloating and gas formation	Gripping pain in lower abdomen
Fever	Uncommon	Present
Tenesmus	Absent	Present
Blood or mucus stools	Not common	Common
Inflammatory cells and red blood cells in stools	Absent	Present

Management (Flowchart 5.4)

Q. IM16.14 Describe and enumerate the indications, pharmacology and side effects of pharmacotherapy for bacterial and viral diarrhea.

Most of acute diarrhea is self-limited and fluid and electrolyte replacement are of most important in all cases

Rehydration

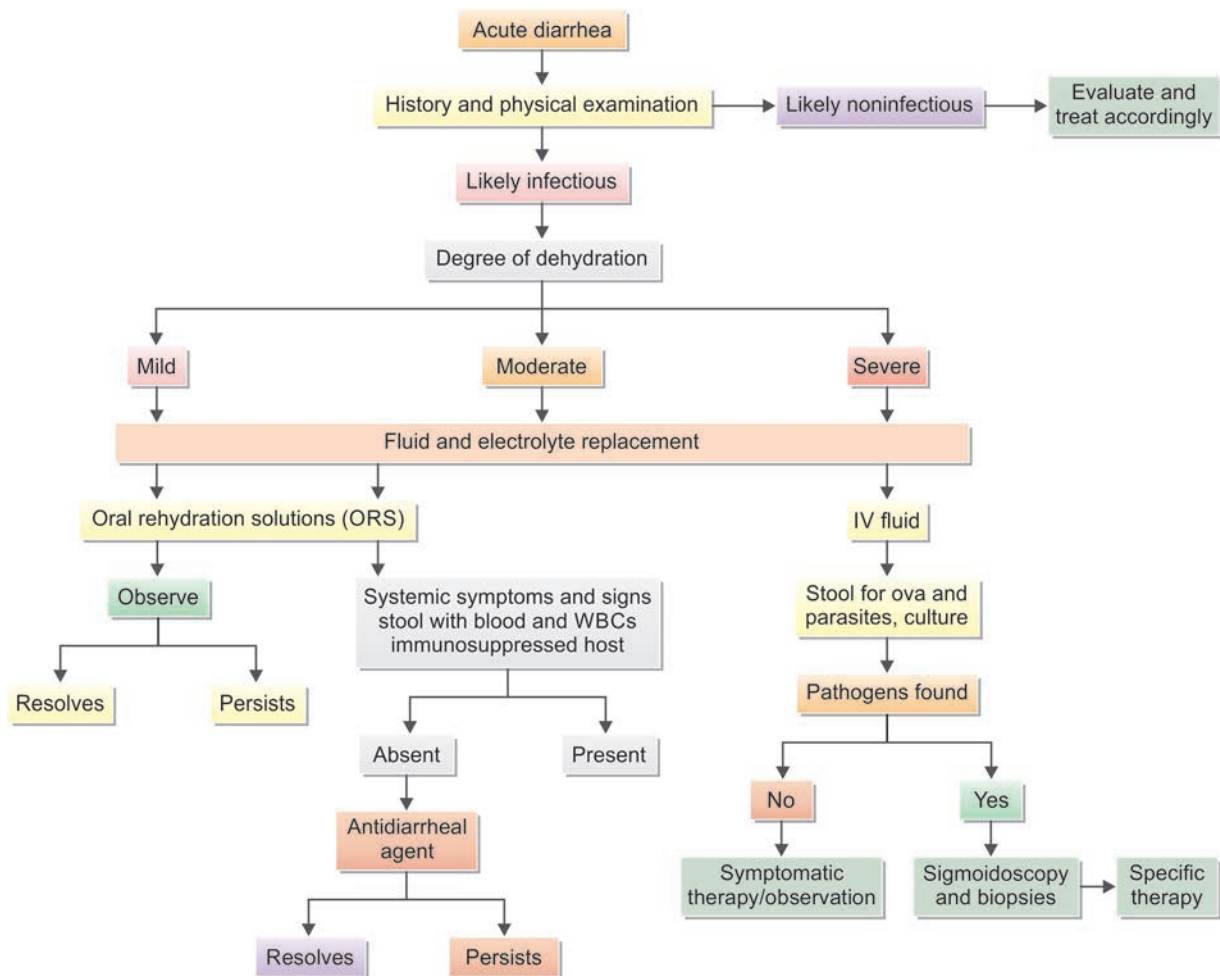
- Severe diarrhea produces dehydration especially in the very young and very old
- Toxin-induced diarrhea produces stools that are usually isotonic. Concentrations of sodium and chloride are slightly less than that of plasma while bicarbonate concentration is double that of plasma. The stools also contain significant amount of potassium
- In inflammatory diarrhea the electrolyte loss is of less compared to toxigenic diarrhea

Oral rehydration solution (ORS)

Used for mild-to-moderate dehydration

Intravenous fluids

- Used when there is moderate-to-severe dehydration
- Usually Ringer's lactate is administered
- In severe cases, fluids are administered at a rate of 20 mL/kg/hour for the first 2–3 hours. If the patient improves, reduce the rate to 10 mL/kg/h for the next 2–4 hours
- Concurrently start oral rehydration therapy

FLOWCHART 5.4: Approach to the management of acute diarrhea.**Absorbents**

Kaolin absorbs the toxin and may be of use in few patients. However, it does not affect the course of the disease

Antimotility drugs

- Used for symptomatic treatment of toxin-induced diarrhea in adults only. It should not be used in young children and elderly
- It can be given in inflammatory diarrhea along with antibiotics. These include:
 - Opiates (e.g., morphine and codeine): They may cause respiratory depression
 - Diphenoxylate/atropine combination: May cause respiratory depression and anticholinergic side effects
 - Loperamide: Dose is two tablets of 4 mg each initially, then 2 mg after each unformed stool, not to exceed 16 mg/day for ≤ 2 days
 - Bismuth subsalicylate: It acts as an antisecretory agent. Dose is one tablet every 30 minutes for a total of eight doses or 60 mL every 6 hourly

Antisecretory agents: Racecadotril

- Reduces the hypersecretion of water and electrolytes into the intestinal lumen
- Inhibits enkephalinase (an enzyme that degrades enkephalins)
- Dose: 100 mg thrice daily. To be given to patients with acute, watery diarrhea only
- Contraindication: Renal insufficiency, pregnancy, and breastfeeding

Antispasmodics

Mild antispasmodics such as dicyclomine, hyoscine may be used in patients with significant abdominal cramps

Antibiotics: Indications

- Symptomatic patients with inflammatory diarrhea (high fever, toxicity, and abdominal pain)
- Acute febrile dysentery illness
- Diarrhea caused by *Campylobacter jejuni*. Early use of erythromycin or azithromycin limits the duration of illness
- Commonly used antibiotics: Quinolones (norfloxacin 400 mg and ciprofloxacin 500 mg both given twice daily or levofloxacin 500 mg given once a day) for 3–5 days
- Cholera: Doxycycline in a dose of 300 mg as single dose. Alternatives include trimethoprim-sulfamethoxazole, furazolidone, and norfloxacin

Specific treatment once cause identified

Nonspecific drug therapy for chronic diarrhea is presented in **Table 5.26**.

Traveler's Diarrhea**Q. Write short note on traveler's diarrhea.**

Traveler's diarrhea is the leading cause of illness in travelers. Usually a short-lived and self-limited.

Etiology

- Causative agents: Various pathogens causing travelers' diarrhea are listed in **Table 5.27**. Most important pathogens are *Escherichia coli* and Enteroaggregative *E. coli*: rotavirus and norovirus.
- Source of infection: Main sources of infection are food and water contaminated with fecal matter.

Clinical Features

- Usually involves intercontinental travelers.
- Symptoms: Abrupt in onset, watery diarrhea lasting 2–5 days, abdominal cramps, nausea, vomiting, anorexia, and fever.
- Signs: Diffuse tenderness over abdomen.

TABLE 5.26: Nonspecific drug therapy for chronic diarrhea.

Drug class	Agent
Opiates (mu opiate receptor selective)	Codeine, diphenoxylate, loperamide, morphine, and tincture of opium
Enkephalinase inhibitor (delta opiate receptor effects)	Racecadotril
Alpha-2 adrenergic agonist	Clonidine
Somatostatin analog	Octreotide
Bile acid-binding resin	Cholestyramine, colestevlam, and colestipol
Fiber supplements	Calcium polycarbophil, psyllium

TABLE 5.27: Various causes of traveler's diarrhea.

Bacteria (50–75%)	Viruses (0–20%)	Parasites (0–10%)
Enterotoxigenic <i>Escherichia coli</i>	Norovirus (winter vomiting bug)	<i>Giardia lamblia</i>
Enterotoxigenic <i>E. coli</i>	Rotavirus	<i>Cryptosporidium</i>
<i>Campylobacter jejuni</i>		<i>Entamoeba histolytica</i>
<i>Shigella</i>		
<i>Salmonella</i> (nontyphoid)	No pathogens (10–50%)	

DISEASES OF THE ESOPHAGUS**Dysphagia****Q. Discuss causes and investigation (evaluation) and management of dysphagia.**

Dysphagia, from the Greek dys (difficulty, disordered) and phagia (to eat), refers to the sensation that food is hindered in its passage from the mouth to the stomach.

Causes of Dysphagia*Oropharyngeal dysphagia*

Processes that affect the mouth, hypopharynx, and upper esophagus. The patient often is unable to initiate a swallow and repeatedly has to attempt to swallow. Patients frequently describe coughing or choking when they attempt to swallow. Causes of oropharyngeal dysphagia are listed in **Table 5.28**.

Treatment

- Usually self-limited and requires no treatment
- Dehydration is corrected by oral rehydration supplements
- Drugs:
 - Antidiarrheal agents and antibiotics are only rarely required
 - If there is fever or bloody diarrhea advise norfloxacin or ciprofloxacin. Azithromycin is used in patients who are allergic to quinolones
 - Rifaximin (a poorly absorbed rifampicin derivative) is very effective against noninvasive bacterial pathogens

Prevention

- Drugs: Doxycycline 100 mg/day for a few weeks. Norfloxacin/ciprofloxacin/rifaximin once a day
- Probiotics may be useful

TABLE 5.28: Causes of oropharyngeal dysphagia.

Neuromuscular causes	Structural causes
➤ Amyotrophic lateral sclerosis (ALS) ➤ Multiple sclerosis ➤ Muscular dystrophy ➤ Myasthenia gravis ➤ Parkinson's disease ➤ Polymyositis or dermatomyositis ➤ Stroke ➤ Thyroid dysfunction	➤ Carcinoma ➤ Infections of pharynx or neck ➤ Osteophytes and other spinal disorders ➤ Prior surgery or radiation therapy ➤ Proximal esophageal web ➤ Plummer-Vinson syndrome ➤ Thyromegaly ➤ Zenker's diverticulum

TABLE 5.29: Common causes of esophageal dysphagia.

Motility (neuromuscular) disorders	Structural (mechanical) disorders
Primary disorders ➤ Achalasia ➤ Diffuse esophageal spasm ➤ Hypertensive LES ➤ Ineffective esophageal motility ➤ Nutcracker (high-pressure) esophagus Secondary disorders ➤ Chagas' disease ➤ Reflux-related dysmotility ➤ Scleroderma and other rheumatologic disorders	Intrinsic ➤ Carcinoma and benign tumors ➤ Diverticula ➤ Eosinophilic esophagitis ➤ Esophageal rings and webs (other than Schatzki ring) ➤ Foreign body ➤ Lower esophageal (Schatzki) ring ➤ Medication-induced stricture ➤ Peptic stricture Extrinsic ➤ Mediastinal mass ➤ Spinal osteophytes ➤ Vascular compression

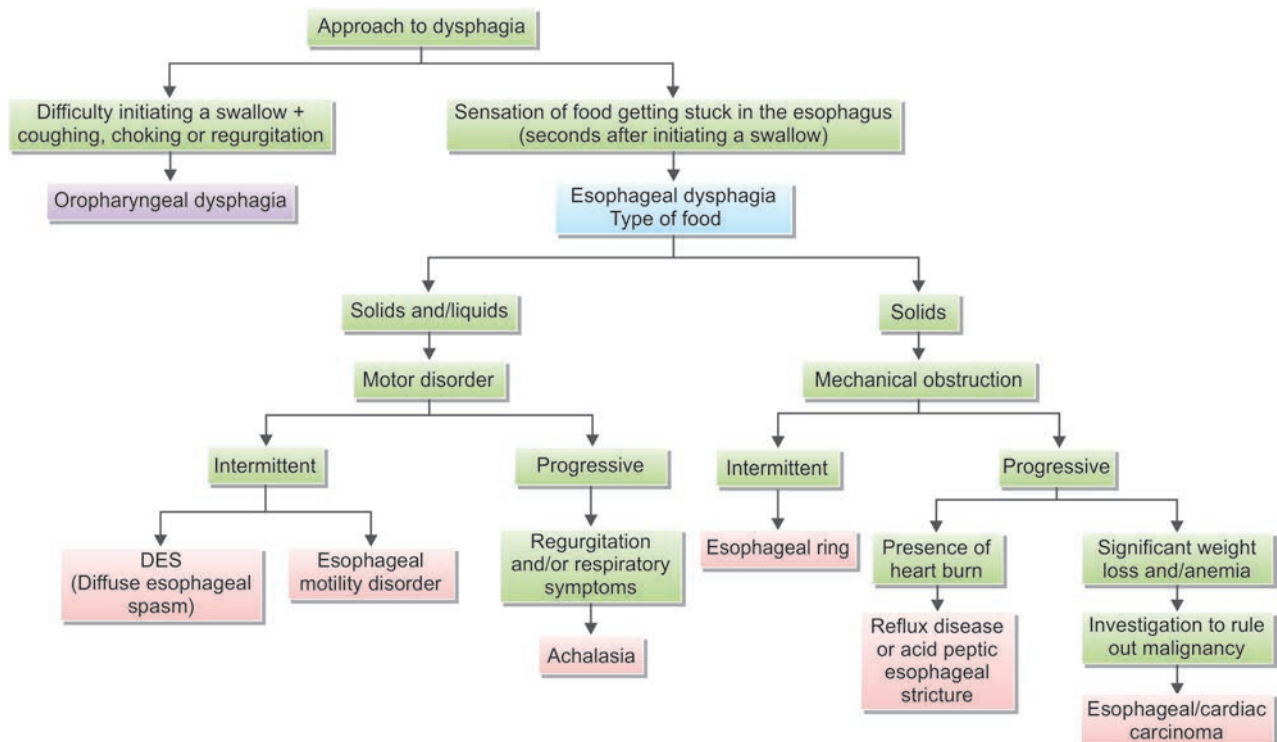
Esophageal dysphagia (Table 5.29)

Differential Diagnosis of Dysphagia

Approach to the patient with dysphagia is presented in **Flowchart 5.5**.

Investigations

- Hemoglobin and peripheral smear for anemia
- Chest radiograph detects retrosternal goiter, mediastinal lymph nodes, aortic aneurysms, and primary and secondary malignancy of lungs.
- Esophagoscopy allows removal of foreign body, visualization and biopsy of tumors, ulcers, strictures, etc.
- Computed tomography (CT) scan of thorax
- Biopsy from the growth, ulcer, or inflamed mucosa
- Barium swallow detects tumors as filling defects or strictures (rat tail appearance).
- Esophageal motility studies (esophageal manometry)

FLOWCHART 5.5: Approach to the patient with dysphagia.

Odynophagia

Q. Write short note on the causes for odynophagia.

Odynophagia, or painful swallowing, is specific feature for esophageal involvement. It usually reflects an inflammatory process in the esophageal mucosa. Its severity varies. It may be a dull retrosternal ache on swallowing to a stabbing pain with radiation to the back so severe that the patient cannot eat or even swallow their own saliva.

Causes of Odynophagia

See Table 5.30.

Plummer-Vinson Syndrome

Discussed in Chapter 9.

TABLE 5.30: Causes of odynophagia.

Caustic ingestion: Acid, alkali Pill-induced injury ➤ Alendronate and other bisphosphonates ➤ Aspirin and other NSAIDs ➤ Iron preparations ➤ Potassium chloride (especially slow-release form) ➤ Tetracycline and its derivatives ➤ Quinidine ➤ Zidovudine	Infectious esophagitis ➤ <i>Viral:</i> Cytomegalovirus, Epstein-Barr virus, herpes simplex virus, and human immunodeficiency virus ➤ <i>Bacteria:</i> Mycobacteria (tuberculosis or <i>Mycobacterium avium</i> complex) ➤ <i>Fungal:</i> <i>Candida albicans</i>, histoplasmosis ➤ <i>Protozoan:</i> <i>Cryptosporidium</i> and <i>Pneumocystis</i> Severe reflux esophagitis Esophageal carcinoma
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Treatment

- Iron deficiency anemia is treated with iron. It may resolve dysphagia
- Endoscopy:
 - Dysphagia may require endoscopic dilatation
 - Follow-up endoscopy at regular intervals to detect development of carcinoma

Gastroesophageal Reflux Disease

Q. Describe the etiopathogenesis, complications, and management of gastroesophageal reflux disease (GERD).

Q. Write short essay on reflux esophagitis and its management.

Definition

- Gastroesophageal reflux disease (GERD) is a consequence of the failure of the normal antireflux barrier to protect against frequent and abnormal amounts of gastroesophageal reflux (GER; i.e., gastric contents moving retrograde effortlessly from the stomach to the esophagus).
- Spectrum of injury to the esophagus includes *esophagitis, stricture, Barrett's esophagus, and adenocarcinoma*.

Pathophysiology

Normal defense mechanisms preventing reflux and reflux esophagitis

Several defense mechanisms prevent the reflux of gastric contents into the esophagus.

- **Antireflux barrier at the gastroesophageal junction (Fig. 5.3):** This consists of (1) lower esophageal sphincter (LES), at the lower end of esophagus, below the diaphragm, (2) striated muscles of the crural diaphragm, (3) phrenoesophageal ligament, (4) oblique entrance of the esophagus into the stomach (angle of His), and (5) attachment of the lower esophageal sphincter (LES) to the crural diaphragm: Intra-abdominal pressure reinforces the LES tone.
- **Esophageal clearance mechanisms:** Reflux of gastric contents into the esophagus occurs in healthy persons and is normally cleared by esophagus in a two-step process:
 1. Volume clearance by peristaltic function: After acid reflux from stomach, the esophageal peristalsis returns the refluxed fluid to the stomach.
 2. Neutralization of acid by bicarbonate in the swallowed saliva: The small amounts of residual acid refluxed into the esophagus are neutralized by weakly alkaline (bicarbonate) contained in swallowed saliva.
- **Epithelial defensive factors:** The esophageal mucosa contains mainly three lines of defense.
 1. **Pre-epithelial barrier:** Consisting of (1) small unstirred water layer, (2) bicarbonate from swallowed saliva, and (3) secretions of submucosal glands.
 2. **Epithelial defense:** Consisting of cell membranes and tight intercellular junctions, buffers, and ion transporters.
 3. **Postepithelial defense:** Consists of the blood supply to the esophagus.

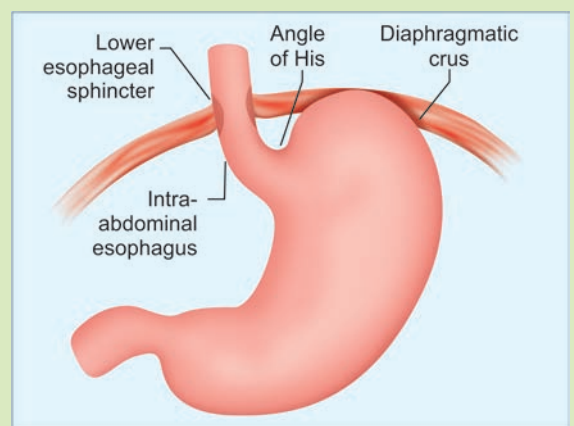


FIG. 5.3: Anatomy of normal antireflux barrier at the lower esophagus.

Causes of Disruption of Normal Defense Mechanisms (Fig. 5.4)

- **Defective antireflux barriers**
 - **Hiatus hernia (sliding type):** This is characterized by sliding of the esophagogastric junction through the diaphragm. This results in increased exposure of the esophagus to acid and may lead to esophagitis, Barrett's esophagus or peptic strictures.
 - **Abnormalities of lower esophageal sphincter (LES):** Transient relaxation and reduced tone of LES can result in regurgitation, especially when intra-abdominal pressure is increased.
 - ◆ Cigarette smoking, chocolate, alcohol, fatty foods, and caffeine cause relaxation and reduction of tone of LES.
 - ◆ Cardiomyotomy and vagotomy reduce the efficiency of the LES.
 - ◆ Drugs (aminophylline, β -agonists, nitrates, and calcium channel blockers) reduce the tone of LES.
 - Crural diaphragm
 - **Increased the intra-abdominal pressure:** It may occur during pregnancy, obesity, ascites, weight lifting, and straining.
- **Prolonged/delayed esophageal clearance of refluxed acid:** It may be due to (1) impaired peristalsis, (2) reduced salivation, and (3) body position. Poor esophageal clearance leads to increased acid exposure time. Impaired production of saliva may be observed in smokers and Sjogren's syndrome.
- **Defective gastric emptying:** It increases the gastric content available for reflux. It may be due to gastric outlet obstruction, anticholinergic drugs, and fatty food.

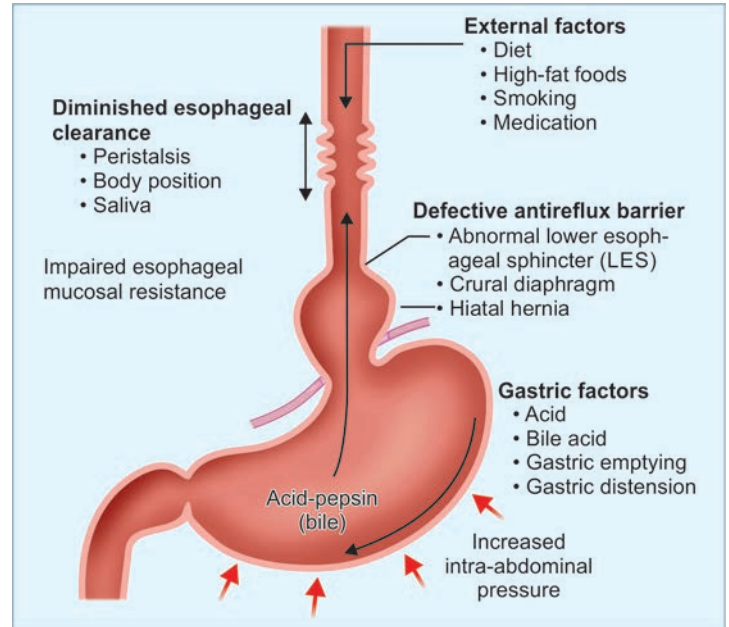


FIG. 5.4: Pathophysiological factors associated with the development of gastroesophageal reflux disease (GERD).

Clinical Features

Classical triad of symptoms is (1) heartburn, (2) acid regurgitation, and (3) epigastric pain.

- **Heartburn:** It is the classic symptom of GERD. Patients usually complain of a burning feeling, rising from the stomach or lower chest and radiating toward the neck, throat, and occasionally the back. It occurs postprandially, particularly after large meals or after ingesting spicy foods, citrus products, fats, chocolates, and alcohol. The supine position and bending over may exacerbate heartburn.
- **Acid regurgitation:** Effortless regurgitation of acidic fluid, especially after meals and worsened by stooping or the supine position is highly suggestive of GERD.
- **Epigastric pain:** Sometimes radiating through to the back.

Other symptoms

- **Odynophagia** (painful swallowing)
- **Dysphagia:** Dysphagia is reported by more than 30% of individuals with GERD. Transient to solids (due to esophageal spasm) or persistent dysphagia to solids (due to strictures).
- Less common symptoms associated with GERD include water brash, burping, hiccups, nausea, and vomiting.
- Iron deficiency anemia may occur due to blood loss.

Extraesophageal symptoms

- Atypical chest pain which may be severe and can mimic angina.
- Upper respiratory tract: Hoarseness, sore throat, sinusitis, otitis media, chronic cough, and laryngitis
- Pulmonary: The prevalence of GERD in asthmatics is estimated between 34% and 89%. Other pulmonary diseases associated with GERD include aspiration pneumonia, interstitial pulmonary fibrosis, chronic bronchitis, and bronchiectasis.

Alarm symptoms or signs (Table 5.31)

Associated conditions: Pregnancy (reducing LES pressure due to the effects of estrogen and progesterone and possibly mechanical factors from the gravid uterus), scleroderma, achalasia cardia, and Zollinger-Ellison syndrome.

TABLE 5.31: Alarm symptoms or signs of GERD.

➤ Dysphagia	➤ Persistent heartburn
➤ Weight loss	➤ Odynophagia
➤ Gastrointestinal bleeding	➤ Anemia
➤ Nausea and vomiting	➤ Family history of cancer early satiety

TABLE 5.32: Complications of GERD.

➤ Esophagitis	➤ Adenocarcinoma esophagus
➤ Hemorrhage and bleeding (hematemesis, melena)	➤ Aspiration pneumonia
➤ Esophageal ulceration	➤ Iron deficiency anemia
➤ Esophageal strictures	➤ Dental caries
➤ Barrett's esophagus	

Complications of GERD

See Table 5.32.

Investigations (Table 5.33)

Diagnostic endoscopy should be performed in patients who fail to respond to therapy or have alarm symptoms or signs.

- **Endoscopy:** It is useful for the detection of:
 - **Erosive esophagitis** which would be visualized and confirmed by biopsy.
 - **Complications:** Peptic stricture and Barrett's esophagus (confirmed by biopsy)
- **Barium radiography:** It has no role in the diagnosis of GERD. Barium swallow and meal examination is useful for appreciation of gastroesophageal anatomy and may be important in the diagnosis of hiatus hernia.
- **24-hour pH recording:** It is the "gold standard" for diagnosis of GERD. Prolonged measurement of pH is the most accurate method for the diagnosis of gastroesophageal reflux **particularly** in patients with atypical symptoms and with normal endoscopies.
- **Esophageal manometry:** Useful to exclude achalasia and other motility disorders.
- **Resting ECG and stress ECG:** To rule out ischemic heart disease (IHD).

TABLE 5.33: Diagnostic tests for gastroesophageal reflux disease.

Tests for reflux

- Intraesophageal pH monitoring (catheter or catheter-free system)
- Ambulatory impedance and pH monitoring (nonacid reflux)
- Barium esophagogram

Tests to assess symptoms

- Empirical trial of acid suppression
- Intraesophageal pH monitoring with symptom analysis

Tests to assess esophageal damage

- Endoscopy
- Wireless capsule endoscopy
- Esophageal biopsy

Tests to assess esophageal function

- Esophageal manometry
- Esophageal impedance

Q. Write short note on management/treatment of gastroesophageal reflux disease (GERD)/reflux esophagitis.

Treatment

General measures

These are lifestyle modifications as listed in **Box 5.6**.

Medical treatment

Inhibition of gastric acid secretion is the cornerstone of the treatment of acute GERD.

Antacids: Liquid antacids buffer acid and increase LESP. They are used in the dose of 10–15 mL, 1 and 3 hours after meal and at bedtime or as needed. They relieve heart burn in mild cases.

Histamine (H₂)-receptor antagonists: They decrease acid secretion. These drugs include cimetidine (800 mg bid, 400 mg qid) or ranitidine (150 mg qid), or famotidine (20–40 mg bid) daily to be given with meals and before bed time, for at least 6 weeks (in mild cases).

Proton-pump inhibitors (PPIs): They decrease acid secretion and gastric volume. They are superior to histamine (H₂)-receptor antagonists. These include omeprazole (20–40 mg/day), lansoprazole (15–30 mg/day), pantoprazole (40 mg/day), esomeprazole (20–40 mg/day), and rabeprazole (10–20 mg/day). Maintenance doses may be necessary for 6–8 months.

- Present evidence does not support the common practice of using metoclopramide or domperidone 10 mg thrice daily either as monotherapy or an adjunct to acid suppression therapy. Its significant adverse effects argue against the use of this drug in GERD.
- *Helicobacter pylori* eradication does not have any therapeutic value.
- Dilatation of esophageal strictures.
- Anemia is treated with oral iron or blood transfusion.

Surgical treatment

Indications: (1) failure to respond to medical therapy, (2) patients not willing to take long-term PPIs or intolerant to PPIs, (3) patients with severe symptom, and (4) patients with regurgitation.

Box 5.6: Lifestyle modifications in the treatment of GERD.

- Avoid foods such as fatty food, alcohol, mint, tomato-based foods, spicy foods, coffee, tea, and acidic foods
- Avoid late night meals before retiring
- Avoid weight lifting, stooping, and bending at waist
- Elevation of the head of the bed in patients with regurgitation or heartburn during night
- Weight reduction
- Stop smoking, alcohol
- Frequent feeds of small volume

Surgical measures

- **Surgical resection of esophageal strictures:** For patients with PPI-refractory GERD, antireflux mucosectomy (ARMS) or antireflux mucosal ablation (ARMA) can be tried.
- **Antireflux surgery:** Laparoscopic fundoplication for patients with PPI-refractory GERD, antireflux mucosectomy (ARMS) or antireflux mucosal ablation (ARMA) can be tried. (additional valve mechanism) yields results comparable to continued PPI therapy.

Barrett's Esophagus**Q. Write short answer on Barrett's esophagus.**

Barrett's esophagus is a **premalignant condition**, in which the **normal squamous lining of the lower esophagus is replaced by columnar mucosa** (columnar lined esophagus; CLO). The columnar mucosa may show areas of intestinal metaplasia (with goblet cells). It is an adaptive response to chronic gastroesophageal reflux and is often asymptomatic. The risk of esophageal cancer depends on the severity and duration of reflux and it may be detected when the patient presents with esophageal cancer. It is more common in men, obese, and above the age of 50. It is weakly associated with smoking but not with alcohol intake.

Diagnosis: It needs multiple systematic biopsies which may show intestinal metaplasia and/or dysplasia.

Management

Regular endoscopic surveillance to detect dysplasia at an early stage. Currently recommendation are as follows:

- Barrett's esophagus with intestinal metaplasia, but without dysplasia: Should undergo endoscopy at 3–5 yearly intervals, if the length of the Barrettic segment is 3 cm and at 2–3 yearly intervals if the length is >3 cm.
- **Low-grade dysplasia:** It should be endoscoped at 6-monthly intervals. Treatment is indicated only for symptoms of reflux or complications (e.g., stricture).

Endoscopic therapies: Such as radiofrequency ablation or photodynamic therapy, can induce regression. However, they are used only for patients with dysplasia or intramucosal cancer.

High-grade dysplasia or intramucosal carcinoma: Treatment options are either esophagectomy or endoscopic therapy, with a combination of endoscopic resection of any visibly abnormal areas and radiofrequency ablation of the remaining Barrett's mucosa, as an "organ-preserving" alternative to surgery.

Hiatus Hernia**Q. Write short note on hiatus hernia.**

It is the herniation of elements of the abdominal cavity (part of stomach) through the diaphragm into the thoracic cavity.

Types (Figs. 5.5A to D)

- **Sliding or type I hiatus hernia:** It is the most common type. In this type, the gastroesophageal junction and the fundus of stomach slide upward above the diaphragmatic hiatus.
- **True paraesophageal (rolling) or type II hiatus hernia:** It is uncommon. In this type, location of gastroesophageal junction is in its normal position, but the fundus and parts of the greater curvature of the stomach herniate into the mediastinum alongside the esophagus.
- **Mixed paraesophageal hernia or type III:** In this type, gastroesophageal junction and a large part of the stomach herniate into the mediastinum.

Predisposing factors: Obesity, pregnancy and ascites. Occurs in 33% of normal adults and 50% of elderly.



FIGS. 5.5A TO D: Types of hiatus hernia. (A) Normal esophagus and stomach; (B) type I (sliding); (C) type II (paraesophageal/rolling); (D) type III (mixed paraesophageal) hiatus hernia.

Clinical Features

- Majority of hiatus hernia are asymptomatic. Hiatus hernia predisposes to gastroesophageal reflux disease (GERD), and hence, symptoms of GERD may be present.
- Type I: Usually asymptomatic or present with symptoms of heartburn or acid regurgitation.
- Type II and III may present with epigastric pain, chest pain, substernal fullness, shortness of breath, nausea, or vomiting.

Investigations

- **Plain radiograph of the chest:** Hernia may be visible as a gas bubble, often with a fluid level behind the heart.
- **Barium swallow/meal:** It is the best method of diagnosis and demonstrates the presence of gastroesophageal junction in the thorax.
- **Endoscopy**

Management

- Asymptomatic hiatus hernias do not require any treatment. Surgical repair of hernia is required in selected cases with gastroesophageal reflux symptomatic rolling hiatus hernias require surgical repair because it is potentially liable to undergo volvulus as a dangerous complication.
- **Surgical treatment:** (1) Repair of the diaphragmatic defect and (2) fixing the stomach in the abdominal cavity (fundoplication) combined with an antireflux procedure.

Achalasia of the Esophagus

Q. Write short note on achalasia cardia and its diagnosis/investigations.

Achalasia is characterized by **esophageal aperistalsis** and results from progressive degeneration of ganglion cells in the myenteric plexus in the esophageal wall, causing failure of relaxation of the hypertonic lower esophageal sphincter in response to the swallowing wave. Failure of propagated esophageal contraction results in progressive dilatation of the gullet.

Cause: Unknown. Autoimmune, neurodegenerative, and viral etiologies have been suggested. Reduction in nitric oxide synthase containing neurons is detected by immunohistochemical staining in the lower esophageal sphincter. There is degeneration of ganglion cells within the sphincter and the body of the esophagus. Infection with *Trypanosoma cruzi* in Chagas' disease causes a syndrome that is clinically similar to achalasia.

Clinical Features

- **Age and gender:** Occurs equally in males and females and at all ages but is rare in childhood.
- **Dysphagia:** It develops slowly, is initially intermittent and characteristically for both liquids and solids from the onset (worse for solids). Dysphagia is eased by drinking liquids, and by standing and moving around after eating.
- **Episodes of chest pain:** Spontaneous chest pain occurs due to esophageal spasm which may be misdiagnosed as cardiac.
- **Regurgitation of food from the dilated esophagus:** As the disease progresses, dysphagia worsens and caused poor emptying of the esophagus. This may produce nocturnal pulmonary aspiration and aspiration pneumonia.
- Heartburn does not occur because the closed esophageal sphincter which prevents gastroesophageal reflux.
- Achalasia **predisposes to squamous carcinoma of the esophagus.**

Investigations

- **Chest X-ray:** It shows a dilated esophagus. Sometimes fluid level may be observed behind the heart. There is **absence of fundic gas shadow.**
- **Barium swallow:** There is absence of peristalsis and often synchronous contractions in the body of the esophagus. There is tapered narrowing of the lower esophagus producing a "**bird's beak**" appearance due to failure of the sphincter to relax.
- **Manometry** shows high-pressure, nonrelaxing lower esophageal sphincter with poor contractility of the esophageal body.
- **Endoscopy:** It should be performed because carcinoma of the cardia can mimic the presentation and radiological and manometric features of achalasia ("pseudo-achalasia").

Management

Treatment for achalasia is palliative

Endoscopic

Endoscopic dilatation of the lower esophageal sphincter (LES)

Endoscopically directed injection of botulinum toxin into the LES (intersphincteric injection)

Surgical

- **Surgical myotomy (Heller's operation)**
- **Peroral endoscopic myotomy (POEM)**

Eosinophilic Esophagitis

Q. Write short note on eosinophilic esophagitis.

- Eosinophilic esophagitis is characterized by eosinophilic infiltration of esophageal mucosa.
- Atopy (allergic rhinitis, eczema, and asthma) is common, and food allergens may trigger the process.
- Dysphagia is prominent, but symptoms can also mimic GERD.
- Endoscopy reveals—include corrugated mucosa, longitudinal mucosal furrows, fixed esophageal rings or **trachealization** (Fig. 5.6) whitish mucosal plaque or exudate, stricture, and mucosal friability giving the appearance of crepe paper.
- Biopsy ≥ 15 eosinophils per HPF
- Treatment:**
 - First-line therapy consists of PPIs, which also treats concomitant GERD.
 - Topical steroids (swallowed fluticasone, 880–1,760 $\mu\text{g}/\text{d}$ in two to four divided doses, or swallowed budesonide, 1–2 mg/d in two to four divided doses) are options; prednisone is an alternate option if topical steroids are ineffective.
 - Six food elimination diet (SFED) (eggs, milk, soy, gluten, tree nuts, and seafood).
- If no response higher doses of topical steroids, systemic steroids, elimination diet trials, or cautious esophageal dilation.

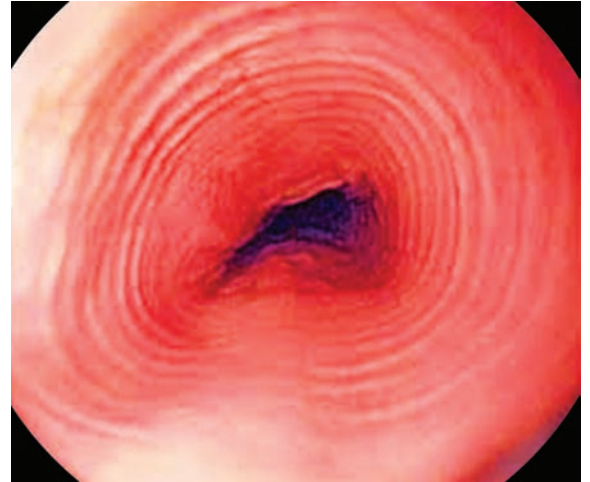


FIG. 5.6: Endoscopy showing trachealization.

DISEASES OF THE STOMACH AND DUODENUM

Peptic Ulcer Disease

Q. PA24.2 Discuss the etiology, pathogenesis, pathology, clinical features, investigations, and management of peptic ulcer disease or acid peptic disease.

Definition

An ulcer in the gastrointestinal (GI) tract may be defined as a break in the lining of the mucosa, with appreciable depth at endoscopy or histologic evidence of involvement of the submucosa.

Sites of Peptic Ulcer

Any portion of the GI tract exposed to acidic gastric juices

- Duodenum:** More common in the first portion of the duodenum (anterior or posterior wall) than in the stomach. Occasionally occurs at both anterior and posterior sites (“kissing” ulcers)
- Stomach:** Lesser curvature near the junction (transitional zone) of the body and antrum
 - Proximal ulcers: Located in the body of the stomach
 - Distal ulcers: Located in the antrum and angulus of the stomach
- Gastroesophageal junction of esophagus**
- Anastomotic site:** Occurs in patients who have undergone a distal gastric resection. Occur at margins of the gastroduodenal anastomosis/ gastrojejunostomy (anastomotic ulcer)
- Multiple ulcers:** In the duodenum, stomach, and/or jejunum in Zollinger-Ellison syndrome
- At metaplastic or heterotopic gastric mucosa,** e.g., Meckel diverticulum within an ileum having ectopic gastric mucosa

Incidence: ~12% in males and 10% in females.

Etiology

Q. Write short essay/note on risk factors for peptic ulcer disease.

Normal process in the stomach

Two opposing sets of forces keep stomach in a normal state: (1) damaging forces and (2) defensive forces.

- Damaging forces:** Capable of inducing mucosal injury are two gastric secretory products: (1) hydrochloric acid, and (2) pepsinogen.
- Defensive forces** is a three-level barrier composed of pre-epithelial, epithelial, and subepithelial elements.

Pre-epithelial barrier is a mucus-bicarbonate layer of the stomach.

- **Surface mucus secretion:** Mucin is secreted by surface foveolar cells. Actions of mucus are: (1) Mucus layer promotes formation of an “unstirred” protective layer of fluid on the mucosa. (2) Prevents the direct contact of large food particles with the epithelium. (3) Impedes the diffusion of ions and molecules such as pepsin.
- **Bicarbonate secretion into mucus** by surface epithelial cells → diffuse into the unstirred mucus → buffer the hydrogen ions entering from the luminal aspect → result in a pH gradient, ranging from 1 or 2 at the gastric luminal surface, and reaching to a neutrality of 6–7 along the epithelial cell surface.

Epithelial barrier: Consists of surface epithelial cells act through several factors, such as (1) production of mucus, (2) epithelial cell ionic transporters that maintain intracellular pH, (3) bicarbonate production, and (4) intracellular tight junctions.

Subepithelial barrier

- **Rich gastric mucosal blood flow:** Provides (1) bicarbonate (HCO_3^-), which neutralizes the acid generated by parietal cell, (2) an adequate supply of nutrients and oxygen, and (3) removes toxic metabolic by-products.

Pathogenesis of Peptic Ulcer (Fig. 5.7)

Q. Write short essay/note on *Helicobacter pylori* and its role in pathogenesis (pathophysiology) of gastric ulcer.

Q. **MI3.6** Describe the etiopathogenesis of acid peptic disease (APD) and the clinical course. Discuss the diagnosis and management of the causative agent of APD.

The imbalances between mucosal defensive forces (disruption of any of protective mechanisms) **and damaging forces** (direct mucosal injury) **cause chronic gastritis** and also **PUD**.

Direct mucosal injury/increased damage

Majority of PUD (both gastric and duodenal ulcers) can be attributed to NSAIDs and *H. pylori*.

- *Helicobacter pylori* is a gram-negative spiral bacteria with multiple unipolar flagella that allow them to move freely through the gastric mucus layer, where they remain protected from low gastric pH. It is one of the most important, common, primary causes of PUD. It is associated with ~85–90% of duodenal and ~65% of gastric ulcers. *Helicobacter pylori* infection remains one of the most common chronic bacterial infections in humans more than 50% of the world's population is infected with the bacterium. *H. pylori* also plays a role in the development of gastric and duodenal ulcer, gastritis, MALT (mucosal-associated lymphoid tissue) lymphoma, and gastric adenocarcinoma.
- **Mode of spread:** Oral-oral or feco-oral route either by kissing or ingestion of contaminated vomitus.

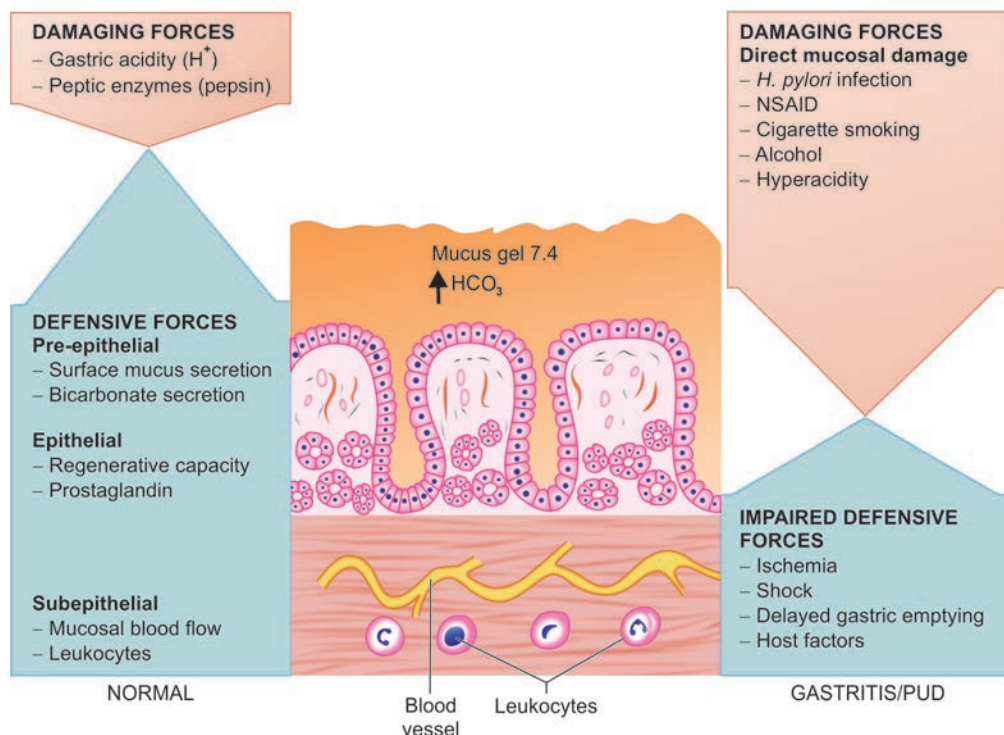


FIG. 5.7: Components involved in mucosal defense and repair in normal (left side) and in acute or chronic gastritis. Gastric mucus barrier consists of viscid mucus (which forms an unstirred layer between the epithelium and the gastric lumen) and bicarbonate.

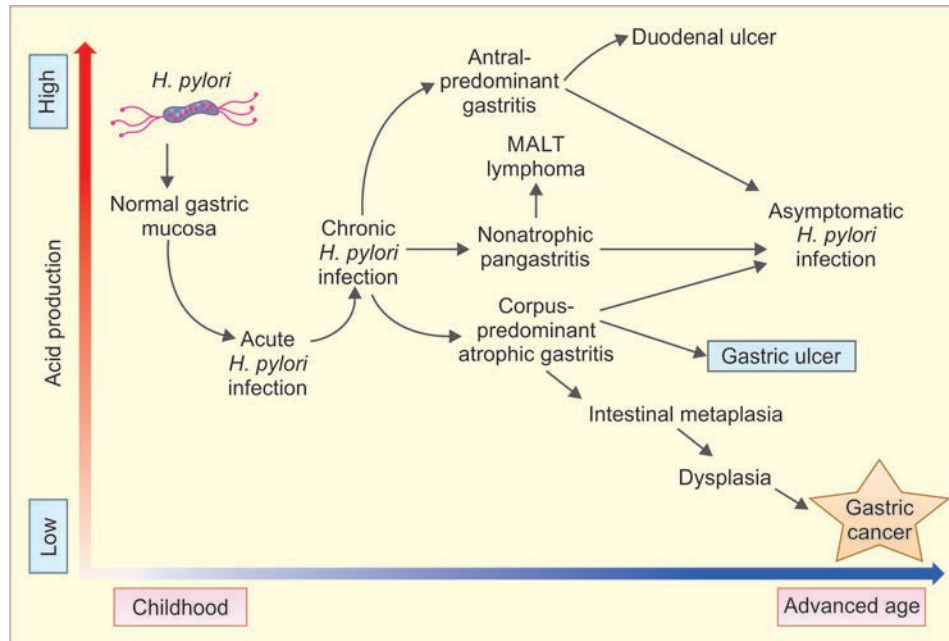


FIG. 5.8: Natural history of *H. pylori* infection.

- **Lesions produced:** *H. pylori* may attach to gastric epithelium causing damage to the mucosa. Causes chronic antral gastritis with high acid production → may progress to pangastritis → resulting multifocal atrophic gastritis → increased risk of gastric adenocarcinoma. Natural history of *H. pylori* infection is shown in **Figure 5.8**.

Nongastric diseases and *H. pylori* infection: Raynaud's, scleroderma, idiopathic urticaria, acne rosacea, migraine, thyroiditis, Guillain-Barré syndrome, coronary artery disease, immune thrombocytopenic purpura.

- **Mechanism of action by *H. pylori*:**

- ♦ **Flagella:** It makes them motile, allows them to burrow and live beneath the mucus layer above the epithelial surface.
- ♦ **Urease:** It is produced by *H. pylori* → converts urea into ammonia (strong alkali) → raises the local gastric pH, acts on the antral G cells → release of gastrin → hypergastrinemia → result in hypersecretion of gastric acid.
- ♦ **Adhesion molecule:** It helps it to bind to gastric epithelial (surface foveolar) cells.
- ♦ **Enzymes:** These include proteases and phospholipases act on the mucous gel → reduce the mucosal defense.
- ♦ **Cytotoxins:** Two genes namely cytotoxin-associated gene A (*CagA*) and vacuolating agent (*vacA*) gene cause gastritis, peptic ulceration, and cancer.
- ♦ **Cytokine induces inflammatory response:** Normally *H. pylori* does not invade the cells/tissues. It causes increased production of proinflammatory cytokines [interleukin (IL)-1, IL-6, tumor necrosis factor (TNF) and IL-8] by the mucosal epithelial cells → activation of neutrophils and macrophages (inflammatory response to gastric mucosa) → release of lysosomal enzymes, leukotrienes, and reactive oxygen species → impairs mucosal defense. The cytokines also stimulate gastrin release → **increased acid production**.
- **Nonsteroidal anti-inflammatory drugs (NSAIDs) and aspirin:** Causes (1) direct chemical irritation of mucosa, (2) suppresses mucosal prostaglandin synthesis, and (3) reduces the bicarbonate secretion.
- **Cigarette smoking** impairs blood flow to the mucosa and healing of mucosal damage.
- **Alcohol, radiation therapy, and chemotherapy:** Direct injury to mucosal cells.
- **Ingestion of chemicals:** Such as acids or bases cause direct injury.
- **Gastric hyperacidity:** It is induced by *H. pylori* infection, parietal cell hyperplasia, and Zollinger-Ellison syndrome.
- **Others:** High-dose corticosteroids that suppress prostaglandin synthesis and impair healing, other drugs (e.g., bisphosphonates, cocaine, and amphetamines), hypercalcemia, psychological stress, duodenal gastric reflux, Crohn's disease, and systemic mastocytosis.
- **Stress ulcers:** They occur with **shock, sepsis, or severe trauma**.
- **Curling ulcers:** They develop in the proximal duodenum with **severe burns or trauma**.

- **Cushing ulcers:** They develop in the stomach, duodenum, and esophagus in patients with **intracranial disease**. Highly prone for perforation.

Clinical Features

Recurrent episodes of abdominal pain: It is the **most common presentation** and has three notable characteristics:

1. **Localization to the epigastrium:** Pain is referred to epigastrium and the patient will be able to localize the site with one finger (pointing sign). The characteristics of pain are:
 - **Nature:** Usually burning in character or gnawing discomfort.
 - **Radiating pain:** May radiate to the back, thorax, and other parts of abdomen.
 - **Nocturnal:** Pain in duodenal ulcer occurs 90 minutes to 3 hours after a meal. May occur at night (most specific) and wakes the patient from sleep between midnight and 3 AM, and is relieved by food, milk, or antacids.
 - **Hunger pain:** Pain occurs on empty stomach (painful hunger) relieved by food or antacids.
2. **Relationship to food**
 - Pain is usually **relieved by food, milk, antacids**, belching, or vomiting in duodenal ulcer.
 - In contrast, in few patients with gastric ulcer food may precipitate the pain.
3. **Periodicity (episodic pain)**
 - In untreated patients, pain tends to occur in episodes. Each episode consisting of daily pain **lasting 2–8 weeks**, separated by prolonged asymptomatic intervals.
 - Between episodes (periods of remission) patient may be perfectly well and may be able to eat even heavy or spicy meals without apparent discomfort.
 - During the initial stages, the episodes tend to be of short in duration and less frequent. Later, the episodes become longer in duration and more frequent. More symptomatic during winter and spring. Relapses are more common in smokers compared to nonsmokers.

TABLE 5.34: Difference between gastric ulcer and duodenal ulcer.

Features	Gastric ulcer	Duodenal ulcer
Commonest site	Along the lesser curvature	First part of duodenum
Incidence	Less common	More common
Age	Beyond 6th decade, M > F	Between 25 and 50 years, M > F
Association with <i>H. pylori</i> infection	Less common	Strong association
Acid level	Usually normal	High
Clinical features		
Relationship of pain to antacids	Relief of pain not consistent	Prompt relief of pain
Relationship of pain to food	Aggravates the pain	Relieves the pain
Night pain	Not observed	Common
Heartburn	Not common	Common
Hematemesis/melena	Hematemesis more common	Melena more common
Vomiting	Common	No vomiting
Weight loss	Present	Absent
Anorexia and nausea	More common	Less common
Duration of episodes of pain	Relatively longer in duration	Relatively shorter in duration
Course of the illness	Less remission	More remission
Complication	Rarely undergo malignant change	No malignant change

Other symptoms

Retrosternal burning (heartburn), water-brash (excessive salivation), loss of appetite, and acidic regurgitation into the throat and vomiting. Persistent daily vomiting suggests gastric outlet obstruction. Fullness, bloating, anorexia, nausea, and dyspepsia. Tarry stools or coffee-ground vomitus indicate bleeding. Rarely, with anemia due to chronic blood loss, abrupt hematemesis, acute perforation, or gastric outlet obstruction.

Difference between gastric ulcer and duodenal ulcer is shown in **Table 5.34**.

Complications (Box 5.7)

Q. List complications of peptic ulcer disease.

Investigations/Diagnosis

Q. Discuss diagnosis of peptic ulcer disease.

Anatomic diagnosis: Documentation of a peptic ulcer needs either a radiographic (barium study) or an endoscopic examination.

- **Endoscopy (Fig. 5.9):** It is most sensitive and specific for the detection of ulcer disease of the upper GI tract. Typical location

Box 5.7: Complications of peptic ulcer.

- Upper gastrointestinal bleeding (20%)
- Perforation of ulcer (6–7%)
- Gastric outlet obstruction 1–2% (with fluid and electrolyte imbalance), gastrocolic or duodenocolic fistulas
- Rarely malignant transformation
- Rarely pancreatitis due to posterior penetration of ulcer

of peptic ulcer is duodenal bulb and lesser curvature of stomach. Advantages of endoscopy are: (1) direct visualization of mucosa (to determine if an ulcer is a source of blood loss) and the ulcer (even lesions too small to detect by radiographic examination), (2) useful for photographic documentation of a mucosal defect, and (3) biopsy can be taken to rule out malignancy (about 10% of gastric ulcers are malignant) or *H. pylori*.

- **Endoscopic ultrasound:** It may be useful in detecting an unsuspected submucosal component or enlarged lymph nodes (e.g., in gastric malignancies such as lymphoma and linitis plastica).
- **Etiologic diagnosis:** The cause of the ulcer must be established. The major risk factor for peptic ulcers is either *H. pylori* or NSAID.
- **Tests for *Helicobacter pylori*** (Table 5.35)

Q. Discuss the investigation done in *Helicobacter pylori* infection.



FIG. 5.9: Endoscopic picture of a benign gastric ulcer.

TABLE 5.35: Diagnostic tests for *Helicobacter pylori* (HP).

	Advantages	Disadvantages
Nonendoscopic tests		
Serology [qualitative or quantitative immunoglobulin G (IgG)]	Widely available, economical. Good (negative predictive value (NPV)	Poor positive predictive value (PPV) if HP prevalence is low. Not useful after treatment
Urea breath test (^{13}C or ^{14}C)	Detects active infection. Accuracy (PPV, NPV) not affected by <i>H. pylori</i> prevalence. Useful both before and after treatment	Accuracy affected by PPI and antibiotic use, small radiation dose with ^{14}C test
Stool antigen test	Detects active infection. Accuracy (PPV, NPV) not affected by <i>H. pylori</i> prevalence. Useful both before and after treatment (monoclonal test)	Accuracy affected by PPI and antibiotic use
Endoscopic tests		
Microscopic examination	Excellent sensitivity and specificity, especially with special and immune stains; provides additional information about gastric mucosa	Expensive (endoscopy and histopathology costs), interobserver variability, accuracy affected by PPI and antibiotic use
Rapid urease test	Rapid results, accurate in patients not using PPIs or antibiotics, no added histopathology cost	Requires endoscopy, less accurate after treatment or in patients using PPIs
Culture	Specificity 100%, allows antibiotic sensitivity testing	Difficult and tedious to perform; not widely available; expensive
Polymerase chain reaction (PCR) assay	Excellent sensitivity and specificity, permits detection of antibiotic resistance	Not widely available; technique not standardized; expensive

- **Nonsteroidal anti-inflammatory drugs:** Diagnosis is established based on history of drug use and symptoms of pain.
- **Hypersecretory syndromes:** Zollinger-Ellison syndrome should be considered in patients with multiple ulcers. **Serum gastrin and gastric acid analysis should be performed** in these patients.

Q. **IM15.15** Describe and enumerate the indications, pharmacology and side effects of pharmacotherapy of acid peptic disease including *Helicobacter pylori*.

Treatment/Management

Treatment may be divided into short-term management and long-term management (intermittent, maintenance, and surgical treatment).

A. Short-term management

• General measures (Box 5.8)

• Acid neutralizing/inhibitory drugs

a. Antacids: Neutralizes the secreted acid and rarely used at present.

- ◆ Mainly used for symptomatic relief of dyspepsia.
- ◆ Preparation: Tablet or liquid preparations
- ◆ Dose: 15–30 mL liquid antacid 1 and 3 hours after food and at bedtime for 4–6 weeks.
- ◆ **Commonly used antacids:** Combination of aluminum hydroxide and magnesium hydroxide. Others include calcium carbonate and sodium bicarbonate.
- ◆ **Side effects**
 - Aluminum compounds cause constipation, phosphate depletion and interfere with the absorption of digoxin and tetracycline

Box 5.8: General measures in peptic ulcer disease.

- **Avoid:** Cigarette **smoking**, **aspirin**, and **NSAIDs**
- **Alcohol** to be avoided
- **No special dietary advice** is necessary

- ❑ Magnesium compounds cause diarrhea, hypocalcemia, and hypermagnesemia.
- ❑ Calcium carbonate causes milk-alkali syndrome and sodium bicarbonate produces systemic alkalosis.

b. Histamine H_2 -receptor antagonists:

- ◆ **Drugs:** These include four agents namely cimetidine (400 mg BD or 800 mg at night), ranitidine (150 mg BD or 300 mg at night), famotidine (20 mg BD or 40 mg at night), and nizatidine (150 mg BD or 300 mg at night). All are equally effective.
- ◆ **Mechanism of action:** Inhibit acid and pepsin secretion by blocking H_2 -receptors.

Duration of treatment:

- ◆ **Duodenal ulcer:** Usually for 4 weeks. Smokers and patients with recent major complications (e.g., hematemesis, perforation), treatment is prolonged to 6–8 weeks.
- ◆ **Gastric ulcer:** For 6 weeks, followed by endoscopy and further treatment if necessary.

c. Proton pump (H^+ , K^+ -ATPase) Inhibitors (PPIs)

- ◆ These agents are substituted benzimidazole derivatives that covalently bind and irreversibly inhibit H^+ , K^+ -ATPase.
- ◆ They include omeprazole (20 mg/d), esomeprazole (20–40 mg/d), lansoprazole (15–30 mg/d), rabeprazole (20 mg/d), and pantoprazole (40 mg/d). All have similar efficacy in the treatment of various acid-peptic disorders.
- ◆ **Mechanism of action**
 - ❑ Proton-pump inhibitors are lipophilic compounds that cross the parietal cell membrane and enter the acidic parietal cell canaliculus.
 - ❑ Upon entering the acidic parietal cell, the PPIs are protonated, and trapped within the acid environment of the tubulovesicular and canalicular system. They become activated and bind covalently with the H^+ / K^+ ATPase enzyme and potentially inhibit all phases of gastric acid secretion by the proton pump.
- ◆ **Side effects:** Headache, diarrhea, abdominal pain, and nausea. The use of PPI may predispose to an increased risk of *Clostridium difficile* infection, community acquired pneumonia, hip fracture, and vitamin B_{12} deficiency.
- ◆ **Advantages:** Superior healing rates, shorter healing time, and faster relief of symptom compared to H_2 -blockers.

◆ Indications (Box 5.9)

Box 5.9: Indications for proton pump inhibitors (PPIs).

- GERD and reflux esophagitis
- Peptic ulcer not responding to other medical measures.
- As an adjunct to anti-*H. pylori* treatment.
- Zollinger-Ellison syndrome

d. Cytoprotective agents

- ◆ **Sucralfate:** It is a complex sucrose salt insoluble in water and becomes a viscous paste within the stomach and duodenum. It binds to sites of active ulceration. Sucralfate acts as a protective barrier, over the ulcer and increases the mucosal defense and repair. Standard dose 1 g qid.
- ◆ **Bismuth-containing preparations:** Colloidal bismuth subcitrate (CBS) and bismuth subsalicylate are used to induce healing of peptic ulcers. Side effects include black stools, constipation, darkening of the tongue, and neurotoxicity. They are commonly used as one of the agents in an anti-*H. pylori* regimen.
- ◆ **Prostaglandin analogs:** They enhance mucosal defense and repair and useful in preventing NSAID-induced mucosal injury. Dose (e.g., Misoprostol) is 200 μ g qid.

Q. Discuss management of *Helicobacter pylori* infection/*Helicobacter pylori* eradication regimens.

e. Treatment for *H. pylori* infection (Tables 5.36 and 5.37; Flowchart 5.6)

- ◆ **Indications:** Consensus of opinion is that all patients with proven acute or chronic duodenal ulcer and those with gastric ulcers who are *H. pylori*-positive should be administered drugs against *H. pylori* (even without documenting the presence of bacteria).
- ◆ **Advantages:** It reduces the risk of recurrence of ulcer.
- ◆ **Type of therapy:** Triple or quadruple therapy (Tables 5.36 and 5.37).

B. Long-term management

1. **Intermittent treatment:** When the symptoms relapses < four times a year, 4 weeks course of one of the ulcer-healing agents are prescribed.
2. **Maintenance treatment**
 - Continuous maintenance treatment is not required after successful eradication of *H. pylori*.
 - If symptoms relapses for more than four times per year or history of complications (e.g., repeated bleeding or perforation) require the lowest effective dose of PPI.
 - Long-term maintenance is with H_2 -receptor antagonists (cimetidine 400 mg at night, ranitidine 150 mg at night, famotidine 20 mg at night or nizatidine 150 mg at night).

New Treatments

- Cholecystokinin 2 receptor antagonists (CCK2): Itriglumide
- Potassium competitive acid blockers (P-CABs): Revaprazan

3. Surgical treatment

- Most of peptic ulcers are cured by *H. pylori* eradication therapy and by acid-suppressing drugs. Elective surgery is reserved for the treatment of medically refractory disease (recurrence of ulcer following surgery, gastric outlet obstruction), or urgent/emergency surgery for the treatment of an ulcer-related complication (e.g., perforation and hemorrhage).
- **Indications for surgery**
 - ◆ **Chronic nonhealing gastric ulcer:** Persistent ulceration despite adequate medical therapy. The procedure of choice is partial gastrectomy with a Billroth I anastomosis, in which the ulcer and the ulcer-bearing area of the stomach are resected.
 - ◆ **Gastric outflow obstruction**
 - ◆ **Recurrent ulcer following gastric surgery**
 - ◆ **Duodenal ulcer:** Most commonly performed procedures are:
 - Vagotomy and drainage (by pyloroplasty, gastroduodenostomy, or gastrojejunostomy)
 - Highly selective vagotomy (which does not require a drainage procedure)
 - Vagotomy with antrectomy
 - ◆ As an emergency for complications namely **perforation and hemorrhage**.

TABLE 5.36: First-line treatment of *Helicobacter pylori* infection.

Treatment regimen	Duration	Eradication rate
PPI (omeprazole/lansoprazole/pantoprazole/rabeprazole/esomeprazole), clarithromycin 500 mg, amoxicillin 1,000 mg (each twice daily)	10–14 days	70–85%
PPI, clarithromycin 500 mg, metronidazole 500 mg (each twice daily)	10–14 days	70–85%
Sequential therapy PPI, amoxicillin 1,000 mg (each twice daily) for 5 days followed by PPI, clarithromycin 500 mg, tinidazole 500 mg (each twice daily) for next 5 days	10 days	90%
Bismuth subsalicylate 525 mg, metronidazole 500 mg, tetracycline 500 mg (each four times daily) plus PPI or H ₂ RA (Ranitidine twice daily)	10–14 days	75–90%

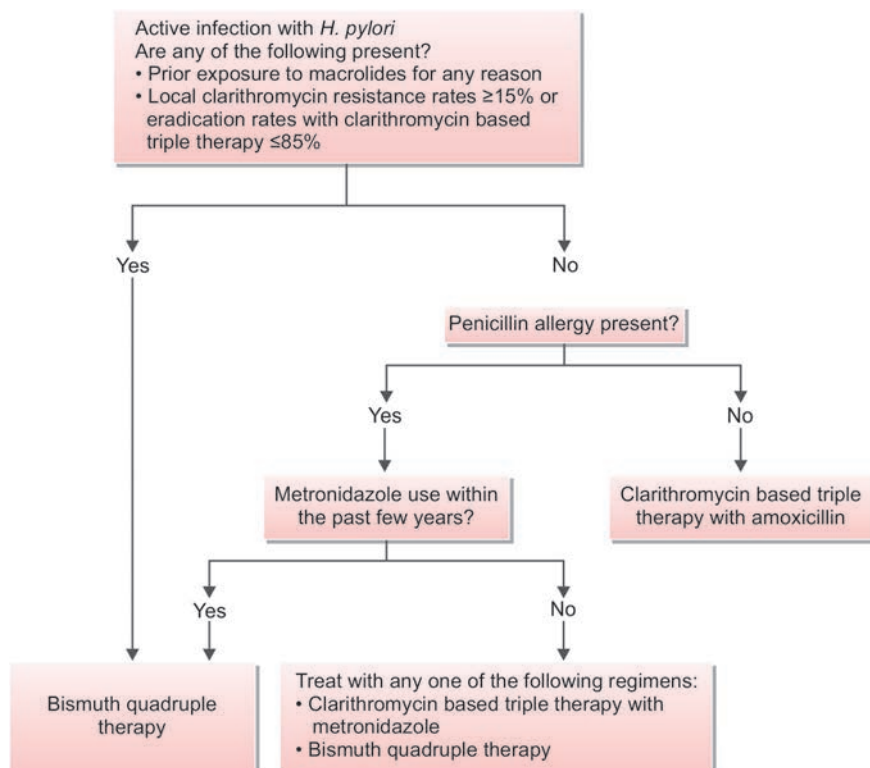
TABLE 5.37: Rescue treatment for persistent *Helicobacter pylori* infection.

Regimen	Duration	Eradication rate
Quadruple therapy: Bismuth subsalicylate 525 mg, metronidazole 500 mg, tetracycline 500 mg (each four times daily) plus PPI or H ₂ RA (twice daily)	14 days	70%
PPI, amoxicillin 1,000 mg, levofloxacin 250 mg (each twice daily)	10–14 days	57–91%
PPI amoxicillin 1,000 mg, rifabutin 150 mg (each twice daily)	14 days	60–80%

(H₂RA: Histamine; H₂-receptor antagonists; PPI: Proton pump inhibitors)

(H₂RA: Histamine; H₂-receptor antagonists; PPI: Proton pump inhibitor)

FLOWCHART 5.6: Initial approach to antibiotic treatment for *H. pylori* infection.



Dumping Syndrome

Q. Write a short note on dumping syndrome.

It is a series of vasomotor and GI signs and symptoms that occur when food reaches the small bowel too rapidly and usually develops in patients who have undergone vagotomy and drainage (especially Billroth procedures).

Triggering factor: It usually occurs after meals rich in simple carbohydrates (especially sucrose) and high osmolarity. Ingestion of large amounts of liquids may also contribute.

Phases: There are two phases of dumping namely early and late.

1. Early phase of dumping:

- **Time of occurrence:** Occurs 15–30 minutes after meals.
- **Signs and symptoms:** Consists of crampy abdominal discomfort, nausea, diarrhea, belching (bloating), borborygmi, tachycardia, palpitations, diaphoresis, lightheadedness, and rarely, syncope.
- **Mechanism:** The signs and symptoms are due to rapid emptying of hypertonic gastric contents into the small intestine that draws the fluid into lumen of the gut and leads to distension of small intestine. This leads to reduced volume of plasma and acute intestinal distention. Release of vasoactive GI hormones (vasoactive intestinal polypeptide, neurotensin, and motilin) may also play a role.

2. Late phase of dumping:

- **Time of occurrence:** Occurs 90 minutes to 3 hours after meals.
- **Signs and symptoms:** Consists of vasomotor symptoms such as lightheadedness, diaphoresis/sweating, palpitations, tachycardia, and occasionally syncope.
- **Mechanism:** Possibly secondary to hypoglycemia from excessive insulin release.

Provocative Test

- After an overnight fast, a solution of 50–75 g glucose given orally.
- Measure blood glucose, hematocrit, pulse rate, and blood pressure at 30 minutes intervals, immediately before and up to 180 minutes after ingestion of glucose.
- Positive test: Characterized by an early (30 minutes) increase in pulse rate (>10 /minutes) or hematocrit $>3\%$ and late (120–180 minutes) hypoglycemia.

Treatment

- Patients should be asked have small, multiple (six) meals and avoid simple carbohydrates and liquids during meals.
- Antidiarrheals and anticholinergic drugs are complementary to diet.
- Guar and pectin: They increase the viscosity of intraluminal food contents, thereby slowing down gastric emptying.
- Acarbose: It is an α -glucosidase inhibitor that delays digestion of ingested carbohydrates. It appears beneficial in the treatment of the late phases of dumping.
- Octreotide: It is a somatostatin analog given subcutaneously (50 g tid), before the meal may be used in diet-refractory cases. A long-acting octreotide may be given once every 28 days.

DISEASES OF THE INTESTINE

Malabsorption Syndrome

Q. PE25.1 Discuss the classification, etiology, clinical features, investigations (diagnosis), and management of malabsorption syndrome.

Q. List the investigation of a case of steatorrhea.

Definition: Malabsorption is defined as defective/diminished intestinal absorption of one or more dietary nutrients.

Classification and Etiology (Fig. 5.10)

Drugs causing malabsorption and systemic diseases associated with malabsorption are presented in **Table 5.38**.

TABLE 5.38: Drugs causing malabsorption and systemic diseases associated with malabsorption.

Drugs	Systemic diseases
Colchicine, neomycin, methotrexate, cholestyramine, and laxatives	Addison's disease, thyrotoxicosis, hypothyroidism, diabetes mellitus, and collagen vascular disease

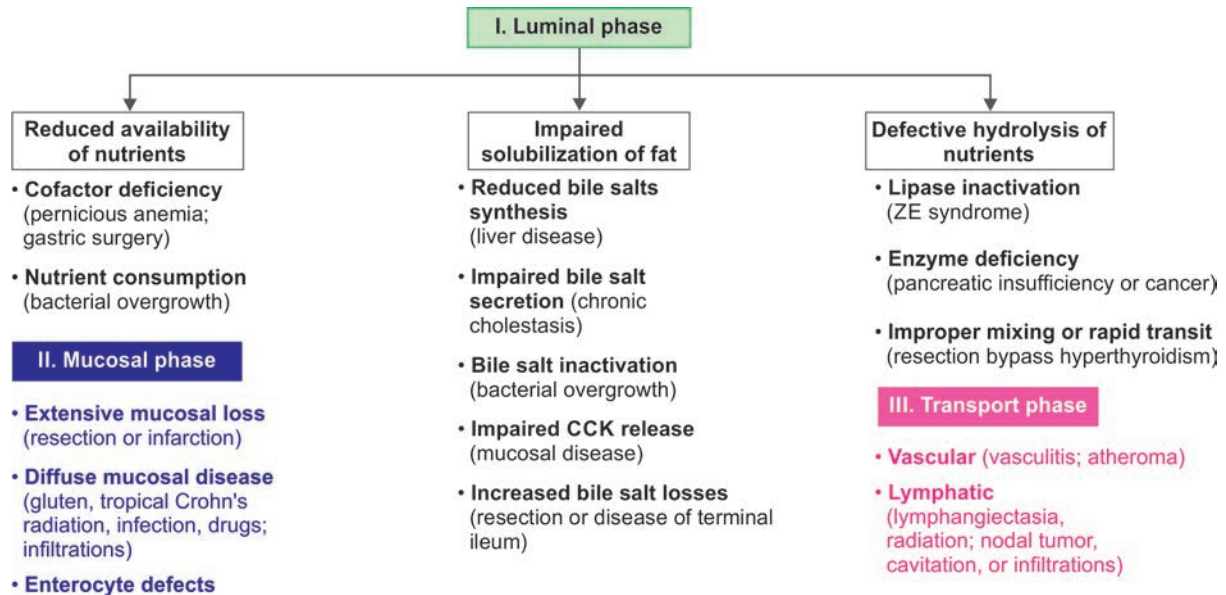


FIG. 5.10: Classification and causes of malabsorption.

Clinical Features

- Onset: Insidious and gradually progresses.
- General features: Diarrhea, abdominal pain, distension, loss of weight, and anemia.
- Symptoms and signs and relevant pathophysiology are presented in **Table 5.39** and **Figure 5.11**.

Q. IM16.3 Describe and discuss the chronic effects of diarrhea including malabsorption.

TABLE 5.39: Signs and symptoms of malabsorption and relevant pathophysiology.

Symptom or sign	Pathophysiology
Gastrointestinal	
Diarrhea	➢ Osmotic activity of carbohydrates or short-chain fatty acids ➢ Secretory effect of bile acids and fatty acids ➢ Decreased absorptive surface ➢ Intestinal loss of conjugated bile acids
Abdominal distention, flatulence	Bacterial gas production from carbohydrates in colon, small intestinal bacterial overgrowth
Foul-smelling flatulence or stool	Malabsorption of proteins or intestinal protein loss
Pain	Gaseous distention of intestine
Ascites	Protein loss or malabsorption
Musculoskeletal	
Tetany, muscle weakness, and paresthesias	Malabsorption of vitamin D, calcium, magnesium, and phosphate
Bone pain, osteomalacia, and fractures	Protein, calcium, or vitamin D deficiency; secondary hyperparathyroidism
Cutaneous and Mucosal	
Easy bruisability, ecchymosis, and petechiae	Vitamin K deficiency and vitamin C deficiency
Glossitis, cheilosis, and stomatitis	Vitamin B complex, vitamin B ₁₂ , folate, or iron deficiency
Edema	Protein loss or malabsorption
Ichthyosis, acrodermatitis, and scaly dermatitis	Zinc and essential fatty acid deficiency
Follicular hyperkeratosis	Vitamin A deficiency
Hyperpigmented dermatitis	Niacin deficiency (pellagra)
Koilonychia	Iron deficiency
Perifollicular hemorrhage	Malabsorption of vitamin C
Other	
Weight loss and hyperphagia	Nutrient malabsorption
Growth and weight retardation, infantilism	Nutrient malabsorption in childhood and adolescence

Contd...

Contd...

Symptom or sign	Pathophysiology
Anemia	Iron, folate, or vitamin B ₁₂ deficiency
Kidney stones	Increased colonic oxalate absorption
Amenorrhea, impotence, and infertility	Multifactorial (including protein malabsorption, secondary hypopituitarism, and anemia)
Night blindness, xerophthalmia	Vitamin A deficiency
Peripheral neuropathy	Vitamin B ₁₂ or thiamine deficiency
Fatigue, weakness	Calorie depletion, iron and folate deficiency, and anemia
Neurologic symptoms, ataxia	Vitamin B ₁₂ , vitamin E, or folate deficiency

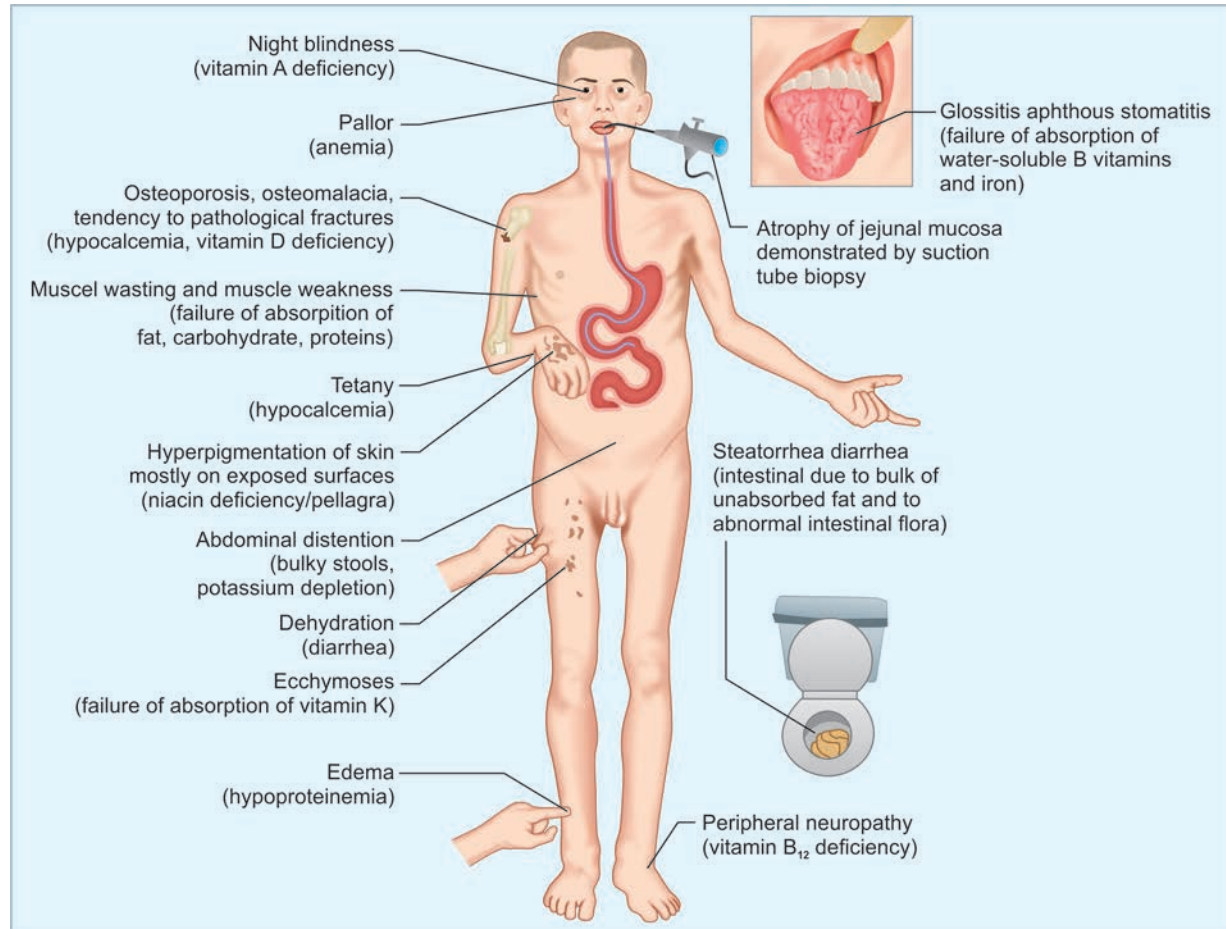


FIG. 5.11: Signs and symptoms of malabsorption.

Clinical features of specific malabsorption disorders are presented in **Table 5.40**.

TABLE 5.40: Clinical features of specific malabsorption disorders.

Disorder	Cardinal clinical features
Adrenal insufficiency	Skin darkening, hyponatremia, and hyperkalemia
Amyloidosis	Renal disease, nephrotic syndrome, cardiomyopathy, neuropathy, carpal tunnel syndrome, macroglossia, and hepatosplenomegaly
Carcinoid syndrome	Flushing, cardiac murmur
Celiac disease	Variable symptoms: Dermatitis herpetiformis, alopecia, aphthous mouth ulcers, arthropathy, neurologic symptoms, and (life-threatening) malnutrition; abnormal liver function test results, and mild iron deficiency
Crohn's disease	Arthritis, aphthous mouth ulcers, episcleritis, uveitis, pyoderma gangrenosum, erythema nodosum, abdominal mass, fistulas, primary sclerosing cholangitis (PSC), and laboratory signs of inflammation
Cystic fibrosis	Chronic sinopulmonary disease, meconium ileus, distal intestinal obstruction syndrome (DIOS), and elevated sweat chloride
Diabetes mellitus	Long history of diabetes and diabetic complications

Contd...

Contd...

Disorder	Cardinal clinical features
Glucagonoma	Migratory necrolytic erythema
Hyperthyroidism, hypothyroidism	Symptoms and signs of thyroid disease
Lymphoma	Enlarged mesenteric or retroperitoneal lymph nodes, abdominal mass, abdominal pain, and fever
Pancreatic insufficiency	History of pancreatitis, abdominal pain; or alcoholism; large-volume fatty, and oily stools
Parasitic infection	History of travel to endemic areas
Primary biliary cirrhosis	Jaundice, itching
Scleroderma	Dysphagia, Raynaud's phenomenon, and skin tightening
Tropical sprue	History of travel to endemic area
Tuberculosis	Specific history of exposure, living in or travel to endemic area, immunosuppression, abdominal mass or intestinal obstruction, and ascites
Whipple's disease	Lymphadenopathy, fever, arthritis, cerebral symptoms, and heart murmur
Zollinger-Ellison syndrome	Peptic ulcers and diarrhea

Investigations (Table 5.41 and Flowchart 5.7)

TABLE 5.41: Laboratory tests useful in patients with suspected malabsorption.

Test	Comment
Peripheral blood findings	
Hematocrit, hemoglobin	Decreased in iron, vitamin B ₁₂ , and folate malabsorption or with blood loss
Mean corpuscular hemoglobin or mean corpuscular volume	Decreased in iron malabsorption; increased in folate and vitamin B ₁₂ malabsorption
White blood cells, differential	Decreased in vitamin B ₁₂ and folate malabsorption
Biochemical tests (serum)	
Triglycerides	Decreased in severe fat malabsorption
Cholesterol	Decreased in bile acid malabsorption or severe fat malabsorption
Albumin	Decreased in severe malnutrition, protein-losing enteropathy
Alkaline phosphatase	Increased in calcium and vitamin D malabsorption (severe steatorrhea); decreased in zinc deficiency
Calcium, phosphorus, and magnesium	Decreased in extensive small intestinal mucosal disease, after extensive intestinal resection, or in vitamin D deficiency
Zinc	Decreased in extensive small intestinal mucosal disease or intestinal resection
Iron and ferritin	Decreased in celiac disease, in other extensive small intestinal mucosal diseases, and with chronic blood loss
Other serum tests	
Prothrombin time	Prolonged in vitamin K malabsorption
Folic acid	Decreased in extensive small intestinal mucosal diseases, with anticonvulsant use, in pregnancy; may be increased in small intestinal bacterial overgrowth
Vitamin B ₁₂	Decreased after gastrectomy, in pernicious anemia, terminal ileal disease, and small intestinal bacterial overgrowth
Endoscopy and biopsy stool tests	
Fat	Qualitative or quantitative increase in fat malabsorption
Elastase, chymotrypsin	Decreased concentration and output in exocrine pancreatic insufficiency
pH	<5.5 in carbohydrate

Tests for fat absorption

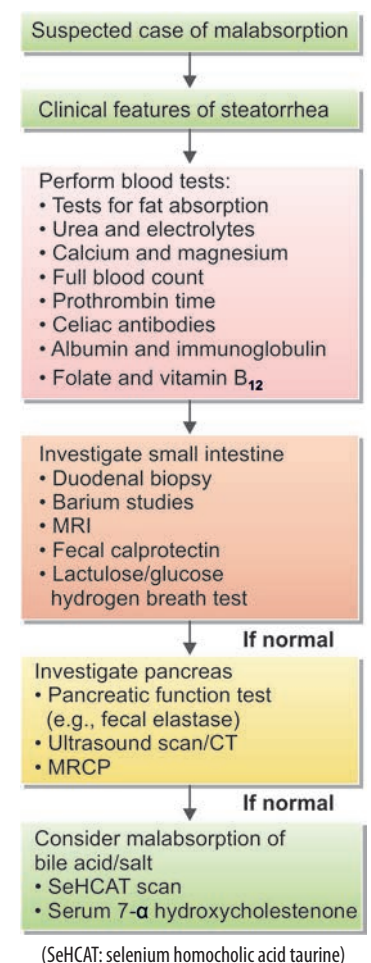
Quantitative tests

- 72-hour-stool fat collection: Gold standard
- 6 g/day: Pathologic, patients with steatorrhea >20 g/day.

Qualitative tests

- **Sudan III stain** for fat globules in stool: Detect clinically significant steatorrhea in >90% of cases. Acid steatocrit: A gravimetric assay. Sensitivity—100%, specificity—95%, PPV—90%
- **NIRA (near infrared reflectance analysis)**: Equally accurate with 72 hours stool fat test. Allows simultaneous measurement of fecal fat, nitrogen, and CHO (carbohydrate).

FLOWCHART 5.7: Step-wise investigation in a suspected case of malabsorption.



- Measurement of blood levels of fat-soluble vitamins (A, D, E, and K); prothrombin time.
 - Plasma vitamin A level after 2–3 days of oral retinol: It will be lower than normal.

Tests for carbohydrate absorption

- **Glucose tolerance test:** It reveals absence of rise in blood glucose levels.
- **Blood sugar level:** Low
- **D-xylose absorption test:** D-xylose is a pentose monosaccharide absorbed exclusively at the proximal small intestine. This test is used to assess proximal small intestine mucosal function. The procedure is as follows:
 - After overnight fast, administer 25 g D-xylose perorally.
 - Urine collected for next 5 hours and urinary excretion of D-xylose is estimated.
 - Abnormal test: <4.5 g is abnormal and suggest intestinal malabsorption (duodenal/jejunal mucosal disorders) and excludes pancreatic cause.
- **Lactose tolerance test (LTT):**
 - Orally 50 g of lactose is given.
 - Blood glucose levels measured at 0, 60, and 120 minutes.
 - Normally, blood glucose shows a rise of >20 mg/dL.
 - Blood glucose <20 mg/L + development of symptoms favor diagnosis.
- **Hydrogen breath test:** It is used for the diagnosis of lactase deficiency. 50 g lactose given orally and breath hydrogen measured every hour for 4 hours. Normal value is <10 ppm increase above baseline.

Tests for protein absorption

- **Serum albumin:** Low
- **Nitrogen in the 24-hour stool:** >2.5 g.
- **α_1 -antitrypsin content in stool:** Normally no antitrypsin in stool. A 3 days collection of stools is estimated for α_1 -antitrypsin content.

Test for bile acid malabsorption

Selenium-homocholic acid taurine (SeHCAT) test: In this test, Se-labeled bile acid is given orally. After 7 days, total body retention is measured with a gamma camera. Retention value of <10% is abnormal.

Tests for absorption of other substances

- Serum B₁₂ level: Reduced
- **Schilling test for absorption of B₁₂:** To determine the cause of cobalamin (B₁₂) malabsorption (discussed in Chapter 9).

Test for bacterial overgrowth

- Quantitative bacterial count from aspiration from small intestine. Normal count is <10 U/mL (jejunum), >10 U/mL (ileum).
- **Breath tests:** Choly-14-C-glycine, e.g., bacterial overgrowth. Lactose H₂ for lactose intolerance, e.g., in bacterial overgrowth.
- **Small intestinal biopsy (duodenal or jejunal):** Lesions seen can be classified into three:
 1. Diffuse and specific, e.g., Whipple's disease
 2. Patchy and specific, e.g., Crohn's disease, lymphoma, and infectious causes
 3. Diffuse and nonspecific, e.g., celiac sprue, tropical sprue, and autoimmune enteropathy

Barium studies: Important information about the gross anatomy and morphology of small intestine.

Treatment of Malabsorption Syndrome

- **General measures**
 - Dehydration and electrolyte deficiency is treated by intravenous infusion
 - Replacement therapy for anemia, bone disease and coagulation defects. Vitamin D, B complex, and calcium supplements
- **Specific therapy depending on the causes**
 - Celiac disease: Gluten-free diet
 - Pancreatic insufficiency: Pancreatic supplements
 - Bile acid deficiency/malabsorption: Low-fat diet and cholestyramine (bile acid binder)

Celiac Disease

Q. Write short essay/note on celiac disease (nontropical sprue/gluten-induced enteropathy).

Definition

Celiac disease is characterized by small intestinal malabsorption of nutrients after the ingestion of wheat gluten or related proteins from rye and barley, villus atrophy of the small intestinal mucosa, prompt clinical and histologic improvement following strict adherence to a gluten-free diet, and clinical and histologic relapse when gluten is reintroduced.

Etiology

Exact etiology is not known, but environmental, immunologic, and genetic factors play an important role.

- **Environmental factor:** Gluten is the protein present in the cereals such as wheat, barley and rye. **Gliadin** is a component of gluten. There is a clear-cut association of the celiac disease with gliadin. Gluten restriction results in morphologic changes within hours in patient with celiac disease.
- **Immunologic factor:** It involves both adaptive and innate immune responses. Presence of serum antibodies such as IgA anti-gliadin, IgA anti-endomysial (90–95% sensitivity and specificity), and IgA anti-tTG antibodies in patients with celiac disease. It is not known whether they represent primary or secondary to the tissue damage.
- **Genetic factor(s):** Increased incidence of celiac disease is observed within families but the exact mode of inheritance is unknown. All patients express the HLA-DQ2 or HLA-DQ8 allele.

Pathology

- **Site of lesion:** Biopsy from the **second portion of the duodenum or proximal jejunum** (which are exposed to the highest concentrations of dietary gluten) is diagnostic.
- **Microscopic changes:** Characterized by **increased numbers of intraepithelial CD8+ T lymphocytes** (intraepithelial lymphocytosis), crypt hyperplasia, and **villous atrophy**. Increased numbers of **plasma cells, mast cells, and eosinophils** within the upper part of the **lamina propria**.

Clinical Features

Age and gender: It can manifest at any age. Usually manifests early in childhood at about 2 years of age (after wheat is introduced into the diet), or later in the second to fourth decades of life with a female preponderance.

Symptoms: Many patients are asymptomatic (silent). The symptoms are very variable and often nonspecific.

- Symptoms due to malabsorption of multiple nutrients, with diarrhea, steatorrhea, weight loss, and the consequences of nutrient depletion (i.e., anemia and metabolic bone disease).
- Symptoms due to the depletion of a single nutrient (e.g., iron or folate deficiency or vitamin B₁₂ deficiency, osteomalacia, edema from protein loss, and tetany).

Associated diseases

See **Box 5.10**.

Complications

See **Box 5.11**.

Investigations and Diagnosis

- Small-intestinal biopsy: Duodenal/jejunal biopsy is required and is considered as “gold standard” to establish a diagnosis.
- Serologic tests (**Box 5.12**)
- Tests for malabsorption of proteins, carbohydrate, fat, and vitamins
- HLA typing. HLA-DQ2 is observed in 90–95% of patients and HLA-DQ8 in about 8%.

Box 5.10: Diseases associated with celiac disease.

- Insulin-dependent diabetes mellitus (2–8%)
- Thyroid disease (5%)
- Primary biliary cirrhosis (3%)
- Sjögren's syndrome (3%)
- IgA deficiency (2%)
- Pernicious anemia

Box 5.11: Complications of celiac disease.

- Gastrointestinal malignancies: An increased incidence of both gastrointestinal (e.g., *carcinoma of the esophagus*) and non-gastrointestinal neoplasms as well as *enteropathy associated T cell lymphoma (EATCL)*
- Development of intestinal ulceration independent of lymphoma (ulcerative jejunitis), refractory sprue, and collagenous sprue
- Nonresponsive celiac disease: Few patients do not improve on a strict diet
- Refractory celiac disease (RCD)
- Pneumococcal infections
- Dermatitis herpetiformis
- Osteomalacia

Box 5.12: Serologic tests in celiac disease.

Serologic tests

- IgA anti-endomysial antibody
- IgA antitissue transglutaminase antibody (anti-tTG)
- IgA anti-gliadin antibody
- IgG anti-gliadin antibody

Treatment/Management of Celiac Disease

- **Diet: Lifelong gluten-free diet.** Avoid wheat, rye, and barley. Oats not contaminated with flour are tolerated by most patients. Rice, meat, dairy products, vegetables, and fruits are safe. Completely avoid beer and whisky obtained from malt distillation procedures. Lactose-free diet is advisable because of secondary lactase deficiency associated with celiac disease.
- **Supplementation of vitamins and minerals** in cases patients with these deficiencies, folic acid supplements for women of childbearing age.
- **Pneumococcal vaccinations** (because of splenic atrophy) once in every 5 years.
- **Corticosteroids:** They are rarely required. Used in critically ill-patients who present with acute celiac crisis characterized by severe diarrhea, dehydration, weight loss, acidosis, hypocalcemia, and hypoproteinemia. May be used in gliadin shock after a gluten challenge.

Tropical Sprue

Q. Write short essay on tropical sprue (idiopathic tropical malabsorption syndrome).

Definition: Tropical sprue is a condition presenting with chronic diarrhea and malabsorption (of two or more substances) that occurs in residents or travelers to certain, but not all tropical areas (especially India and Southeast Asia) in the absence of other intestinal diseases or parasites.

Etiology

- Etiology and pathogenesis of tropical sprue are unknown.
- **Role of infectious agents:** Because tropical sprue responds to antibiotics, it is possible that it may be caused by one or more infectious agents (e.g., *Klebsiella pneumoniae*, *Enterobacter cloacae*, or *E. coli*) or a toxin produced by these agents.
- **Role of folic acid deficiency:** Folic acid is absorbed in the duodenum and proximal jejunum, and most patients with tropical sprue show evidence of folate malabsorption and depletion.

Clinical Features

- Clinical features vary in different parts of the world.
- **Intensity:** Vary and consist of diarrhea, anorexia, abdominal distension fatigue, and weight loss.
- **Onset:**
 - **Acute:** Sometimes acute with sudden severe diarrhea and accompanied by fever. It occurs either a few days or many years after being in the tropics.
 - **Chronic phase:** Onset may also be insidious, with chronic diarrhea and evidence of nutritional deficiency. Features of megaloblastic anemia (folic acid malabsorption) and other deficiencies, including ankle edema, glossitis, and stomatitis, are common. Remissions and relapses may occur.

Treatment

- **Broad-spectrum antibiotics:** Tetracycline or oxytetracycline 1 g daily in four divided doses for up to 6 months and produces improvement within 1–2 weeks
- **Folic acid:** Dose of 5 mg daily along with tetracycline is most often curative. It induces a hematologic remission and improvement in appetite, weight gain, and also morphological changes in small intestinal biopsy
- **Correction of deficiencies:** Severely ill patients require resuscitation with fluids and electrolytes for dehydration. Nutritional deficiencies should be corrected such as vitamin B (1,000 µg)
- **Symptomatic treatment for diarrhea**
- **Prognosis:** Excellent. Mortality is usually due to water and electrolyte depletion

Diagnosis and Investigations

- **Abnormal jejunal mucosal biopsy:** It shows partial villous atrophy and mononuclear cell infiltrate in the lamina propria. The lesion is less severe than that of celiac disease.
- **Stool examination**
- **Demonstration of malabsorption**

Lactose Intolerance

Q. Write short note on lactose intolerance.

Etiology

- Lactose is the disaccharide present in milk and it is digested to monosaccharides (glucose and galactose) by the brush border lactase.
- **Classification:** There are two types of lactase deficiency namely primary and secondary.
 - Primary lactase deficiency:* It is due to genetic deficiency or absence of lactase and both intestinal absorption and brush border enzymes are normal.
 - Secondary lactase deficiency:* It occurs in association with both structural and functional disorders of brush border enzymes and transport processes, e.g., celiac disease, tropical sprue, and Crohn's disease.
- Lactose intolerance develops due to the deficiency of lactase and lactose present in the milk cannot be hydrolyzed. Hence, lactose gets fermented by bacteria causing gastrointestinal symptoms.

Clinical Features

- Diarrhea, abdominal pain, cramps, and/or flatus develops after ingesting milk or milk products.
- Symptoms improve on withdrawal of milk or milk products.

Investigations

- **Hydrogen breath test:** This is used as a screening test to measure transit time and detect bacterial overgrowth in the small intestine. Appearance of breath hydrogen peak after oral lactulose indicates that the undigested lactose is fermented by the colonic microflora and produces hydrogen which can be detected in expired air.
- Measurement of lactase activity in a jejunal biopsy.

Treatment

- Lactose free diet. Avoid milk and dairy products. However, it may lead to nutritional deficiencies, particularly calcium deficiency.
- **Administration of exogenous β -galactosidase**
 - Enzyme replacement therapy with microbial exogenous lactase (obtained from yeasts or fungi)
 - Enzymes supplement to milk and dairy products
 - Use of yogurt and probiotics (source of β -galactosidase)

Whipple's Disease

Q. Write short note on Whipple's disease and its important clinical features.

Whipple's disease is a chronic, rare, multisystemic infectious bacterial disease caused by *Tropheryma whipplei* (gram-positive).

Clinical Features

- Gender: More common in males than in females.
- Whipple's disease is a chronic multisystem disease associated with diarrhea, steatorrhea, weight loss, arthralgia, and central nervous system (CNS) and cardiac problems.
- Migratory arthralgias of the large joints
- Dementia and other CNS findings (such as supranuclear ophthalmoplegia, nystagmus, and myoclonus)
- Cardiac disease (dyspnea, pericarditis, and culture-negative endocarditis)
- Pleuropulmonary (pleural effusion)

Investigations

Biopsy: Jejunal biopsy and biopsy of other involved tissues show numerous **PAS—positive macrophages with small bacilli**. In the small intestine, these macrophages produce lymphatic blockade in the lamina propria causing malabsorption.

Treatment: Trimethoprim-sulfamethoxazole (double strength tablet): Dose twice a day for 1 year. This should be preceded by a 2-week course of parenteral therapy with ceftriaxone (2 g daily) or by meropenem.

Giardiasis

Q. Write short essay/note on clinical features, diagnosis, and treatment of giardiasis.

- Giardiasis is an infection caused by flagellate *Giardia lamblia* (*Giardia intestinalis*).
- Giardiasis is found worldwide and is one of the most common parasitic diseases in both developed and developing countries, causing intestinal disease, and diarrhea.
- It mainly affects children, tourists and immunosuppressed individuals (agammaglobulinemia).
- **Incubation period:** 1–3 weeks
- **Mode of infection:** By ingesting **contaminated water**. In cystic form, it is viable in water for up to 3 months.
- Malabsorption may develop due to loss of brush-border enzyme activities and few patients may show flattening of villi.

Clinical Features

- Range from asymptomatic carrier stage to fulminant diarrhea and malabsorption.
- After ingestion of flagellar trophozoite form, it attaches to the mucosa of duodenum and jejunum and causes inflammation. It produces disease of small intestine.
- **Acute giardiasis:** Starts as **diarrhea, abdominal pain**, nausea, vomiting, anorexia, belching, flatus, weakness, and abdominal pain. Fever and blood in stool are rare. The duration is usually >1 week.
- **Chronic giardiasis:** May present with or without having experienced antecedent acute episode. Diarrhea may not be a prominent symptom and they may have increased flatus, and loose stools. It may produce steatorrhea; malabsorption if illness is prolonged may result in marked weight loss. Symptoms may be continuous or episodic and can persist for years. Chronic giardiasis in children from developing countries may develop retardation of growth.
- **Severe disease:** Giardiasis may be severe in patients with hypogammaglobulinemia, resulting in malabsorption, weight loss, growth retardation, and dehydration.
- Physical examination may reveal abdominal distension and tenderness.
- **Extraintestinal manifestations:** These include urticaria, anterior uveitis, and arthritis.

Investigations/Diagnosis

- **Stool examination: Demonstration of protozoal parasite**
- **Duodenal or jejunal aspiration** by endoscopy or with a luminal capsule gives a higher diagnostic yield.
- **String test:** In this one, end of a piece of string is passed into the duodenum by swallowing and retrieved after an overnight fast. The fluid is expressed from this and examined for the presence of *G. lamblia* trophozoites.
- **Biopsy:** Biopsy and histological examination of jejunal mucosa may show *G. lamblia* on the epithelial surface.
- **Stool antigen detection tests: Detection of parasite antigens in the feces** is sensitive and specific and easier to perform.
- **Other tests:** Chronic cases may show steatorrhea, malabsorption of xylose, and vitamin B₁₂ and lactose intolerance.

Treatment/Management

- **Tinidazole** 40 mg/kg (2 g/day) once by mouth and repeated after 1 week
- **Metronidazole** 2 g daily for 3 days or metronidazole 200 mg thrice daily for 5–7 days is usually curative in >90% of cases
- **Nitazoxanide** 500 mg twice a day for 3 days. Other drugs include mepacrine and albendazole
- **Paromomycin** is an oral aminoglycoside and is not well absorbed. It can be given to symptomatic pregnant patients
- **Refractory cases:** Prolonged therapy with metronidazole 800 mg thrice daily for 3 weeks

Carcinoid Tumors (Neuroendocrine Tumors)

Q. Write short note on carcinoid tumors (neuroendocrine tumors) and carcinoid syndrome.

- Neuroendocrine cells are amine- and acid-producing cells located throughout the body.
- Neuroendocrine tumors (NETs) are tumors derived from the diffuse neuroendocrine system of the GI tract.
- Location of carcinoid tumors: Appendix (25%), small intestine (25%), and rectum (15%). Stomach (5%) and lung (15% of all carcinoid tumors).

Classification of Carcinoid Tumors

According to their anatomic area of origin:

- Foregut (lungs, bronchi, and stomach)
- Midgut (small intestine, appendix, and proximal large bowel)
- Hindgut (distal large bowel, rectum).

Ultrastructurally, they possess electron-dense neurosecretory **granules which may contain various substances. These include: 5-hydroxytryptamine (serotonin), histamine, tachykinins, motilin, prostaglandins, bradykinins, adrenocorticotrophic hormone, and corticotropin-releasing factor.**

Carcinoid Syndrome

It is symptom complex in which systemic symptoms are produced when the vasoactive substances are secreted by carcinoid tumors into the systemic circulation. Carcinoid syndrome occurs in only 8–10% of patients.

- **Midgut carcinoids** (usually ileal carcinoid) with hepatic metastasis can produce carcinoid syndrome. Because, normally liver is able to metabolize these secretory products into inactive forms and with hepatic metastasis the product can reach systemic circulation.
- **Foregut (bronchial and extraintestinal) carcinoids** can present with carcinoid syndrome without hepatic metastases. This is because bioactive products can bypass liver and can be released directly into systemic circulation.
- **Hindgut carcinoids** seldom produce syndrome since they do not secrete these products.
- Intestinal carcinoids have an increased tendency for liver metastasis.

Clinical Features (Table 5.42)

Characterized by cutaneous flushing, involving head and neck (blush area) associated with lacrimation, sweating, bronchospasm, colicky abdominal pain, diarrhea, and right-sided cardiac valvular fibrosis.

Diagnosis

Diagnosed by measuring urinary or plasma serotonin or its metabolites in the urine.

- **Measurement of 5-hydroxyindoleacetic acid (5-HIAA):** Increased (>9 mg) urinary excretion in 24-hour sample collection.
- **Serum chromogranin A levels** are elevated in 56–100% of patients.
- Serotonin level in blood and platelets is high, because it is overproduced in 92% of patients.

TABLE 5.42: Clinical features of carcinoid tumors.

Intestinal obstruction by tumor	Intestinal ischemia due to either mesenteric infiltration or vasospasm
Hepatic metastases may produce pain and jaundice	Features of carcinoid syndrome: Flushing, wheezing, diarrhea, etc.
Cardiac involvement (e.g., tricuspid regurgitation, pulmonary stenosis) leading to heart failure	Facial telangiectasia

- Plasma neuron-specific enolase levels are increased but are less sensitive than chromogranin A.
- **Others:** CT, MRI, and somatostatin receptor scintigraphy, PET (positron emission tomography) with radiolabeled 5-hydroxytryptophan may be helpful.

Ischemic Colitis

Q. Write short note on ischemic colitis.

Etiology

- **Transient nonocclusive hypoperfusion of a segment of the colon**, e.g., systemic hypoperfusion
- **Occlusion of the inferior mesenteric artery or its branches**, e.g., atherosclerotic or thrombotic occlusion
- **Less common causes**, e.g., hypercoagulable states (in young persons), iatrogenic ligation of the inferior mesenteric artery (e.g., with aortic surgery), embolism, vasculitis, and colonic obstruction (e.g., colonic cancer).
- **Drugs and chemicals:** Medications, illicit drugs, and chemicals may also produce chemical picture identical or similar to ischemic colitis (e.g., digitalis, vasopressin, pseudoephedrine, amphetamines, cocaine, estrogens, ergot alosetron, etc.).

Clinical Features

- **Age:** Occur in persons older than 60 years without any apparent cause.
- **Onset:** Many cases are acute and self-limited.
- **Symptoms:** Colicky lower abdominal pain (mostly left lower quadrant), diarrhea, and passage of bright red blood per rectum, nausea, and vomiting. Though the blood loss is not usually enough to require transfusion, some patients may progress to shock because of loss of blood and fluid.
- **Physical examination:** Abdominal tenderness and guarding over the affected portion of the colon (especially left iliac fossa), and abdominal distention fever and tachycardia.

Laboratory Findings/Investigations

- Nonspecific findings such as leukocytosis and hemoconcentration
- Plain radiographs of the abdomen: It may show “**thumbprinting**” at splenic flexure and descending colon or may be normal.
- Computed tomography (CT) scanning of abdomen: Useful to exclude other disorders.
- Double contrast barium enema demonstrates involvement of splenic flexure and descending colon. Mucosal abnormalities are **thumbprinting** and ulceration.
- Colonoscopy has replaced barium studies as it is more sensitive. Sigmoidoscopy shows normal rectal mucosa and bleeding descending from above.
- Arteriography confirms the diagnosis in patients with obstructive lesions.

Pseudomembranous Colitis

Q. Write short note on pseudomembranous colitis (antibiotic-associated colitis).

Pseudomembranous colitis also referred to as antibiotic-associated colitis or antibiotic-associated diarrhea, generally caused due to toxin produced by *Clostridium difficile*, when the normal bacterial flora is altered or suppressed by antibiotics.

- Antibiotic-associated diarrhea may (**Table 5.43**) also be caused by other organisms such as *Salmonella*, *C. perfringens* type A, or *Staphylococcus aureus* but only *C. difficile* causes pseudomembranous colitis.
- In the colon, the vegetative form of *C. difficile* multiplies and produces **two major toxins** namely toxin A (an enterotoxin) and toxin B (a cytotoxin).

Treatment of Carcinoid Tumors

- Avoid conditions and diets that precipitate flushing and diarrhea
- Supplementation of diet with nicotinamide
- Wheezing is treated with oral bronchodilators
- Treatment of heart failure with diuretics
- Control of the diarrhea with antidiarrheal agents (e.g., loperamide and diphenoxylate). If not controlled advise serotonin receptor antagonists (cyproheptadine, methysergide, and ondansetron) to control diarrhea
- Analogs of somatostatin (e.g., octreotide and lanreotide) to control flushing
- Surgical resection of the carcinoid tumor
- Excision of liver metastases or hepatic artery embolization with or without chemotherapy

Treatment

- **For right-sided ischemic colitis:** It requires visceral angiography which is needed for both diagnosis and intra-arterial administration of vasodilators (e.g., papaverine as a 60-mg intravenous bolus followed by an infusion of 30–60 mg/hour). Some patients may need urgent surgery
- **Left-sided acute ischemic colitis:** It accounts for most cases and most cases resolve within hours to a few days with supportive therapy. Therapy includes volume replacement, correction of any low-flow state, broad-spectrum antibiotics (similar to those recommended earlier for patients with small bowel ischemia), avoidance of vasoconstrictive medications, and rarely, blood transfusion. Surgery is required only in patients with signs and symptoms of transmural necrosis, perforation, or massive bleeding
- **Conservative management:** Intravenous fluids, hemodynamic stabilization, discontinuation or avoidance of vasoconstrictive agents, bowel rest, and empiric antibiotics
- **Surgical treatment:** Peritonitis and strictures

Clinical Features

- **Age:** Pseudomembranous colitis usually develops in adults and about 80% of cases occur in people over 65 years of age.
- Patient is on antibiotics or has received antibiotics within last 8 weeks. Clinical symptoms range from asymptomatic to severe and sometimes life-threatening fulminant pseudomembranous colitis.
- **Diarrhea:** It is the most common manifestation.
 - Stool: Soft and unformed to watery or mucoid in consistency and has a characteristic odor. Stools are almost never grossly bloody.
 - Frequency: Patients may have as many as 20 bowel movements/day.
 - Profuse watery diarrhea with abdominal cramps.
 - Substantial fluid and protein losses combined with fever, cramps, hypoalbuminemia, leukocytosis, and hypotension.
- **Laboratory findings:** Leukocytosis is common (observed in up to 50% of patients) and is a marker of severe disease. Leukocytosis often $>15,000/\text{mm}^3$ white blood cells (WBCs)/L. Extremely high level ($>50,000/\text{mm}^3$ indicates fulminant and potentially fatal illness).
- **Complications:** Dehydration, electrolyte disturbances, hypoalbuminemia, toxic megacolon, bowel perforation, hypotension, renal failure, hypotension, lactic acidosis sepsis, and death.

Investigations and Diagnosis

- Enzyme immunoassay for *C. difficile* glutamate dehydrogenase
- Enzyme immunoassay for *C. difficile* toxins A and B
- Real-time PCR tests detect the critical gene is *tcdB*.
- Stool culture for *C. difficile*
- **Sigmoidoscopy or colonoscopy:** Endoscopy is a rapid diagnostic tool in seriously ill patients and characteristically shows erythema, white plaques or an adherent pseudomembrane (Fig. 5.12). Biopsy can be done.

TABLE 5.43: Antimicrobial agents that predispose to *Clostridium difficile*-associated diarrhea and colitis.

Most frequently	Less frequently
➤ Ampicillin and amoxicillin	➤ Macrolides (including erythromycin)
➤ Cephalosporins	➤ Other penicillins
➤ Clindamycin	➤ Sulfonamides
➤ Fluoroquinolones	➤ Trimethoprim/sulfamethoxazole

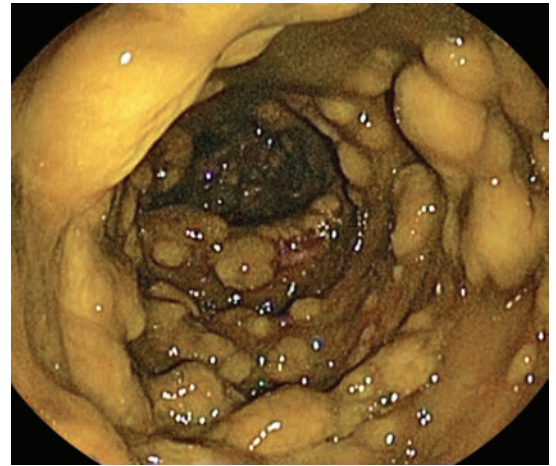


FIG. 5.12: Endoscopic appearance of pseudomembranous colitis.

Treatment/Management

Treatment consists of (1) discontinuation of the implicated antibiotic, (2) supportive care, and (3) avoidance of antiperistaltic agents and opiates

- **Withdraw/discontinue** the precipitating antibiotic/antimicrobial drug and isolate the patient
- **Supportive care:** Oral or intravenous rehydration
 - Mild-to-moderate cases:** Oral metronidazole 500 mg thrice daily for 10–14 days. Rifaximin can be added
 - **Severe cases:** Oral vancomycin 125 mg 6 hourly for 14 days
 - **Severe complicated or fulminant cases:** Vancomycin (500 mg orally or via nasogastric tube) plus intravenous metronidazole (500 mg q8h) *plus consider* rectal instillation of vancomycin (500 mg in 100 mL of normal saline as a retention enema q6–8h). Intravenous vancomycin not effective. Newer drug fidaxomicin (200 mg twice daily) is as efficacious as vancomycin. If ileus develops vancomycin administered by enema
 - ◆ **Oral probiotic therapy:** Use of live nonpathogenic bacteria to re-establish the gut flora often used in resistant or relapsed cases. Probiotics use organisms resistant to gastric acid. However, its clinical benefits are not proved
- **Avoidance of antiperistaltic agents and opiates:** Antiperistaltic drugs such as loperamide or diphenoxylate should not be administered because they can precipitate toxic megacolon
- Colectomy may be necessary in severely ill and refractory cases
- Newer treatment modalities: **Fecal microbiota transplantation** (fecal enemas or infusion of donor feces through a nasoduodenal tube)

Psychosomatic Disorders of the Gastrointestinal Tract

Classification (Table 5.44)

TABLE 5.44: Classification of the functional disorders of gastrointestinal tract.

➤ Bad taste (cacogeusia)—foul smell of breath (halitosis)	➤ Globus hystericus
➤ Functional heartburn—noncardiac chest pain due to esophageal dysmotility	➤ Unspecified excessive belching
➤ Functional dyspepsia—functional bloating aerophagia	➤ Functional or psychogenic vomiting and cyclic vomiting syndrome
	➤ Irritable bowel syndrome
	➤ Functional diarrhea and functional constipation

Irritable Bowel Syndrome

Q. Write short essay/note on irritable bowel syndrome, its clinical features and management.

Definition

- Irritable bowel syndrome (IBS) is a functional disorder of the gastrointestinal tract characterized by abdominal pain or discomfort and altered bowel habits in the absence of detectable structural, infective or biochemical abnormalities.
- Irritable bowel syndrome is benign, chronic symptom complex of altered bowel habits, and abdominal pain.

Pathophysiology

Exact cause not known and pathogenesis of IBS is poorly understood. However, the following factors have been proposed.

Gastrointestinal motor abnormalities: GI motility is quantitatively different in 25–75% of IBS patients compared with healthy controls.

- Patients with diarrhea as a predominant symptom exhibit rapid jejunal contraction waves, rapid intestinal transit, an increased number of colonic contractions (colonic transit is accelerated), and rectosigmoid motor activity.
- Patients with predominant constipation have decreased orocecal transit and a reduced number of colonic contraction waves.

Visceral hypersensitivity: IBS is associated with increased sensory responses to visceral stimulation induced by intestinal distension. The mechanism for visceral hypersensitivity is not known. This is more common in females and in diarrhea-predominant IBS.

Central neural dysregulation: There is a clinical association of emotional disorders and stress with symptom exacerbation in IBS. MRI has shown that in response to distal colonic stimulation, the mid-cingulate cortex (brain region concerned with attention processes and response selection) shows greater activation in IBS patients. This cerebral dysfunction may lead to the increased perception of visceral pain.

Abnormal psychological features: Abnormal psychiatric features are found in up to 80% of IBS patients. These include anxiety, tension, depression, excessive worry, somatization, and neurosis.

Altered intestinal flora: Bacterial overgrowth (shown by positive lactulose hydrogen breath test) in the small bowel has been observed in IBS patients and may contribute to symptoms.

Abnormal serotonin pathways: Serotonin is a neurotransmitter that plays an important role in the regulation of GI motility and visceral perception.

Postinfectious IBS: IBS may be induced by initial GI infection by *Campylobacter*, *Salmonella*, and *Shigella*.

Immune activation and mucosal inflammation: Certain diet/foods may precipitate an attack.

Clinical Features

- **Age:** IBS affects all ages. Most patients present before age 45 (20–40 years).
- **Gender:** More common in women compared to men (3:1).
- **Abdominal pain** is the most common key symptom (prerequisite clinical feature) for the diagnosis of IBS. Characteristics of pain:
 - **Location:** Pain is most typically referred to the lower abdomen (left or right iliac fossa) or hypogastrium.
 - **Aggravation:** Pain may be aggravated by emotional stress or poor sleep, and intake of food.
 - **Relief:** Abdominal symptoms are relieved by defecation or passage of flatus, but this relief may be temporary.
- **Altered bowel habits:** Most consistent clinical feature in IBS. Bowel pattern subtypes/**variants** are:
 - **Irritable bowel syndrome-constipation predominant (IBS-C):** Patient reports that abnormal bowel movements are usually constipation (>25% Bristol stool types 1 or 2; <25% Bristol stool types 6 or 7).
 - **Irritable bowel syndrome-diarrhea predominant (IBS-D):** Patient reports that abnormal bowel movements are usually diarrhea (>25% Bristol stool types 6 or 7; <25% Bristol stool types 1 or 2).
 - **Mixed IBS (IBS-M):** They have features of alternating diarrhea and constipation (>25% Bristol stool types 1 or 2; >25% Bristol stool types 6 or 7).
 - **Irritable bowel syndrome-unclassified (IBS-U)** includes those patients who meet the diagnostic criteria for IBS but cannot be accurately categorized into one of the other three subtypes.

Most common alteration in the bowel habit is constipation alternating with diarrhea.

- **Diarrhea:** Frequent defecation, often painless, but produce low-volume stools and rare/never at night. Diarrhea usually consists of small volumes of loose stools. Sense of incomplete evacuation, leading to repeated attempts at defecation in a short time span.
- **Constipation:** Pass infrequent pellet-like (ribbon-like or pencil-like) stools usually in association with abdominal pain or proctalgia.
- **Stools:** Usually hard with narrowed caliber. May be accompanied by passage of large amounts of mucus.
- **Gas and flatulence:** Abdominal distention and increased belching or flatulence more common among female patients. Postprandial tenesmus is common and is due to an exaggerated gastrocolic reflex.
- **Other symptoms:** Dyspepsia, heartburn, nausea, and vomiting
- **Extraintestinal symptoms:** Headache, back pain, fatigue myalgia, dyspareunia, and urinary frequency are often observed.
- **Physical examination:** Usually unremarkable, except for abdominal tenderness (most often in the left lower quadrant) or a tender, palpable sigmoid colon. In IBS-C, rectal examination may show paradoxical contraction of the puborectalis muscle or decreased descent of the pelvic floor when simulating a bowel movement.

Investigations

Aim of investigations is to exclude organic gastrointestinal diseases.

- **Full blood count**
- **Stool examination:** For parasites, ova, leukocytes, and occult blood.
- **Sigmoidoscopy:** Usually normal. There may be difficulty in negotiating rectosigmoid curve due to spasm. There may be marked motor activity of bowel and may show abundant mucus.
- **Colonoscopy:** Should be performed to exclude colorectal cancer, in older patients and those with history of rectal bleeding.
- **Barium enema:** Usually normal but may show spasticity of sigmoid, exaggerated haustral markings and a tubular appearance of the descending colon.
- In diarrhea-predominant patients investigations to exclude microscopic colitis, lactose intolerance, bile acid malabsorption, celiac disease, and thyrotoxicosis.

Diagnostic Criteria

Rome IV diagnostic criteria for irritable bowel syndrome are presented in **Table 5.45**. Red flag signs of IBS are presented in **Table 5.46**.

TABLE 5.45: Rome IV diagnostic criteria for irritable bowel syndrome.

Recurrent abdominal pain on average at least 1 day per week in the last 3 months (with symptom onset at least 6 months prior to diagnosis) associated with *two or more* of the following:

1. Related to defecation
2. Associated with a change in frequency of stool
3. Associated with a change in form (appearance) of stool

Symptoms that support the diagnosis of irritable bowel syndrome:

- Abnormal stool frequency: ≤ 3 bowel movements per week or > 3 bowel movements per day
- Abnormal stool form: Lumpy/hard stool or loose/watery stool
- Defecation straining
- Urgency
- Feeling of incomplete bowel movement
- Passing mucus
- Bloating or feeling of abdominal distention

TABLE 5.46: Red flag signs of IBS.

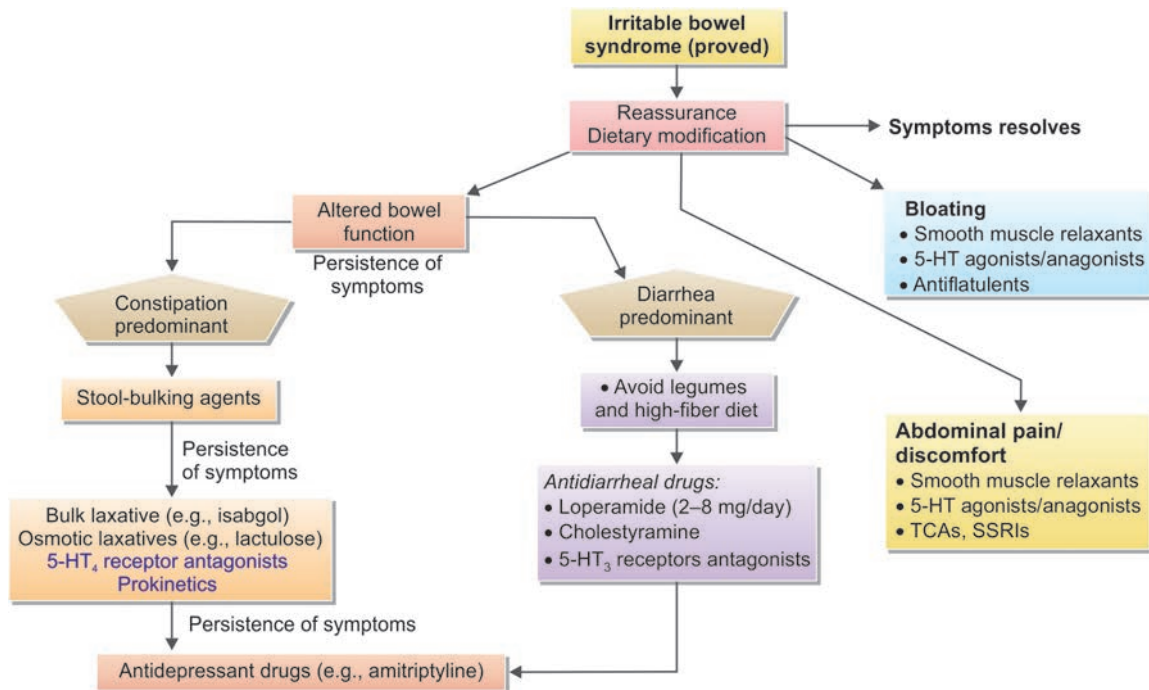
- | | |
|--------------------------------------|--|
| ➤ More than minimal rectal bleeding | ➤ Family history of selected organic diseases including colorectal cancer, inflammatory bowel disease (IBD), or celiac sprue |
| ➤ Weight loss | ➤ Fever |
| ➤ Unexplained iron deficiency anemia | |
| ➤ Nocturnal symptoms | |

Treatment (Flowchart 5.8)

Patient counseling and dietary alterations

- **Reassurance** and thorough careful explanation of the functional nature and benign prognosis of the disorder
- **Dietary modification:** Avoid obvious foods that precipitate symptoms. The role of probiotics is not well-established
 - Many IBS patients have nonspecific intolerance to foods
 - The dietary restriction of fermentable carbohydrates popularly termed the low **FODMAP** (fermentable oligosaccharides, disaccharides, monosaccharides, and polyols) diet
 - Exclusion of gas-producing foods/foods that increase flatulence (e.g., beans, onions, celery, carrots, raisins, bananas, apricots, prunes, brussels sprouts, wheat germ, pretzels, and bagels), alcohol, and caffeine
 - Irritable bowel syndrome patients should avoid foods that trigger an onset of their symptoms, consume a minimum of high fat foods, and take part in regular physical activity

FLOWCHART 5.8: Management of irritable bowel syndrome.



(5-HT: 5-hydroxytryptamine; TCAs: tricyclic antidepressants; SSRIs: selective serotonin reuptake inhibitors)

Constipation-predominant IBS

- **Stool-bulking agents:** High-fiber diets and bulking agents (e.g., bran or hydrophilic colloid) increase the roughage content of the diet (increases the fecal output of bacteria) and delay colonic transit. If symptoms persist add a bulk laxative (methylcellulose, isabgol husk, psyllium fiber supplement, and corn fiber). Osmotic laxatives (e.g., polyethylene glycol, lactulose, and milk of magnesia) or secretory stimulators (e.g., lubiprostone) may be tried, if high-fiber supplementation fails.
- **Prokinetic agents**
 - Tegaserod is a 5-HT₄ receptor agonist and a prokinetic drug, stimulates intestinal peristalsis. However, tegaserod has been withdrawn from the market; because of serious cardiovascular events.
 - Cisapride and mosapride are not useful in most patients.
- **Antispasmodics:** If abdominal pain is not adequately relieved, synthetic anticholinergic agent (dicyclomine 10 mg TID) or an antispasmodic (mebeverine 135 mg TID) may be given.
- **Antidepressant drugs:** Patients with intractable symptoms may be benefited from tricyclic antidepressant drugs such as amitriptyline (10–25 mg orally at bedtime) or desipramine (10–25 mg orally at bedtime).
- **Antiflatulence therapy:** Advised to eat slowly and avoid chew gum or carbonated beverage drinks. Antibiotic rifaximin (400, three times a day) may also help in relieving bloating.
- **Chloride-channel activators:** Lubiprostone activates chloride channels in the intestinal epithelial cells. This in-turn induces passive movement of sodium and water into the lumen of the intestine and improves bowel function.

Diarrhea-predominant IBS

- Advised to avoid legumes and high-fiber diet.
- **Antidiarrheal drugs**
 - Loperamide (2–4 mg up to four times a day maximum of 12 g/day).
 - Cholestyramine resin is a bile acid-binding agent used as a second-line agent (4 g four to six times a day).
- **Serotonin receptor antagonists:** Serotonin acts on 5-HT₃ receptors and increases the sensitivity of afferent neurons of the gut. Alosetron (others drugs include ondansetron, granisetron, alosetron, and cilansetron) is a 5-HT₃ receptor antagonist reduces abdominal visceral pain, induces rectal relaxation, increases rectal compliance, and delays colonic transit. But it is of little use because of danger of producing ischemic colitis.
- **Antispasmodics:** For the control of pain.
- **Antibiotics:** Nonabsorbable antibiotics such as rifaximin 400 mg BID for 5–6 weeks has been tried.
- **Probiotics use:** The bifidobacteria, *Saccharomyces boulardii*, and other combinations of probiotics demonstrate some efficacy in IBS.
- **Asimadoline:** Kappa-opioid receptor agonist has been in trials for IBS.

Abdominal Tuberculosis

Q Describe the etiology, pathology, clinical features, complications, differential diagnosis, investigations, and treatment of abdominal tuberculosis.

It includes gastrointestinal tuberculosis, peritoneum, and abdominal lymph nodes, either individually or in combinations.

Gastrointestinal Tuberculosis

Intestinal tuberculosis commonly involves ileocecal region (about 70% of cases) probably because this region has: (1) increased rate of absorption of fluid and electrolyte; (2) increased physiological stasis; (3) prominent lymphoid tissue; and (4) minimal digestive activity.

Other rare regions: These include ascending colon, jejunum, sigmoid colon, rectum, duodenum, stomach, and esophagus.

Etiology

- **Causative agent:** *Mycobacterium tuberculosis* human or bovine strain.
- **Routes of spread of intestinal tuberculosis**
 - **Hematogenous spread:** Primary or secondary tuberculosis of lung may spread through blood into the intestine.
 - **Ingestion:** Mycobacteria in sputum from active pulmonary focus may be ingested and cause intestinal tuberculosis.
 - **Direct spread:** From nearby or adjacent organs.
 - **Through lymphatics:** From infected lymph nodes
- **Routes of spread of peritoneal tuberculosis:** Infection may spread to peritoneum from lymph nodes or intestinal lesions or tuberculous salpingitis (fallopian tube) in females.

Pathology

Intestinal tuberculosis: Three types:

1. **Ulcerative (60%):** More often observed in malnourished adults.
 - **Gross:** Characteristics of tuberculous ulcers are: (1) multiple superficial ulcers mainly found in the ileocecal region; (2) long axis of ulcers is perpendicular to the long axis of the intestine (in typhoid fever, long axis of ulcers lies parallel to the long axis of the intestine).
 - **Consequences:** Fibrosis occurring during healing of ulcers produces strictures. Rarely, endarteritis may lead to massive bleeding.
2. **Hypertrophic (10%):** This is a variant of secondary tuberculosis observed in ileocecal region. More often found in relatively well-nourished adults.
 - **Gross:** The lesions are seen mainly in cecum and/or ascending colon. These regions become thick-walled due to scarring, fibrosis, and local hypertrophy and show mucosal ulceration.
 - Clinically, the lesion present as a palpable and may be mistaken for carcinoma.
3. **Ulcerohypertrophic (30%):**
 - It represents combination of both ulcerative and hypertrophic types.
 - Right iliac fossa mass due to hypertrophic ileocecal region, and involved mesenteric fat and lymph nodes.

Peritoneal Tuberculosis

- **Gross appearance of peritoneum:** Shows multiple yellow-white tubercles. Peritoneum is thick and hyperemic with a loss of its shiny appearance. Omentum is also thickened.
 - **Forms:**
 - *Wet type:* It presents with ascites.
 - *Encysted (loculated) type:* It presents with a localized abdominal swelling.
 - *Fibrotic type:* It presents with abdominal masses. They consist of thickening of mesentery and omentum with matted bowel loops. Adhesions may produce intestinal obstruction.
- A combination of above types is also can be seen.

Solid Organ Tuberculosis

These include: (1) Spleen, (2) liver, (3) adrenal tuberculosis, and (4) genitourinary tuberculosis.

Tuberculosis of Mesenteric Lymph Nodes

Usually observed in young adults. Involves mesenteric lymph nodes and these nodes may become palpable as rounded mass (tabes mesenterica). May be confused with lymphoma.

Clinical features

- Usually found in young adults.
 - It may present as acute, chronic, or acute on chronic condition.
 - **Common symptoms:** Fever, abdominal pain, anorexia, malaise, weight loss, diarrhea, constipation, alternating constipation and diarrhea, and gastrointestinal bleeding or obstruction.
 - Abdominal pain may be either colicky (due to luminal compromise) or dull and continuous when there is involvement of mesenteric lymph nodes.
- **Physical examination of the abdomen:** It is described as “**doughy**” because matted loops of bowel may be palpable. A well-defined, firm, mobile mass may be found in the right lower quadrant of the abdomen. Mesenteric lymphadenitis produces mass. Abdominal distention due to ascites.

Complications and differential diagnosis are described in **Box 5.13** and **Table 5.47**.

Investigations (Flowchart 5.9)

Hematological investigations: ESR is raised, anemia, and hypoalbuminemia.

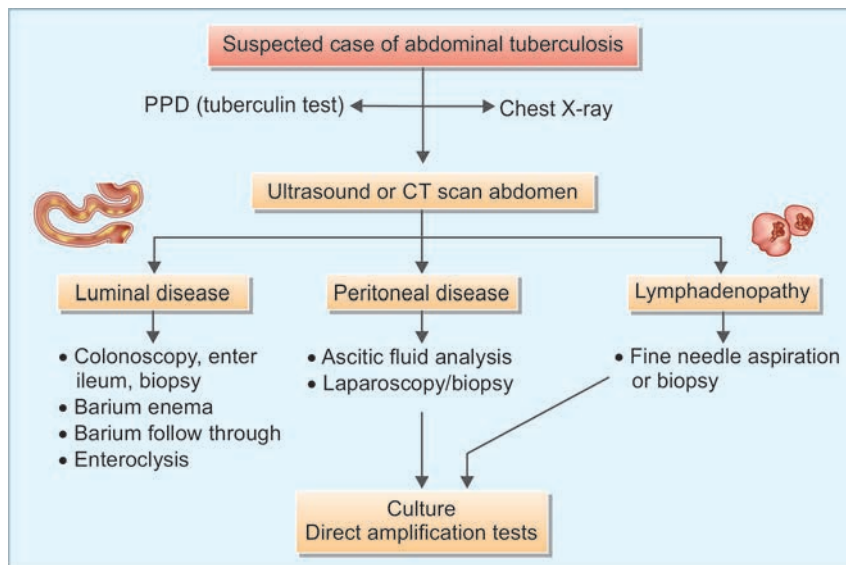
Box 5.13: Complications of intestinal tuberculosis.

- Intestinal hemorrhage
- Subacute intestinal obstruction
- Fistula formation (between skin and intestine or between loops of intestines)
- Intestinal perforation
- **Malabsorption:** Ileocecal tuberculosis is the common cause of malabsorption in India

TABLE 5.47: Differential diagnosis of tuberculosis of intestine.

➤ Tropical sprue	➤ Amebiasis
➤ Intestinal worm infestations	➤ Crohn's disease
➤ Lymphoma	➤ Malignancy of colon

FLOWCHART 5.9: Diagnostic algorithm for abdominal tuberculosis.



Radiological investigations

- Chest X-ray: May show evidence of active or old tuberculosis.
- Plain X-ray of abdomen: It may reveal calcified lymph nodes.
 - **If there is intestinal obstruction:** Dilated bowel loops with multiple air-fluid levels may be seen.
 - **If intestinal perforation:** Air under diaphragm and dilated loops.
- **Barium meal**
 - May show hypermotility (accelerated intestinal transit)
 - Hypersegmentation of the barium column (“chicken intestine”)
 - Precipitation, flocculation, and dilution of the barium
 - Luminal stenosis with smooth but stiff contours (“hour-glass stenosis”)

Multiple strictures with segmental dilatation of bowel loops:

Stierlin sign: A defect characterized by failure of the diseased segment to retain barium that is otherwise normally retained by adjacent uninvolved segments. Appears as a narrowing ileum with rapid emptying into a shortened cecum.

String sign: A thin stream of barium seen in the terminal ileum.

Note: Both Stierlin and String sign can also be seen in Crohn’s disease, and hence, are not specific for tuberculosis.

- **Barium enema**
 - Wide gaping of ileocecal valve with narrowing of the terminal ileum (“Fleischner” or “inverted umbrella sign”). Fold thickening and contour irregularity or terminal ileum shrunken in size (“conical cecum”).
 - Pulled up cecum due to contraction and fibrosis of the mesocolon.
 - Loss of normal ileocecal angle and dilated terminal ileum, appearing suspended from a retracted, fibrosed cecum. Localized stenosis opposite the ileocecal valve with a rounded off smooth cecum and a retracted fibrosed cecum.
- **Abdominal ultrasound**
 - Intra-abdominal fluid (free or loculated; with or without debris and septae).
 - Localized fluid between rapidly oriented bowel loops due to local exudation from the inflamed bowel (interloop ascites; “club sandwich” or “sliced bread” sign).
 - Lymphadenopathy, discrete or matted with heterogeneous echotexture due to caseation.
 - Uniform and concentric bowel wall thickening in the ileocecal region (versus eccentric thickening at the mesenteric border found in Crohn’s disease and variegated appearance of malignancy).
- **Contrast-enhanced CT scan**
 - Symmetric circumferential thickening of cecum and terminal ileum. Ulceration or nodularity within the terminal ileum, along with narrowing and proximal dilatation.
 - *Regional lymph nodes*: Caseating lymph nodes with hypodense centers and peripheral rim enhancement. Retroperitoneal nodes (periaortic and pericaval) almost never seen in isolation, unlike lymphoma.
 - *Mesenteric thickening*: Omental thickening seen as an omental cake appearance. Thickened peritoneum and enhancing peritoneal nodules.
 - Ascitic fluid of high-attenuation value
- **Ascitic fluid examination** lymphocytic ascites with serum-ascites albumin gradient (SAAG) (1.1 g/dL and elevated ascites ADA level)
- **Biopsy of peritoneum**: Punch or laparoscopic biopsy
- **Colonoscopy**: It may show mucosal nodules and ulcers in colon and is used for taking biopsy from the edge of ulcers.
- **Biopsy from other organs involved**: Liver, lymph node.

Treatment

- Antitubercular treatment similar to treatment of pulmonary tuberculosis
- Surgery
- Strictureplasty for strictures that reduce the lumen by 50% or more and that cause proximal dilation
- Resection of segment having multiple strictures

TABLE 5.48: Conditions causing ulcers in intestine.

Predominantly small intestine	Predominantly large intestine	Both small and large intestine
➤ Enteric/typhoid fever ➤ Tuberculosis ➤ Zollinger-Ellison syndrome ➤ Mesenteric artery occlusion	➤ Amebiasis ➤ Ulcerative colitis ➤ Gram-negative bacillary dysentery ➤ Ischemic colitis	➤ Crohn’s disease ➤ Malignant ulcers

Ulcers in Intestine

Q. Write short note on ulcers in intestine.

Conditions causing ulcers in intestine are presented in **Table 5.48**.

INFLAMMATORY BOWEL DISEASE

Q. Describe the etiology, pathology, clinical features, investigations, and management of inflammatory bowel diseases (ulcerative colitis and Crohn’s disease).

Q. IM16.15 Distinguish based on the clinical presentation Crohn’s disease from ulcerative colitis.

Inflammatory bowel disease (IBD) is an immune-mediated chronic intestinal condition. It **results from inappropriate mucosal immune activation**.

These include several conditions:

1. Ulcerative colitis (UC)
 2. Crohn disease (CD)
 3. *Others (uncommon) nonspecific inflammatory bowel disease*: Indeterminate colitis (15% patients with IBD) microscopic ulcerative, microscopic lymphocytic, and microscopic collagenous colitis.
- } Most important

Etiology and Pathogenesis of IBD

Genetic Factors

Genetic predisposition/susceptibility contributes to IBD

1. Familial:

- **Crohn’s disease**: Genetic factors **play a prominent role**. The concordance rate for monozygotic twins is about 50%.
- **Ulcerative colitis**: Genetic factors are **less prominent than in Crohn’s disease**. The concordance of monozygotic twins is only 16%. Concordance for dizygotic twins for both Crohn’s disease and ulcerative colitis is less than 10%.

2. **Susceptibility genes: Genes associated with innate immunity and autophagy (e.g., *NOD2/CARD15*, *ATG16L1*, and *IRGM*):** Actions of *NOD2/CARD15* protein are: (1) **prevents the entry of microbes** into the wall of the intestine; (2) **regulates innate immune responses**; and (3) prevents excessive immune response by luminal microbes.

Environmental Factors

Include both the local microenvironment (intestinal microflora) and the nutritional environment.

1. Intestinal microflora/microbiota:

- **Hygiene hypothesis:** The **gut lumen** contains **abundant commensal bacteria** and its composition within individuals remains stable for several years. This intestinal microflora **can be modified by diet and disease. IBD is associated with an alteration in the bacterial flora.**
- **Improved food storage conditions and decreased food contamination:** It has **reduced frequency of enteric infections**, with “clean” environment in the intestinal lumen → the **immune system may not be exposed to microorganisms** (pathogenic or non-pathogenic) → **inadequate development of regulatory processes** to limit mucosal immune responses.

2. **Smoking:** CD patients are more likely to be smokers, and smoking exacerbates CD. In contrast, an increased risk of UC is observed in non- or ex-smokers.

Host Factors

These include epithelial defects, impaired mucosal barrier, abnormal intestinal defensins, and immune dysregulation.

Psychosocial Factors

It can contribute to worsening of symptoms in IBD. Major life events (e.g., illness or death in the family, divorce, or separation) are associated with an increase in symptoms such as pain, bowel dysfunction, and bleeding.

Ulcerative Colitis

Definition

Ulcerative colitis is a severe-ulcerating inflammatory disease. It is limited to large intestine (the **colon and rectum**). It is characterized clinically by recurrent attacks (exacerbations and remissions) of bloody diarrhea. Pathologically, the inflammatory response is found only in the **mucosa and submucosa** of the intestinal wall and shows chronic destruction of crypts.

Pathology

Site (Fig. 5.13): Ulcerative colitis primarily involves the colonic mucosa. Usually **involves the rectum** (involved in 95% of cases) **and extends proximally** for a variable distance in a continuous fashion to involve part or the entire colon. Skip lesions are not seen in UC.

- **Pancolitis:** If entire colon is involved by the disease, then it is termed **pancolitis**.
- **Left-sided colitis:** When disease involves only the left-side of colon.
- **Extensive colitis:** When disease involves colon up to the hepatic flexure.
- **Proctitis/proctosigmoiditis:** When the disease is limited to the rectum alone, then it is known as **ulcerative proctitis**. When the process involves rectum and sigmoid colon, then it is termed **ulcerative proctosigmoiditis**.
- **Backwash ileitis:** The small intestine is normal. However, in severe cases of pancolitis mild mucosal inflammation of the distal ileum may be present and is termed as **backwash ileitis**.

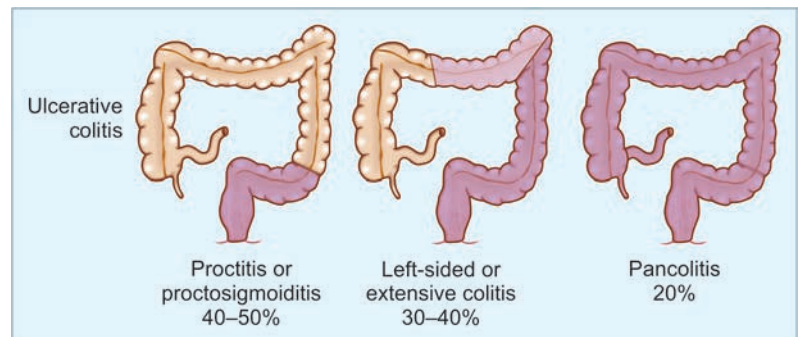


FIG. 5.13: Patterns of distribution in ulcerative colitis.

Microscopy

- **Chronic inflammatory infiltrates: Lymphocytes, plasma cells, and macrophages in the lamina propria.** The density of plasma cells is more in the basilar region of the lamina propria (**basal plasmacytosis**) and extends into the muscularis mucosae.
- **Changes in the crypt:** These include **cryptitis, crypt abscesses, destruction of crypts, and distortion of crypt architecture**. Chronic UC may show epithelial dysplasia.

- **Toxic megacolon:** In **severe cases** (fulminant disease-toxic megacolon), inflammation may damage the muscularis propria and disturbs neuromuscular function and leads to **colonic dilation** and **toxic megacolon** that can undergo perforation. The colon is dilated, walls are thin, and with denuded mucosa and colon may rupture.

Clinical Features of UC

Q. Write short essay/note on clinical features of ulcerative colitis.

Major symptoms of UC: The first attack is usually the most severe and the disease is later characterized by relapses and remissions.

- **Diarrhea:** Colonic motility is altered due to inflammation. When the disease is severe, patients develop diarrhea and pass liquid stool containing blood, mucus, pus, and fecal matter. Diarrhea is often nocturnal and/or postprandial.
- **Rectal bleeding:** Patients with proctitis usually pass fresh blood. When the disease extends beyond the rectum, blood is usually mixed with stool or grossly bloody diarrhea may be noted.
- **Tenesmus** or urgency with a feeling of incomplete evacuation.
- **Passage of mucus**
- **Crampy abdominal pain:** Severe pain is not a prominent symptom, but some patients with active disease may present with vague lower abdominal discomfort or mild central abdominal cramping. Severe cramping and abdominal pain can develop when the attack is severe.
- **Others:** These include anorexia, nausea, vomiting, malaise, fever, and weight loss. There may be symptoms and signs of dehydration and anemia. In severe cases, the patient is toxic with fever, tachycardia, and signs of peritoneal inflammation.

Severity of symptoms: It reflects the extent of disease and the intensity of inflammation.

Duration of symptoms: Although UC can present acutely, symptoms are usually present for weeks to months.

Exacerbations and remissions: They are characteristic features. Relapses are provoked by emotional stress, intercurrent infection, gastroenteritis, antibiotics, or NSAID therapy.

Q. Write short note/essay on extraintestinal manifestations of ulcerative colitis.

Extraintestinal manifestations (refer Table 5.57 and Fig. 5.15).

Risk of carcinoma colon: The incidence is high, especially in patients with total colitis, disease of more than 10 years duration and early age of onset.

Signs:

- Tenderness on palpation directly over the colon, especially in the left iliac fossa.
- Signs of proctitis include tender anal canal and blood on rectal examination.
- *Signs of toxic megacolon:* It is a serious complication and its signs include severe pain and bleeding, and signs of peritonitis, if there is associated perforation.

Investigations

Q. Write short essay/note on investigation of ulcerative colitis.

They are required for confirmation of the diagnosis, know the distribution and activity, and identify complications.

Laboratory findings

- **Anemia:** Decrease in hemoglobin may be due to blood loss or malabsorption (nutritional deficiencies) of iron, folic acid, or vitamin B₁₂. It may be associated with raised platelet count.
- **Raised ESR and acute-phase reactants** [C-reactive protein (CRP)], are nonspecific serum inflammatory markers and are observed during exacerbations. They are sometimes used to monitor the activity of disease.
- **Total WBC count:** Moderate leukocytosis is observed in active disease, but a marked leukocytosis suggests an abscess or another suppurative complication.
- **Hypoalbuminemia:** Reduced serum albumin concentration may be due to protein-losing enteropathy, inflammatory disease, or malnutrition (poor nutrition).
- **Stool examination**
 - *Stool microscopy and culture:* To exclude infective pathology and *C. difficile* toxin.
- Blood cultures in patients with septicemia.
- **Serological tests:** These are supportive and not used independently as markers for diagnose IBD.
 - **Anti-Saccharomyces cerevisiae antibodies (ASCA):** These are antibodies to yeast. They are observed in 40–70% of patients with Crohn's disease and in less than 15% of patients with ulcerative colitis. Presence of raised ASCA immunoglobulin A (IgA) and IgG titers is highly specific for Crohn's disease.
 - **Perinuclear antineutrophil cytoplasmic antibodies (pANCA):** They are detected in 10–20% of colon-predominant Crohn's disease, and in 55–70% of patients with ulcerative colitis.
 - **Antigoblet cell autoantibodies:** Found in 30–40% cases of ulcerative colitis and Crohn's disease.

- **Other tests:**

- **Fecal lactoferrin:** It is a very sensitive and specific marker for detecting intestinal inflammation.
- **Fecal calprotectin:** Its levels correlate with inflammation and helpful in predicting relapses.

Endoscopy

Endoscopy with mucosal biopsy is the gold standard investigation for ulcerative colitis.

Colonoscopy: For mild-to-moderate disease and patient not having an acute flare, colonoscopy is preferable to flexible sigmoidoscopy because it can assess the disease extent and activity. Colonoscopy helps in assessing the extent and activity of ulcerative colitis.

Sigmoidoscopy:

- **During early phase of the mild disease:** (1) Uniform continuous involvement of the mucosa, (2) diffuse mucosal erythema/hyperemia, (3) loss of the normal mucosal vascular pattern, granularity, and edema, and (4) exudate of mucus, pus and blood on the mucosa.
- **As the disease becomes severe:** Mucosa becomes more friable, bleeds easily on touch, and may ulcerate. Pseudopolyps represent epithelial regeneration and may develop after recurrent attacks in patients with long-standing disease.

Biopsy

Histological changes develop more slowly than clinical features. Biopsies help in diagnosis and to know the disease extent. Rectal biopsy shows mucosal inflammation and histological abnormalities are most severe in the distal colon and rectum.

Chromoendoscopy: Used for diagnosis of dysplasia in patients with long-standing UC.

Radiological features

- **Plain abdominal X-ray:** Useful to exclude toxic dilatation of colon.
- **Barium enema in ulcerative colitis:**
 - Earliest change in UC: Fine mucosal granularity
 - Later, the mucosa shows thickening and superficial ulcers. Deep ulcers appear as 'collar-button' ulcers, which indicate that the ulcer has penetrated the mucosa.
 - Haustral folds may be normal in initial phase of the disease, but later become edematous and thickened.
 - Long-standing disease may show loss of haustration, pseudopolyps, narrowing and shortening of the colon.
- **CT scan:** It is not so helpful when compared to endoscopy and barium enema in UC. CT scanning shows: mild mural thickening (<1.5 cm), inhomogeneous wall density, no thickening of small bowel, increased perirectal and presacral fat and target appearance of the rectum.

Biochemical abnormalities

- Electrolyte abnormalities
- Abnormal liver function tests

Q. Discuss severity staging of ulcerative colitis.

Q. Write short essay/note on management of ulcerative colitis.

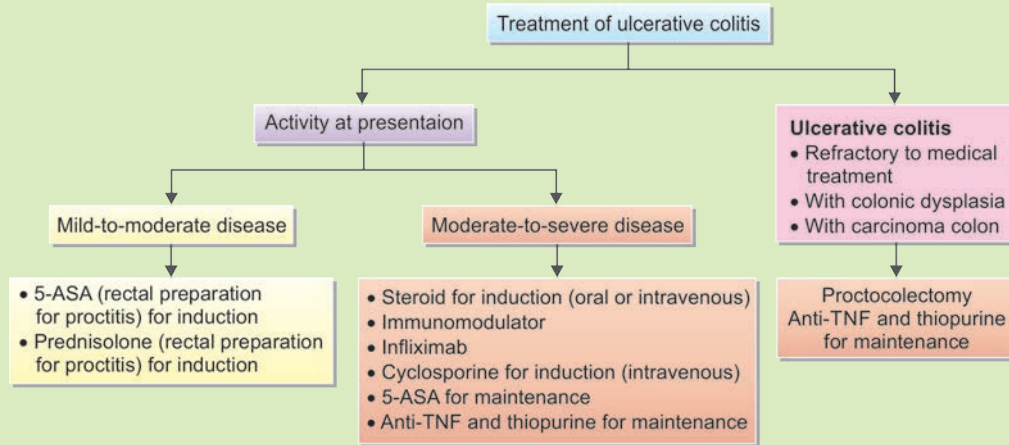
Q. IM16.16 Describe and enumerate the indications, pharmacology, and side effects of pharmacotherapy in UC, including immunotherapy.

Treatment (Flowchart 5.10) of Ulcerative Colitis

Whenever possible, IBD should be approached by a multidisciplinary team consisting of physicians, surgeons, radiologists, and dietitians. IBDs are life-long conditions and have psychosocial implications. Truelove and Witts' severity index of ulcerative colitis is presented in Table 5.49.

TABLE 5.49: Truelove and Witts' severity index of ulcerative colitis.

Features	Mild	Moderate	Severe
Bowel movements (number per day)	Fewer than 4	4–6	6 or more plus at least one of the features of systemic upset (marked with * below)
Blood in stools	No more than small amounts of blood	Between mild and severe	Visible blood
Pyrexia (temperature greater than 37.8°C)*	No	No	Yes
Pulse rate greater than 90 bpm*	No	No	Yes
Anemia*	No	No	Yes
Erythrocyte sedimentation rate (mm/hour)*	30 or below	30 or below	Above 30

FLOWCHART 5.10: Algorithm for treatment of ulcerative colitis.

(5-ASA: 5-aminosalicylate agents; TNF: Tumor necrosis factor)

Medical Management of UC

- Aims of medical therapy (**Box 5.14**)
- **5-Aminosalicylate (5-ASA) agents (Table 5.50):**
 - Available as oral tablets or topical (enema/suppository) preparation (for rectal and sigmoid disease).
 - These agents include 5-aminosalicylic acid (5-ASA) or mesalazine alone, or combination of 5-ASA with a carrier which releases 5-ASA after splitting by bacteria in colon (sulfasalazine, olsalazine, and balsalazide).
 - Topical mesalazine is the initial preferred agent in mild-to-moderate. Ulcerative proctitis/proctosigmoiditis, for induction as well as maintenance of remission. It acts as a topical anti-inflammatory within the lumen of the intestine, controls acute exacerbation, maintains remission, and prevents relapses. Maintenance therapy may decrease the risk of colorectal cancer.
 - Patients who are unwilling or unable to tolerate topical mesalazine can be started an oral 5-ASA medication. Oral 5-ASA should also be added in those patients who do not show remission after 4 weeks of topical therapy.
 - For patients with left-sided or extensive mildly-to-moderately active UC a combination of an oral 5-ASA agent plus rectal mesalazine is used. **Sulfasalazine** and high dose **mesalazine** are the most frequently used oral agents. Sulfasalazine is the combination of a sulfapyridine (acting as a “carrier” that allows 5-ASA to be delivered into the colon) with 5-ASA (active agent). Side effects include: nausea, dyspepsia, hair loss, headache, worsening diarrhea, and hypersensitivity reactions.
 - Sulfa-free aminosalicylate preparations (e.g., olsalazine and balsalazide): They deliver higher amounts of the active ingredient of sulfasalazine (5-ASA, mesalamine) to the site of active disease in the bowel and have limited systemic toxicity.
 - Remission is usually within 2–4 weeks of therapy. If remission is not induced with oral/topical ASA combination within 2–4 weeks, then a topical steroid (Budesonide) is added.

Box 5.14: Aims of medical therapy in ulcerative colitis.

- Treat acute attacks and reduce inflammation
- Prevent relapses and maintain clinical remission
- Detect carcinoma at an early stage
- Select patients for surgery

TABLE 5.50: Various oral 5-ASA (5-aminosalicylate agents) preparations used in ulcerative colitis.

Preparation	Dosage
Azo-bond	
Sulfasalazine	3–6 g (acute) 2–4 g (maintenance)
Olsalazine	1–3 g
Balsalazide	6.75–9 g
Delayed-release	
Mesalamine	2.4–4.8 g (acute) 1.6–4.8 g (maintenance)
Controlled-release	
Mesalamine	2–4 g (acute) 1.5–4 g (maintenance)
Delayed and extended-release	
Mesalamine	1.5 g (maintenance)

Corticosteroids

- Oral or parenteral corticosteroids are beneficial in inducing remission in moderate-to-severe ulcerative colitis.
- They have no role in maintaining remissions. During clinical remission, steroids should be tapered according to the clinical activity.

Local Treatment

- Topical applications of steroid are also beneficial for distal colitis and as an adjunct in patients with involvement of rectal involvement associated with involvement of proximal colon.
- Hydrocortisone or prednisolone enemas, suppositories or foam may control active disease. Budesonide-MMX uses colonic release technology to extend the application of budesonide throughout the colon.
- **Duration of treatment:** 3–6 weeks.

Systemic Treatment

- **Oral:** Active UC that is unresponsive to 5-ASA therapy is treated with oral prednisone at doses of 40–60 mg/day for 3–6 weeks.

- **Parenteral:**

- **Intravenous:** Hydrocortisone 300 mg/day or methylprednisolone, 40–60 mg/day are given as a constant infusion in severe cases.
- **Intramuscular or subcutaneous injection:** Long-acting corticotrophin is used in the treatment of relapses.
- **Tapering of dose:** Steroids once started are gradually tapered and withdrawn, normally at a rate of no more than 5 mg/week.

Antibiotics

No role in the treatment of active or quiescent UC.

Immune Modulators

Indications:

- Patients who remain symptomatic despite 5-ASA therapy.
- Patients who have moderate-to-severe ulcerative colitis.

Drugs

1. Azathioprine and 6-mercaptopurine (6-MP):

- *Usefulness are as follows:*
 - ♦ Patients who require two or more corticosteroid courses within a year.
 - ♦ Relapse of disease as the dose of prednisolone is reduced below 15 mg.
 - ♦ Relapse within 6 weeks of stopping corticosteroid.
- *Dosage:* Azathioprine 2–3 mg/kg/day and 6-mercaptopurine 1.5 mg/kg/day.
- *Disadvantage:* Slow clinical response and may not be evident for as long as 12 weeks.
- *Side effects:* These include allergic reactions, pancreatitis, myelosuppression, infections, hepatotoxicity, and malignancy (lymphoma).

2. Methotrexate: It is a folic acid antagonist useful in patients who do not respond to azathioprine.

3. Cyclosporine (CSA): It inhibits both the cellular and humoral immune systems. CSA blocks the production of IL-2 by T-helper cells. CSA prevents activation of T cells by inhibiting calcineurin and indirectly inhibits B cell function by blocking helper T cells.

4. Tacrolimus: It is a macrolide antibiotic with immunomodulatory properties similar to CSA.

Biologics

Used to treat moderately to severely active ulcerative colitis that has responded inadequately to conventional therapy including corticosteroids and mercaptopurine or azathioprine, or who cannot tolerate, or have contraindications for, such therapies. Common side effects include headaches, nausea, vomiting, diarrhea, skin rash, weight gain from fluid retention, muscle aches, etc. Another important side effect is reactivation of TB. Hence, prior screening for dormant/active TB is mandatory. Screening is using Mantoux test or Quantiferon-TB Gold assay (in WHO—BCG Vaccine countries).

Drugs

1. Anti-TNF agents: Infliximab (given IV) is the most commonly used agent. Antitumor necrosis factor (anti-TNF) therapy can prevent or reduce rates of colectomy. Other agents include adalimumab and golimumab (both given subcutaneously).

2. Vedolizumab—MAB against α 4/ β 7 integrin. Preferred in elderly (>65 years), and those with history of malignancy or infectious complications.

3. Ustekinumab—MAB against interleukin 12/23.

Small Molecule Drugs

Tofacitinib—oral small molecule Janus kinase inhibitor, used for treating adults with moderate to severe UC who have failed or are intolerant to anti-TNF agent-based therapy.

General supportive measures

- *Parenteral nutrition:* It is necessary in seriously ill patients and parenteral broad-spectrum antibiotics may be required in septicemia.
- Correction of dehydration and electrolyte imbalance
- Antispasmodic medications and antidiarrheal (loperamide) for mild diarrhea (to be avoided in severe colitis).

Surgical management (Table 5.51)

- Surgery involves removal of the entire colon (colectomy) and rectum and is a curative procedure. It is indicated UC does not responded adequately to medical therapy. Emergency colectomy may be necessary for toxic megacolon.
- Elective surgical procedure is either panproctocolectomy with ileostomy, or proctocolectomy with ileal-anal pouch anastomosis.

Follow-up

- Extensive colitis of more than 10 years duration: Colonoscopy with multiple biopsies is recommended every 3 years to assess for dysplasia or malignant changes.
- UC for more than 20 years: Colonoscopy is recommended every 2 years. Biopsies from entire colon and samples from suspicious areas. If the biopsy shows high-grade dysplasia, total colectomy is performed.

Q. IM16.17 Describe and enumerate the indications for surgery in inflammatory bowel disease.

Crohn's Disease

Q. Discuss the etiology, pathology, clinical features, investigations, and management of Crohn's disease.

Etiology and Pathogenesis

Refer etiology and pathogenesis of IBD.

Pathology

Gross

- **Site (Fig. 5.14):** It can involve any area of the gastrointestinal (GI) tract from mouth to anus. Unlike UC, rectum is spared in CD. Most commonly involved sites are:
 - Terminal ileum and right side of colon (40–55%)
 - *Small intestine/bowel disease alone* (30–40%): Terminal ileum alone or ileum and jejunum.
 - Colon alone (15–25%)

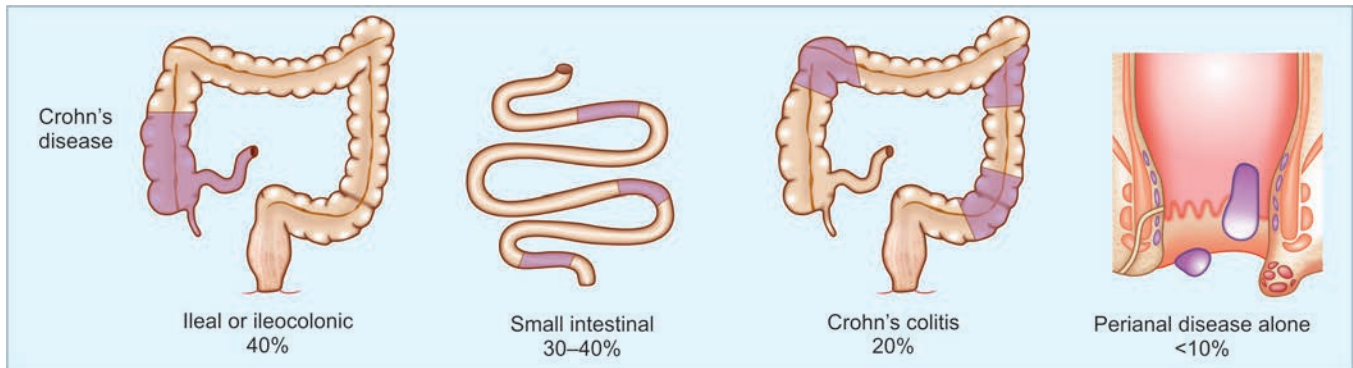


FIG. 5.14: Patterns of distribution in Crohn's disease.

- **Number of lesions:** Usually, the lesions are **multiple**.
- **Skip lesions:** The intestinal involvement is **discontinuous in which inflamed segments of intestine are sharply demarcated/separated by apparently normal intervening bowel/intestine** in between.
- **Intestinal wall:** The involved intestinal wall shows **fibrotic thickening and is rubbery** and appears like a hosepipe and produces a characteristic radiological sign known as the **"string sign"**.
- **Transmural inflammation:** Inflammation involves all layers of the bowel wall.
- **Mucosal lesions:** The earliest lesion is the **aphthous ulcer** which becomes deeper, transmural, and discrete. Multiple lesions may coalesce longitudinally to form linear or **serpentine** (snake-like) ulcers and produce characteristic **"cobblestone"** appearance. Later the ulcer becomes deeper and form linear clefts or **fissures**. Similar to UC, pseudopolyps may also form in CD.
- **External surface:** The fat may encircle around the antimesenteric serosal surface producing a pattern known as **creeping fat**.
- **Adhesion and fistulae:** Involved loops of bowel are often adherent to each other. **Fissures** frequently develop between mucosal folds. They may extend deeply to become **fistula tracts or sites of perforation**. These fistulas may also penetrate from the bowel into other organs, including the bladder, uterus, vagina, and skin.

Microscopic features

- **Chronic inflammation** by **lymphocytes, plasma cells, and macrophages**
- **Mucosal ulcerations:** They are small, superficial (aphthous ulcers).
- **Crypt abscesses:** Clusters of neutrophils within a crypt and are often associated with crypt destruction.
- **Noncaseating granulomas**

Clinical Features of Crohn's Disease

Symptoms due to involvement of small intestine or ileum and right colon (ileocolitis). CD present with one of two patterns of disease: (1) penetrating fistulous pattern or (2) fibrostenotic obstructing pattern. The clinical manifestations depend on the site of disease.

TABLE 5.51: Indications for surgery in ulcerative colitis.

Intractable/extensive colitis	Toxic megacolon	Colonic perforation
Severe colonic hemorrhage	Colon cancer prophylaxis	Colon dysplasia/cancer
Frequent relapses	Chronic damage to bowel-strictures	

- Crohn's disease is a chronic disease with exacerbations and remissions.
- Major symptoms include abdominal pain (typically in the right lower quadrant), diarrhea, and weight loss (due to malabsorption and patients avoid food because eating provokes pain).
- Ileal Crohn's disease may cause subacute or acute intestinal obstruction and patients present with recurrent episodes of colicky abdominal pain, abdominal distention, nausea, vomiting, and excessive borborygmi.
- Sparing of rectum and the presence of perianal disease are observed in about 30% of patients in the form of fistulas, abscesses, fissures, and skin tags. It presents with pain, discharge, and fever (if there is an abscess).

Physical examination

- Signs depend on the location and severity of the disease process.
- Abdominal tenderness in the right lower quadrant (classical), accompanied by fullness, guarding, or a mass depending on the severity of inflammation.
- Palpable mass per abdomen and rectally reflects adherent loops of intestine and abscess.
- Features of malabsorption such as weight loss and anemia (iron, folic acid, and vitamin B₁₂ malabsorption)
- Sodium, potassium, water, magnesium and zinc deficiency may develop due to chronic diarrhea.
- Stool usually does not show frank blood, mucus or pus unless colon is involved.
- Oral ulcers

Investigations

- **Anemia:** Anemia may result from chronic disease, blood loss or nutritional deficiencies of iron, folate, or vitamin B₁₂. It may be normochromic normocytic or macrocytic or hypochromic anemia depending on the deficiency/cause.
- **Leukocytosis:** Modestly raised leukocyte count indicates active disease, but a marked leukocytosis suggests an abscess or other suppurative complication.
- **Raised ESR (erythrocyte sedimentation rate) and C-reactive protein:** They are nonspecific serum inflammatory markers and may be used to monitor the activity of disease.
- Abnormal liver function tests
- **Hypoproteinemia/hypoalbuminemia:** Indication of malnutrition.
- **Malabsorption:** Ileal disease results in malabsorption of vitamin B₁₂ (reduced serum vitamin B₁₂ level) and Schilling test is done to determine the malabsorption of vitamin B₁₂.
- **Serologic markers:** They are supportive but may not independently used for the diagnose IBD.
 - **Anti-Saccharomyces cerevisiae antibodies (ASCA):** These are antibodies to yeast, are present in 40–70% of patients with Crohn's disease (positive in only 10–15% cases of ulcerative colitis).
 - **Perinuclear antineutrophil cytoplasmic antibodies (pANCA):** They are present in 20% of patients with CD (colon predominant disease), and in 55% of patients with ulcerative colitis.
 - Antigoblet cell autoantibodies in 30–40% cases of Crohn's disease.
 - Antigliadin antibodies in about 40–50% cases (e.g., antilaminaribioside, antimannobioside antibodies).
- **Stool examination:** Stool microscopy and culture to exclude infectious causes of diarrhea.
- **Sigmoidoscopy and colonoscopy in Crohn's disease:**
 - Segmental inflammation with normal mucosa between (skip lesions).
 - Discrete, aphthous, or deep ulcers and occasionally connected to one another, forming longitudinal stellate, serpiginous, and linear ulcers.
 - Perianal disease (fissures, fistulas and skin tags) or sparing of the rectum.
 - Strictures are common.
 - Biopsy of involved region (e.g., colonic mucosa, ileal mucosa, anal skin tags, and perianal inflammatory lesions) shows typical noncaseating granulomatous inflammation (tuberculosis must be excluded) and infiltration by lymphocytes and plasma cell.
 - Wireless capsule endoscopy (WCE) allows direct visualization and identification of inflammation in the entire small bowel mucosa. However, it should be avoided in the presence of small bowel strictures.

Radiological findings in Crohn's disease

Barium meal follow thorough and barium enema

- *Early radiographic findings in the small bowel:*
 - Thickened folds and aphthous ulcerations
 - Cobblestone appearance of mucosa resulting from longitudinal and transverse ulcerations.

- *More advanced disease:*
 - Transmural inflammation leads to decreased luminal diameter and limited distensibility.
 - Rigidity of involved segments and multiple strictures.
 - As ulcers progress deeper, they can lead to fistula formation. Radiographic “String sign” is due to long areas of transmural inflammation and fibrosis, producing narrowing of lumen of involved small intestine.
 - *Segmental nature of CD:* It results in wide gaps of normal or dilated bowel between involved segments (skip lesions).
 - Inflammatory masses and abscesses may be found.
 - Fistulous tracts

High-resolution ultrasound and spiral CT scanning

It helps in defining thickness of the bowel wall and intra-abdominal abscesses.

Radionuclide scan

Using gallium-labeled polymorphs or indium-labeled leukocytes identify intestinal and colonic disease and localize extraintestinal abscesses.

Q. Discuss Crohn’s disease severity index.

Crohn’s Disease Activity Index

Clinical and laboratory variables	Weightage
Number of liquid or soft stools each day for 7 days	× 2
Abdominal pain/graded from 0–3 on severity each day for 7 days	× 5
General well-being subjectively assessed from 0 (well) to 4 (terrible) each day for 7 days	× 7
Presence of complications* *One point each is added for each set of complications:	× 20
➤ The presence of joint pains of frank arthritis	
➤ Inflammation of the iris or uveitis	
➤ Presence of erythema nodosum, pyoderma gangrenosum, or aphthous ulcers	
➤ Anal fissures, fistulae or abscesses	
➤ Fever (>100°F) during the previous week	
Taking Lomotil or opiates for diarrhea	× 30
Abdominal mass (0 as none, 2 as questionable, 5 as definite)	× 10
Absolute deviation of hematocrit from 47% in men and 42% in women	× 6
% deviation from standard weight	× 1

Remission of Crohn’s disease is defined as a CDAI of less than 150. Severe disease was defined as a value of greater than 450. Response defined as a fall of the CDAI of greater than 70 points.

Treatment (Flowchart 5.11)

Medical Management of Crohn’s Disease

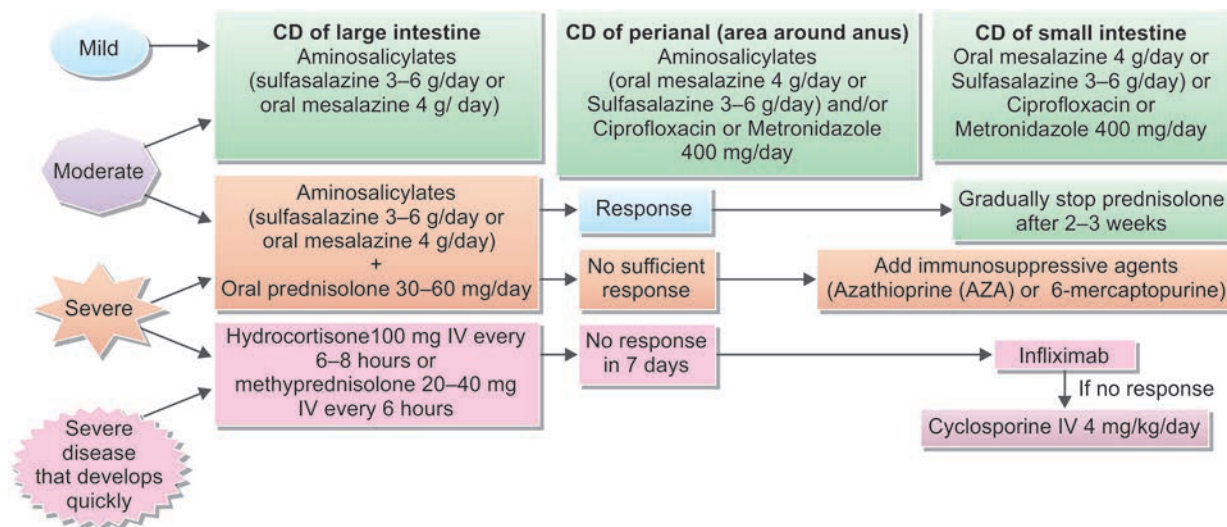
General Considerations

- Stop cigarette smoking
- Diarrhea can be controlled with loperamide, codeine phosphate, or co-phenotrope.
- Long-standing diarrhea may be due to bile acid malabsorption and is treated with bile salt sequestrants (e.g., cholestyramine).
- *Anemia:* Due to vitamin B₁₂, folic acid, or iron deficiency, should be treated accordingly.

Induction and Maintenance of Remission

- **Glucocorticosteroids:** Used to induce remission in moderate and severe Crohn’s disease. Oral prednisolone 30–60 mg/day, which is gradually tapered and withdrawn. Mild-to-moderate ileocecal, CD should be treated by controlled-release steroids such as budesonide (9 mg by mouth once a day).
- **Aminosalicylates:** 5-ASA (e.g., mesalazine 4 g/day or sulfasalazine 3–6 g/day) as maintenance in mild-to-moderate ileocolonic disease. However, they are not effective for maintaining remission.
- **Antibiotics:** Ciprofloxacin and metronidazole 400 mg are used in abscess and perineal Crohn’s disease.
- **Nutritional therapy:** Dietary antigens may stimulate the immune response in the intestinal mucosa. Bowel rest and total parenteral nutrition (TPN) are effective in inducing remission but are not effective as maintenance therapy. Enteral diets with a low fat and a low linoleic acid content (not palatable) may induce remission are similar to steroids.

FLOWCHART 5.11: Treatment approach to Crohn's disease.



- **Immunosuppressive agents:** Azathioprine (AZA) and its metabolite 6-mercaptopurine (6MP) and newer drugs (e.g., mycophenolate mofetil) suppress the proliferation of T- and B-lymphocytes. They are effective in inducing and maintaining remission and have steroid-sparing properties.
- **Biologic therapy:** Anti-TNF therapy is reserved for moderate-to-severe Crohn's disease that failed to respond to other therapies (infliximab, adalimumab, and peg certolizumab) and in CD with refractory perianal and enterocutaneous fistulas. Vedolizumab an integrin antibody is preferred in elderly and in those with malignancies or infectious complications. Another monoclonal antibody (mAb) ustekinumab (anti-IL-12/23) is used when standard anti-TNF therapy fails.

Maintenance of Remission

- Aminosalicylates
- Azathioprine, 6MP, and mycophenolate mofetil

Surgical Management (Table 5.52)

- **Small intestine:** Strictures can be widened (stricturoplasty)/minimal resection and anastomosis.
- **Colorectal:**
 - Subtotal colectomy and ileorectal anastomosis, if rectum is spared.
 - Pancolectomy with ileostomy, if whole colon and rectum are involved

Assessment of severity of Crohn's disease is presented in **Table 5.53**.

TABLE 5.52: Indications for surgery in CD.

Small intestine	Colon and rectum
➤ Stricture and obstruction not responding to medical therapy ➤ Severe hemorrhage ➤ Refractory fistula ➤ Abscess	➤ Intractable/fulminant disease ➤ Perianal disease not responding to medical therapy ➤ Refractory fistula ➤ Colonic obstruction/perforation ➤ Colon dysplasia/cancer

TABLE 5.53: Severity of Crohn's disease.

Mild–moderate	Moderate–severe	Severe/fulminant
CDAI 150–220	CDAI 220–450	CDAI >450
Ambulatory Able to tolerate oral alimentation without manifestations of dehydration Systemic toxicity (high fevers, rigors, and prostration) abdominal tenderness painful mass Intestinal obstruction, or >10% weight loss	Failed to respond to treatment for mild–moderate disease or Has more prominent symptoms of fever, significant weight loss, abdominal pain or tenderness, intermittent nausea or vomiting (without obstructive findings), or significant anemia	Persistent symptoms despite treatment with corticosteroids/biologics as outpatients Or Has high fevers, persistent vomiting, intestinal obstruction, significant peritoneal signs, cachexia or Abscess

(CDAI: Crohn's disease activity index)

Toxic Megacolon

Q. Give a brief account of toxic dilatation of colon (toxic megacolon).

Definition: Toxic megacolon is defined as a transverse or right colon with a diameter of greater than 6 cm, with loss of haustration.

Causes

- **Ulcerative colitis:** One of the most significant complications of UC. It can be triggered by electrolyte abnormalities, narcotics, antidiarrheal (diphenoxylate and loperamide) antispasmodics medications (reduce gastrointestinal motility), and barium enema.
- **Other rare causes:** (1) Diarrheas are due to *Clostridium difficile*, (2) *Campylobacter jejuni* gastroenteritis, (3) Shigellosis, (4) Chagas' megacolon, and (5) amebic colitis.

Clinical Features

- Features of severe colitis
- High fever, tachycardia, dehydration, electrolyte imbalance, and leukocytosis
- **Physical examination:** Patient is toxic and ill, postural hypotension, colonic dilatation, abdominal tenderness over the involved colon, and absent or hypoactive bowel sounds.
- **Plain radiograph of abdomen:** Shows colonic dilatation with transverse colonic diameter greater than 6 cm. Air may be observed in the wall of the colon.

Treatment

- **Medical therapy:** About 50% of cases will resolve with medical therapy alone
 - Nil mouth, and nasogastric suction
 - Intravenous fluids to replace water and electrolyte abnormalities
 - Broad-spectrum antibiotics (e.g., ampicillin/sulbactam) plus metronidazole
- Urgent colectomy may be necessary when there is no improvement with medical therapy

Complications

Perforation is the most dangerous of the local complications.

Complications of Inflammatory Bowel Disease

Q. Write short note on the complications of inflammatory bowel disease.

It may be local and systemic (extraintestinal).

Local Complications (Tables 5.54 and 5.55)

Conditions that mimic inflammatory bowel disease (ulcerative or Crohn's colitis) are presented in Table 5.56.

Extraintestinal Manifestations (Table 5.57 and Fig. 5.15)

Q. Write short note/essay on extraintestinal manifestations of inflammatory bowel disease.

TABLE 5.54: Local complications of Crohn's disease.

- **Fistula formation:** It may develop between loops of intestine and surrounding structures such as urinary bladder, vagina, and abdominal or perianal skin. Perforations and peritoneal abscesses are common
- **Stricture formation:** May produce intestinal obstruction
- **Iron-deficiency anemia** with colonic disease
- **Malabsorption:** Extensive involvement of the small intestine may result in loss of **protein and hypoalbuminemia, generalized nutrient malabsorption, or malabsorption of vitamin B₁₂ and bile salts**
- **Fibrosing strictures** in the terminal ileum are common
- **Acute complications:** Perforation abscesses and hemorrhage
- **Development of carcinoma** is rare. Risk of carcinoma colon is increased in patients with longstanding colonic disease
- Gallstones and **urinary oxalate stones**
- **Systemic secondary amyloidosis** rare

TABLE 5.55: Local complications of UC.

- **Toxic megacolon:** In fulminant cases, the inflammation and inflammatory mediators can damage the muscularis propria and disturb neuromuscular function. This may lead to colonic dilation and toxic megacolon, which may lead to perforation
- **Colorectal cancer:** Longstanding ulcerative colitis may develop colorectal cancer than the general population
- Hemorrhage leading to blood loss
- Electrolyte disturbances due to diarrhea

TABLE 5.56: Conditions that mimic inflammatory bowel disease (ulcerative or Crohn's colitis).

Infective

- **Bacterial:** *Salmonella*, *Shigella*, pseudomembranous colitis, *Campylobacter jejuni*, gonococcal proctitis, *Chlamydia proctitis* *E. coli* O157
- **Viral:** Herpes simplex proctitis, cytomegalovirus
- **Protozoal:** Amoebiasis

Noninfective

Ischemic colitis, collagenous colitis, radiation proctitis, Behcet's disease, diverticulitis, carcinoma colon, and NSAIDs

TABLE 5.57: Extraintestinal manifestations of IBD.

Complications	Crohn's disease	Ulcerative colitis
Ocular: Uveitis, scleritis, and episcleritis	+	+
Arthropathy: Pauciarticular, polyarticular, and ankylosing spondylitis	+	+
Skin: Erythema nodosum, pyoderma gangrenosum	+	+
Primary sclerosing cholangitis	+	+
Thromboembolic disease	+	+
Renal: Nephrolithiasis	+	+
Bone: Osteomalacia, osteoporosis	+	–
Oral ulcers	+	–
Vitamin B ₁₂ deficiency	+	–

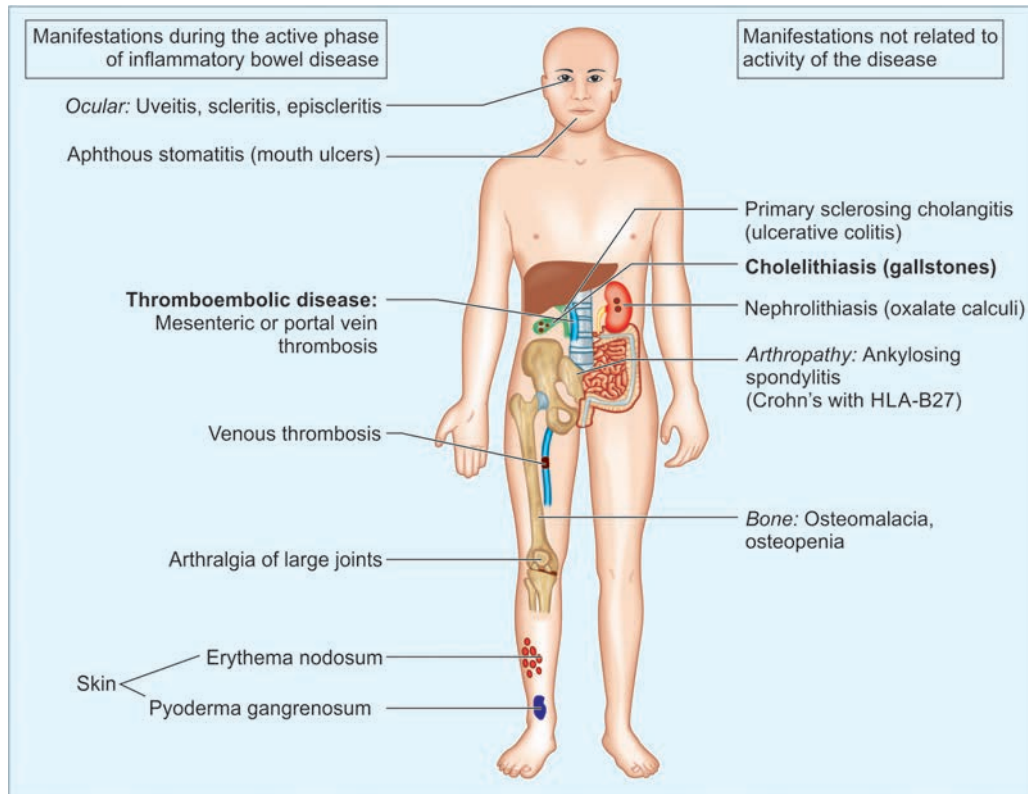


FIG. 5.15: Systemic complications of inflammatory bowel disease.

Differences between Ulcerative Colitis and Crohn's Disease of the Colon (Table 5.58)

Q. Write short note on the differences between ulcerative colitis and Crohn's disease of the colon.

PROBIOTICS AND PREBIOTICS

Q. PE24.5 Discuss briefly about probiotics and prebiotics.

Probiotics

Definition: Probiotics are defined as live microorganisms which are beneficial to its host. Throughout their journey in the digestive tract, they need to be intact, so they can reach the intestines where they act to give their beneficial effects to the body.

Nature

- **Bacteria:** Probiotics are usually bacterial components of the normal intestinal flora of human beings (e.g., *Lactobacilli* and *Bifidobacterium infantis*). They produce lactate and short-chain fatty acids (e.g., acetate and butyrate) as end products of metabolism.
- **Yeast:** *Saccharomyces boulardii* is yeast.

Preparation

- **Yogurt:** It is commonly used probiotic in which is milk fermented by bacteria that convert lactose into lactic acid.

TABLE 5.58: Differences between ulcerative colitis and Crohn's disease of the colon.

Characteristics	Ulcerative colitis	Crohn's disease
Pathological features		
Involvement of bowel	Continuous involvement	Segmental involvement (skip lesions)
Inflammation	Restricted to mucosa and submucosa	Transmural
Noncaseating granulomas	Absent	Common
Crypt abscess	Common	Uncommon
Goblet cell depletion	Seen	Absent
Strictures, fissures and fistulae	Uncommon	Common
Mesenteric and lymph nodal involvement	Absent	Common
Clinical features		
Rectal bleeding	Common	Uncommon
Abdominal pain	Less common	More common
Abdominal masses	May be palpable	Not usually palpable
Fistulae, fissures and perianal skin tags	Uncommon	Common and characteristic
Small bowel involvement	Uncommon	Common
Rectal and colonic involvement	Common	Rare
Recurrence after surgery	Not common	Common
Toxic dilatation	Relatively common	Uncommon

- **Freeze-dried bacteria:** Probiotics are also available as freeze-dried bacteria (in capsule, tablet, or powder form). Their composition varies and most of them contain *Lactobacillus acidophilus* often with bifidobacteria. Other strains include *Bacillus coagulans*, *L. paracasei*, and *L. johnsonii*.

Limitations: Problems with quality control and formulation restrict the clinical availability of some probiotics.

Uses

- **Malnutrition:** Helps in normalizing the nutritional status of malnourished children. WHO suggested the use of yogurt in nutritional recovery.
- **Lactose intolerance:** Yogurt is preferred.
- **Prevention and treatment of antibiotic-associated diarrhea:** Probiotics containing *Saccharomyces boulardii* yeast may be useful to some extent.
- Irritable bowel syndrome (IBS) and colitis
- Improve immune function/immunity
- Necrotizing enterocolitis in neonates
- Speed treatment of certain intestinal infections
- Prevent and treat eczema in children
- Prevent or reduce the severity of colds and flu
- Prevent and treat vaginal yeast infections and urinary tract infections
- *Reduce bladder cancer recurrence:* Probably reduces the development of carcinoma of colon.

Prebiotics

Definition

Prebiotics are referred to as dietary fiber or carbohydrates that act as the food of probiotics. This is why it triggers bacterial growth and activity in the intestinal microflora. Therefore, it promotes good health of the host.

- Chemical substances (usually oligosaccharides) which beneficially nourish and act as a fertilizer to promote growth of the host's intrinsic probiotic bacteria. They encourage their growth.
- Prebiotics are special form of dietary fiber.
- Prebiotics are not digested (nondigestible) by the small intestine and nonmetabolized by nonprobiotic gut flora such as *Bacteroides* species and *Escherichia coli* present in the host.
- Synthetic prebiotics are oligosaccharides containing fructose or galactose, known as fructose oligosaccharides (FOS) and galactose oligosaccharides (GOS), respectively. Lactulose is also a prebiotic.
- These oligosaccharides are enriched with the combination of inulin.
- They may be added to foods, or combined with a probiotic.

Box 5.15: Potential uses of prebiotics.

- Relief of constipation
- Reduce intestinal pH
- Restore intestinal bacterial balance
- Increase in the absorption of calcium and magnesium
- Effect on blood cholesterol level
- Reduction of fasting glucose level in type 2 diabetes patients
- Reduce risk on colorectal cancer
- Effects on the immune system
- Better intestinal flora in infants
 - Lactulose: Useful in hepatic encephalopathy
 - Prevention of osteoporosis: Prebiotics may increase calcium absorption
 - Chronic constipation

Potential Uses of Prebiotics

See Box 5.15.

Synbiotics

- Synbiotics refer to the combination of the prebiotics and probiotics within a single product.
- Prebiotics are targeted to act on the large intestines while the probiotics are most effective in doing its role in the small intestine.
- The two agents have been said to act synergistically. So, the term synbiotics is used.

MISCELLANEOUS TOPICS

Gardner's Syndrome

- Condition in which familial adenomatous polyposis (FAP) is associated with predominant extraintestinal manifestations.
- Autosomal dominant disorder
- *Adenomatous polyps:* Numerous in colon and few in the stomach and duodenum

- **Extraintestinal lesions:** These include:
 - **Osteomas of mandible, skull, and long bones**
 - **Epidermal cysts, desmoid tumors, thyroid tumors, etc.**

Treatment: Prophylactic total colectomy.

Peutz-Jeghers Syndrome

- It is a rare autosomal dominant syndrome.
- Median age of presentation is 11 years
- It is characterized by:
 - **Multiple GI hamartomatous polyps:** Involves the entire GIT but are most common in the small intestine. Usually, present in the dozens.
 - **Mucocutaneous hyperpigmentation:** The melanotic pigmentation is seen in the mouth, buccal mucosa, and genitalia region.
- Peutz-Jeghers syndrome is associated with an **increased risk of several malignancies**. These include cancers of the colon, pancreas, breast, lung, ovaries, uterus, and testicles.
- **Genetics:** Inactivation (loss-of-function mutations) of the *STK11/LKB1* gene (tumor suppressor gene) detected in about 70% of affected families. The **hamartomas are not preneoplastic precursor lesions**.
- **Clinical presentation:** Abdominal pain, rectal bleeding, anemia, intestinal obstruction, and intussusception.
- **Diagnosis and treatment:** Intraoperative enteroscopy (IOE) and remove all small intestinal polyps by double balloon enteroscopy (DBE).

Indications, Contraindications and Complications of Upper Gastrointestinal Endoscopy (Table 5.59)

Q. IM16.12 List indications, contraindications, and complications of upper gastrointestinal endoscopy.

TABLE 5.59: Indications, contraindications, and complications of upper gastrointestinal endoscopy.

Diagnostic indications

- **Upper gastrointestinal bleeding:** Past or present; acute or chronic patients with splenomegaly or suspected chronic liver disease: For presence and grading of esophagogastric varices
- Upper abdominal distress that persists despite an appropriate trial of therapy
- Dysphagia or odynophagia
- Esophageal reflux symptoms that are persistent or recurrent despite appropriate therapy
- Persistent nausea and vomiting of unknown cause surveillance for malignancy, abnormal CT scan or barium meal gastric or esophageal ulcers
- Familial adenomatous polyposis adenomatous gastric polyps Barrett's esophagus
- Occult gastrointestinal bleeding
- After caustic ingestion to assess for acute injury dyspepsia over 55 years of age or with alarm symptoms atypical chest pain
- Weight loss
- Duodenal biopsies in the investigation of malabsorption and to confirm a diagnosis of celiac disease prior to commencement of gluten-free diet

Therapeutic indications

- Treatment of bleeding lesions
- Sclerotherapy or banding of varices
- Removal of foreign bodies
- Removal of selected polypoid lesions or superficial neoplasms
- Placement of feeding or drainage tubes
- Dilation of stenotic lesions
- Palliative treatment of stenosing neoplasms
- Drainage of pancreatic pseudocysts

Contraindications

- **Absolute contraindications**
 - Severe shock
 - Possible visceral perforation
- **Relative contraindications** (can be performed by experienced hands)
 - Recent myocardial infarction, unstable angina, and cardiac arrhythmia
 - Severe respiratory disease
 - Atlantoaxial subluxation

Complications

- Cardiorespiratory depression due to sedation
- Aspiration pneumonia
- Perforation

Indications for Colonoscopy (Table 5.60)

Q. List indications for colonoscopy.

Protein-losing Enteropathy

Q. Write short note on protein-losing enteropathy.

It is characterized by excessive loss of protein into the gut lumen, sufficient to cause hypoproteinemia.

Causes (Table 5.61): Most common in diseases with mucosal ulceration.

Clinical features: Peripheral edema and hypoproteinemia.

Investigations: Liver function tests are normal liver function and no proteinuria. Diagnosis can be confirmed by measuring fecal clearance of $\alpha 1$ -antitrypsin or ^{51}Cr -labeled albumin after intravenous injection. Other investigations depend on the underlying cause.

Treatment of the underlying disorder, with nutritional support and measures to control peripheral edema.

TABLE 5.60: Indications for colonoscopy.

Diagnostic	Therapeutic
➤ Unexplained rectal bleeding ➤ Unexplained lower gastrointestinal symptoms and signs, such as recent onset constipation or feeling of incomplete evacuation, especially in the elderly ➤ Polyp and cancer screening or follow-up ➤ Inflammatory bowel disease ➤ Stricture or colonic narrowing (with and without inflammatory bowel disease) ➤ Diverticular disease ➤ Infectious colitis ➤ Radiation colitis ➤ Ischemic colitis ➤ Endometriosis ➤ <i>Pneumatosis cystoides intestinalis</i>	➤ Polypectomy ➤ Foreign body removal ➤ Dilation of strictures in colon or terminal ileum ➤ <i>Therapeutic:</i> Hemostasis ➤ Tumor resection (palliative) ➤ Colonic decompression (volvulus and pseudo-obstruction)

TABLE 5.61: Causes of protein-losing enteropathy.

With erosions or ulceration of mucosa: Crohn's disease, ulcerative colitis, cancer of esophagus, stomach or colon, radiation damage, and lymphoma

Without erosions or ulceration of mucosa: Menetrier's disease, celiac disease, bacterial overgrowth, eosinophilic gastroenteritis, systemic lupus erythematosus, and tropical sprue

With obstruction of lymphatics: Intestinal lymphangiectasia, lymphoma, Whipple's disease, and constrictive pericarditis

Generalized Abdominal Distension (Box 5.16)

Q. List causes of abdominal distension.

Box 5.16: Causes of abdominal distension (7Fs).

Fat, fluid (ascites), flatus (obstruction/ileus), fetus (pregnancy), feces (constipation), full bladder, fatal neoplasm (organomegaly)

Causes of Oral Ulceration

Q. List causes of oral ulceration and aphthous ulcer.

Aphthous Ulceration

Aphthous ulcers are superficial and painful ulcers which can develop in any part of the mouth. Recurrent ulcers are common in women prior to menstruation. The cause is unknown. Other causes of oral ulceration are presented in **Table 5.62**.

TABLE 5.62: Causes of oral ulcers.

➤ Aphthous: Idiopathic Infection, premenstrual ➤ Infection: Fungal (candidiasis), viral (herpes simplex, HIV), bacterial (e.g., syphilis, tuberculosis) ➤ Drugs: Nicorandil, NSAIDs (nonsteroidal anti-inflammatory drugs), methotrexate, cytotoxic drugs, Stevens–Johnson syndrome ➤ Neoplasia: Carcinoma, leukemia, Kaposi's sarcoma	➤ Gastrointestinal diseases: Crohn's disease, celiac disease ➤ Dermatological conditions: Lichen planus, dermatitis herpetiformis ➤ Systemic diseases: Systemic lupus erythematosus, Behçet's syndrome, vitamin deficiencies ➤ Traumatic ulcers
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